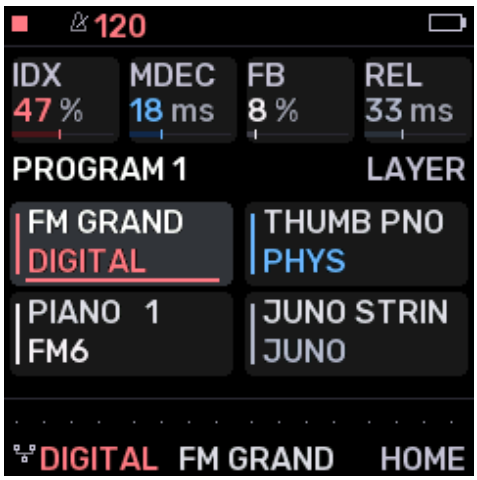
Illustrated user guide

Step-by-step operation, shortcuts, secondary functions and the eight factory programs.



HOME: four loaded sounds, one selected part.

Start with P1 Piano, P2 Epiano, P3 Lead Synth, P4 Oriental Lead, P5 Disco Synth, P6 Brass, P7 Strings or P8 Bass. Every program has four enabled parts with one main instrument.

Installer | Web editor | Source and support

Screens are generated by the production 240 x 240 firmware renderer with simulated panel/USB input. They are not photographs or proof of physical-device validation.

Find your next step

Use the bookmarks or the links below to jump between tasks.

Install and connect / 3

HOME controls and first sound / 5

The eight factory programs / 6

P1-P8: layer roles and playing tips / 7

Layering, mixer, pan and splits / 15

Save, name and load a program / 17

Edit, filters, envelopes, LFO and effects / 18

Scales, roots and custom microtuning / 21

Chords, voicings and strum / 23

USB pedals and MIDI controllers / 24

Scenes and setlists / 26

Arpeggiator and sequencer / 27

Editor, imports and backups / 28

Shortcut reference / 29

Troubleshooting and voice budget / 30

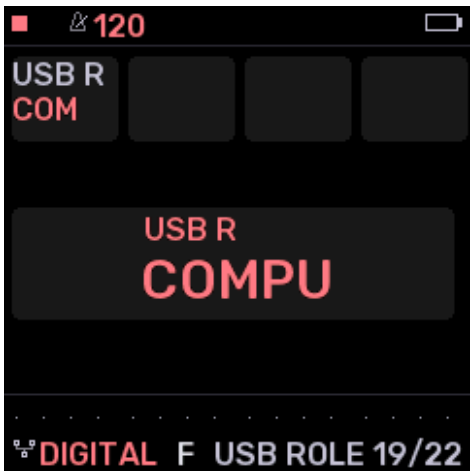
Research, validation and credits / 31

Complete screen-by-screen controls reference | Detailed page finder | Editor reference

Three different things: a sound preset is one sound. A program stores four sounds plus their performance setup. An editor library entry is a browser-side file or record until you send and save it.

Install or recover safely

Make a backup before changing firmware. Keep power connected throughout installation.



usb role

01 Connect the FM-1 to the computer with a USB data cable. If necessary, select GLO > USB ROLE > COMPUTER first.

02 Open the SynthMaven installer in Chrome or Edge. Close other applications that are using the FM-1 MIDI ports.

03 Choose INSTALL and follow the device/port prompts. Wait for completion before unplugging or restarting.

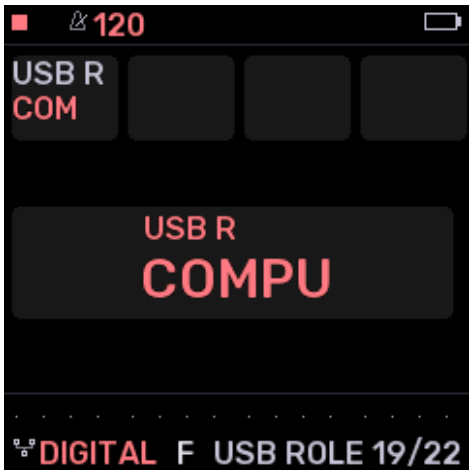
04 Check that the installed firmware reports SYNTHMAVEN 2.2.2. Use the matching editor and this guide.

05 If normal firmware cannot start, hold OCT- while powering on to enter USB rescue, then follow the installer recovery prompts.

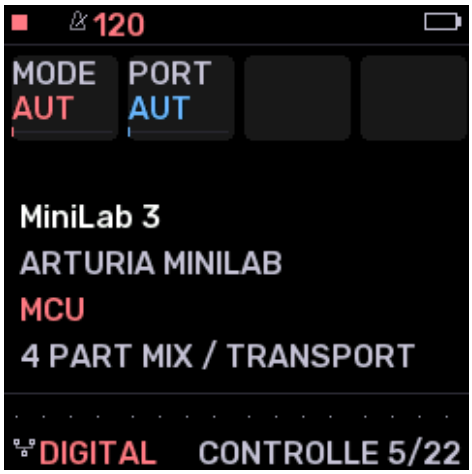
Recovery is an installation path, not a sound-editing shortcut. Keep a separate backup of programs, settings, user sounds and imported source files.

Connect the editor or a controller

The FM-1 USB socket uses one role at a time. Controller identification and USB-role selection are different operations.



usb role



controller

01 For the editor, installer, USB audio or computer MIDI: choose COMPUTER, connect the computer, then click Connect in the editor and allow MIDI/SysEx access.

02 For a directly attached USB MIDI keyboard or pedal: choose PEDAL/HOST and use correctly wired, powered host hardware. This firmware does not supply 5 V VBUS.

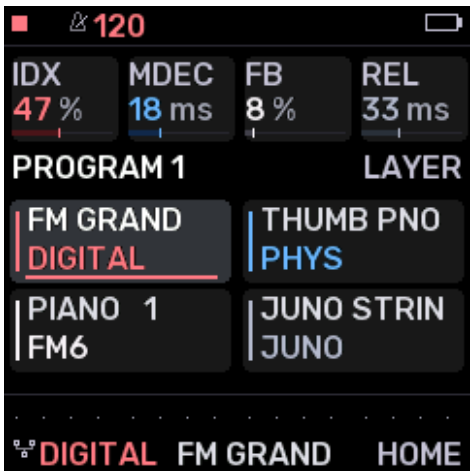
03 The controller profile is detected automatically after host mode is active. Unknown class-compliant devices keep standard MIDI input.

04 Return to COMPUTER before reconnecting the editor or installer. The firmware does not automatically switch USB roles.

The controller screenshot is a simulated MiniLab 3 identification example. USB/controller behavior has software-model coverage; physical compatibility remains pending.

Know the HOME controls

HOME shows the four sounds in the current program. The outline identifies the part being edited.



home

01 Turn SELECT to focus parts 1-4. Focus changes the sound you edit; it does not by itself change the layer mix.

02 Turn ALGORITHM to change the focused part's engine. Turn PRESET/parameter to browse presets within that engine, wrapping at the ends.

03 Turn KNOBS 1-4 to edit the four displayed controls for that part. Their labels depend on the engine.

04 Play the keyboard or MIDI controller. Layering ON uses the enabled parts and their routing; OFF plays the focused part.

> Changing a sound preset replaces the focused sound. It does not select a different four-part factory program. Use SAVE > PROGRAM or the editor factory-program control for that.

Choose a factory program

The main instrument stays in part 1. Three supporting parts add body, articulation or air.

Slot	Main instrument	Program
P1	Piano	STAGE PIANO
P2	Epiano	VELVET EP
P3	Lead Synth	COPPER LEAD
P4	Oriental Lead	ORIENT LEAD
P5	Disco Synth	DISCO POLY
P6	Brass	BRASS STACK
P7	Strings	SILK STRINGS
P8	Bass	NEON BASS

01 On the device, tap SAVE and use SELECT to choose slot 1-8. Turn LOAD twice to confirm. Unused slots provide the matching factory program.

02 Existing saved user programs take precedence in those slots. Working-program autosave also takes precedence at startup. Updating does not replace either.

03 To access any factory program even when your slots are occupied, use Editor > Programs > Factory programs > Load factory program. This replaces working edits but does not overwrite saved slots.

04 Save explicitly if you want the new working program in a device slot. Keep a backup before overwriting a user program.

All eight start with layering ON, chord generation and arpeggiation OFF, equal temperament and master tuning enabled. P1 is a synthesized stage piano rather than a sampled acoustic grand.

P1 - Piano

STAGE PIANO



FOREGROUND + THREE SUPPORTS

Part 1: Main stage piano
DIGITAL | Poly

Part 2: Felt and hammer
PHYS | Mono follower

Part 3: Quiet piano overtones
FM6 | Mono follower

Part 4: Soft string resonance
JUNO | Mono follower

How to play it

Piano remains the foreground instrument. Use three- or four-note voicings; the three quiet mono supports colour the last played note while the main piano stays polyphonic. This is a synthesized stage piano, not a sampled acoustic grand.

Make it your own

01 Keep SELECT on part 1 while shaping the main instrument. Use its four HOME knobs for tone and articulation.

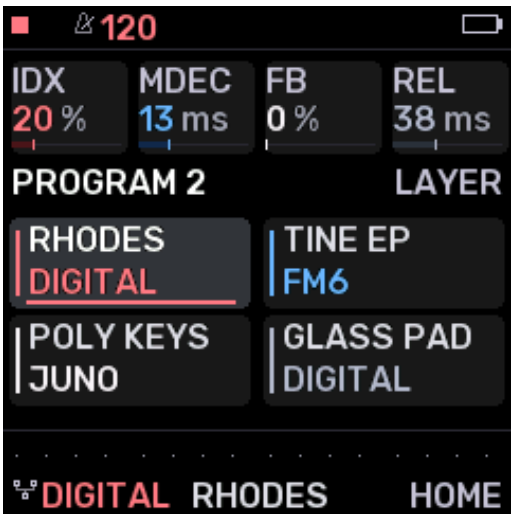
02 Hold EDIT for 3 seconds, then release, to keep the mixer open. Use KNOBS 1-4 to rebalance the four parts. Keep the main instrument audible.

03 Try SCENE 1 MAIN, 2 INTIMATE, 3 AIR or 4 DRY from GLO > SCENES. Save a program after making edits.

Scene 2 reduces ambience and bright support; scene 3 emphasizes the backing texture; scene 4 removes shared effect sends. Scenes retain their musical balance without added loudness normalization. No scene adds voices. Mono supports follow last-held-note behavior, not every chord tone.

P2 - Epiano

VELVET EP



FOREGROUND + THREE SUPPORTS

Part 1: Main electric piano
DIGITAL | Poly

Part 2: Tine sparkle
FM6 | Mono follower

Part 3: Warm poly undertone
JUNO | Poly

Part 4: Upper glass resonance
DIGITAL | Mono follower | +1 octave

How to play it

Rhodes-style electric piano leads. The warm polyphonic undertone follows the chords; the quiet mono tine and upper harmonic supports add articulation. Use up to three-note voicings.

Make it your own

01 Keep SELECT on part 1 while shaping the main instrument. Use its four HOME knobs for tone and articulation.

02 Hold EDIT for 3 seconds, then release, to keep the mixer open. Use KNOBS 1-4 to rebalance the four parts. Keep the main instrument audible.

03 Try SCENE 1 MAIN, 2 INTIMATE, 3 AIR or 4 DRY from GLO > SCENES. Save a program after making edits.

Scene 2 reduces ambience and bright support; scene 3 emphasizes the backing texture; scene 4 removes shared effect sends. Scenes retain their musical balance without added loudness normalization. No scene adds voices. Mono supports follow last-held-note behavior, not every chord tone.

P3 - Lead Synth

COPPER LEAD



FOREGROUND + THREE SUPPORTS

Part 1: Main singing synth
ANALOG | Mono follower

Part 2: Octave sheen
JUNO | Mono follower | +1 octave

Part 3: FM presence
DIGITAL | Mono follower

Part 4: Quiet attack detail
PHYS | Mono follower | +1 octave

How to play it

A singing monophonic synth lead with quiet octave sheen, FM edge and a short modal accent. Use a single-note melody; wheel or pressure adds vibrato.

Make it your own

01 Keep SELECT on part 1 while shaping the main instrument. Use its four HOME knobs for tone and articulation.

02 Hold EDIT for 3 seconds, then release, to keep the mixer open. Use KNOBS 1-4 to rebalance the four parts. Keep the main instrument audible.

03 Try SCENE 1 MAIN, 2 INTIMATE, 3 AIR or 4 DRY from GLO > SCENES. Save a program after making edits.

Scene 2 reduces ambience and bright support; scene 3 emphasizes the backing texture; scene 4 removes shared effect sends. Scenes retain their musical balance without added loudness normalization. No scene adds voices. Mono supports follow last-held-note behavior, not every chord tone.

P4 - Oriental Lead

ORIENT LEAD



FOREGROUND + THREE SUPPORTS

Part 1: Main mizmar reed
VOICE | Mono follower

Part 2: Shabi saw support
JUNO | Mono follower

Part 3: Ney air
FM6 | Mono follower

Part 4: Oud articulation
PHYS | Mono follower

How to play it

Mizmar-style reed leads, with quiet Shabi saw, ney breath and a plucked accent. Starts in equal temperament; select your maqam and root explicitly. All parts follow master tuning by default.

Make it your own

01 Keep SELECT on part 1 while shaping the main instrument. Use its four HOME knobs for tone and articulation.

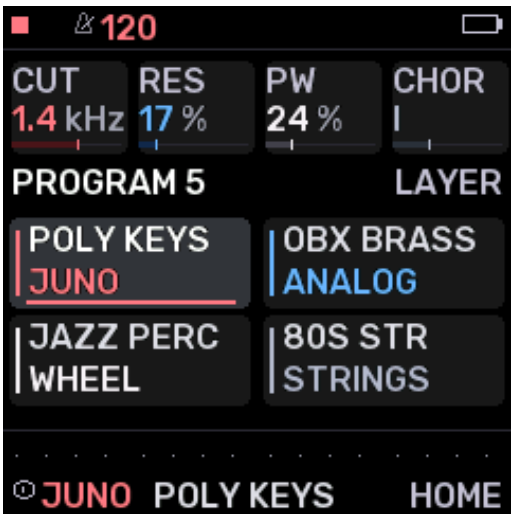
02 Hold EDIT for 3 seconds, then release, to keep the mixer open. Use KNOBS 1-4 to rebalance the four parts. Keep the main instrument audible.

03 Try SCENE 1 MAIN, 2 INTIMATE, 3 AIR or 4 DRY from GLO > SCENES. Save a program after making edits.

Scene 2 reduces ambience and bright support; scene 3 emphasizes the backing texture; scene 4 removes shared effect sends. Scenes retain their musical balance without added loudness normalization. No scene adds voices. Mono supports follow last-held-note behavior, not every chord tone.

P5 - Disco Synth

DISCO POLY



FOREGROUND + THREE SUPPORTS

Part 1: Main disco poly
JUNO | Poly

Part 2: Brass accent
ANALOG | Mono follower

Part 3: Percussive organ
WHEEL | Mono follower

Part 4: Quiet string lift
STRINGS | Mono follower

How to play it

Punchy poly-synth keys are dominant. Use rhythmic three- or four-note stabs; short mono brass and organ accents add definition without taking four voices per chord note.

Make it your own

01 Keep SELECT on part 1 while shaping the main instrument. Use its four HOME knobs for tone and articulation.

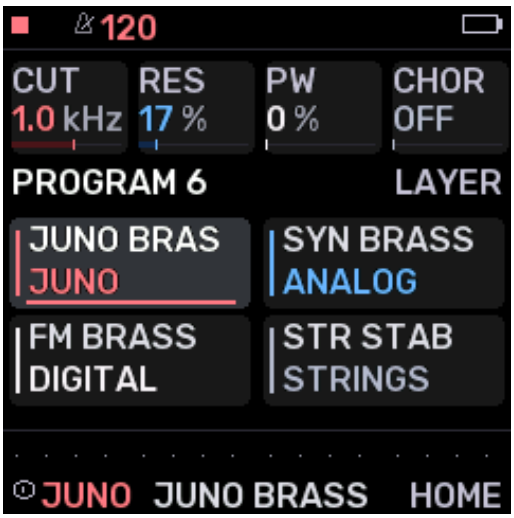
02 Hold EDIT for 3 seconds, then release, to keep the mixer open. Use KNOBS 1-4 to rebalance the four parts. Keep the main instrument audible.

03 Try SCENE 1 MAIN, 2 INTIMATE, 3 AIR or 4 DRY from GLO > SCENES. Save a program after making edits.

Scene 2 reduces ambience and bright support; scene 3 emphasizes the backing texture; scene 4 removes shared effect sends. Scenes retain their musical balance without added loudness normalization. No scene adds voices. Mono supports follow last-held-note behavior, not every chord tone.

P6 - Brass

BRASS STACK



FOREGROUND + THREE SUPPORTS

Part 1: Main brass section
JUNO | Poly

Part 2: Lower brass body
ANALOG | Mono follower | -1 octave

Part 3: FM brass edge
DIGITAL | Mono follower

Part 4: Upper string accent
STRINGS | Poly | +1 octave

How to play it

A bold synth-brass section with lower-octave body, FM bite and upper string attack. Use open two- or three-note voicings.

Make it your own

01 Keep SELECT on part 1 while shaping the main instrument. Use its four HOME knobs for tone and articulation.

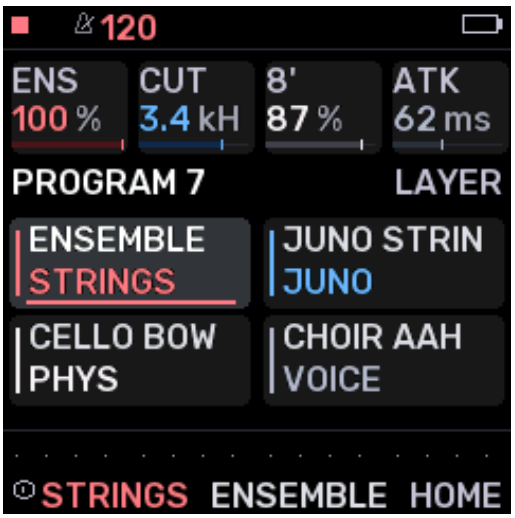
02 Hold EDIT for 3 seconds, then release, to keep the mixer open. Use KNOBS 1-4 to rebalance the four parts. Keep the main instrument audible.

03 Try SCENE 1 MAIN, 2 INTIMATE, 3 AIR or 4 DRY from GLO > SCENES. Save a program after making edits.

Scene 2 reduces ambience and bright support; scene 3 emphasizes the backing texture; scene 4 removes shared effect sends. Scenes retain their musical balance without added loudness normalization. No scene adds voices. Mono supports follow last-held-note behavior, not every chord tone.

P7 - Strings

SILK STRINGS



FOREGROUND + THREE SUPPORTS

Part 1: Main string ensemble
STRINGS | Poly

Part 2: Warm string double
JUNO | Poly

Part 3: Low bowed body
PHYS | Mono follower | -1 octave

Part 4: Quiet choral air
VOICE | Mono follower | +1 octave

How to play it

An ensemble string sound stays in front. The second polyphonic string layer blends below it; mono cello and choir supply quiet body and air. Use open two- or three-note voicings.

Make it your own

01 Keep SELECT on part 1 while shaping the main instrument. Use its four HOME knobs for tone and articulation.

02 Hold EDIT for 3 seconds, then release, to keep the mixer open. Use KNOBS 1-4 to rebalance the four parts. Keep the main instrument audible.

03 Try SCENE 1 MAIN, 2 INTIMATE, 3 AIR or 4 DRY from GLO > SCENES. Save a program after making edits.

Scene 2 reduces ambience and bright support; scene 3 emphasizes the backing texture; scene 4 removes shared effect sends. Scenes retain their musical balance without added loudness normalization. No scene adds voices. Mono supports follow last-held-note behavior, not every chord tone.

P8 - Bass

NEON BASS



FOREGROUND + THREE SUPPORTS

Part 1: Main bass fundamental
ANALOG | Mono follower

Part 2: Pulse definition
JUNO | Mono follower

Part 3: Upper midrange
VECTOR | Mono follower | +1 octave

Part 4: Plucked edge
PHYS | Mono follower

How to play it

The centred analogue bass is the main instrument. Pulse definition, a quiet upper-octave vector layer and a plucked edge support it. All four parts are monophonic.

Make it your own

01 Keep SELECT on part 1 while shaping the main instrument. Use its four HOME knobs for tone and articulation.

02 Hold EDIT for 3 seconds, then release, to keep the mixer open. Use KNOBS 1-4 to rebalance the four parts. Keep the main instrument audible.

03 Try SCENE 1 MAIN, 2 INTIMATE, 3 AIR or 4 DRY from GLO > SCENES. Save a program after making edits.

Scene 2 reduces ambience and bright support; scene 3 emphasizes the backing texture; scene 4 removes shared effect sends. Scenes retain their musical balance without added loudness normalization. No scene adds voices. Mono supports follow last-held-note behavior, not every chord tone.

Toggle layers and open the mixer

The two-second preview and three-second latch are different actions.



layered toggle



layer mixer

01 Hold GLO for 2 seconds. The display replaces the current page with Layered ON or Layered OFF. It toggles once per hold.

02 Release GLO to return. GLO never latches. A normal GLO tap opens its pages without flashing the held screen.

03 Hold EDIT for 2 seconds to preview the layer mixer. Continue to 3 seconds, then release, to keep it open.

04 KNOBS 1-4 set the four part volumes. SELECT chooses the part whose pan and split are shown. Tap EDIT or HOME to exit.

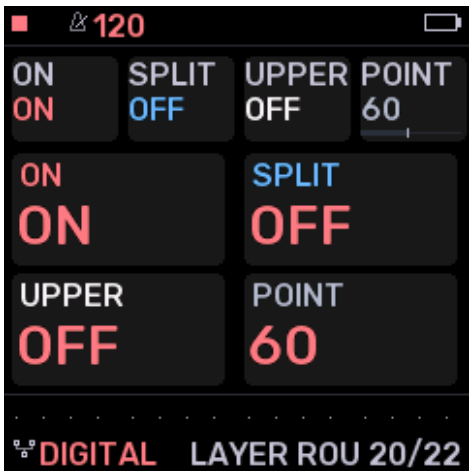
Four layers do not mean 32 voices. All four parts share 8. The factory voicings deliberately use mono support parts where useful to leave more room for the foreground instrument.

Pan and split the keyboard

The supplied bank starts with full-keyboard layering. Splits remain available for your own programs.



layer mixer



layer route

01 In the latched EDIT mixer, turn SELECT to choose a part. Turn PRESET/parameter to set that part's pan.

02 Turn ALGORITHM to set its split point from A1 to A6. The display shows note and octave. Changing it enables the part's current split direction.

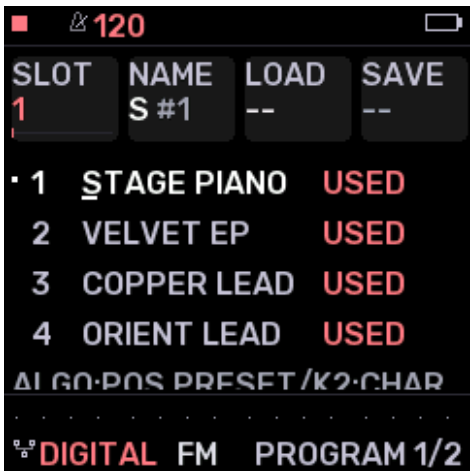
03 Use GLO > LAYER ROUTE or Editor > Sound > Layers to choose Full, Lower or Upper. Lower includes the displayed boundary; Upper starts on the next note.

04 For an upper range starting at C3, use boundary B2 with UPPER enabled. Check both sides of the split, then save the program.

A split can reduce the number of active parts per key and leave more shared voices for chords. Pan alone does not change voice count.

Save, name and reload

A slot is not overwritten until you confirm SAVE. Browsing a slot or editing its name does not save.



save program

01 Stop playback, release keys and sustain, and allow sounding tails to finish. Tap SAVE to open PROGRAM.

02 Turn SELECT or KNOB 1 to choose slot 1-8. ALGORITHM chooses one of the 12 name positions; PRESET or KNOB 2 changes its character.

03 Turn SAVE twice to confirm storing the complete working program and name. Check the on-screen confirmation.

04 To reload, select the slot and turn LOAD twice. The current four sounds, routing, mix, scenes and selected tuning are replaced.

05 Autosave resumes your last working state after power-up. It is separate from deliberately saving a named slot. Export backups regularly.

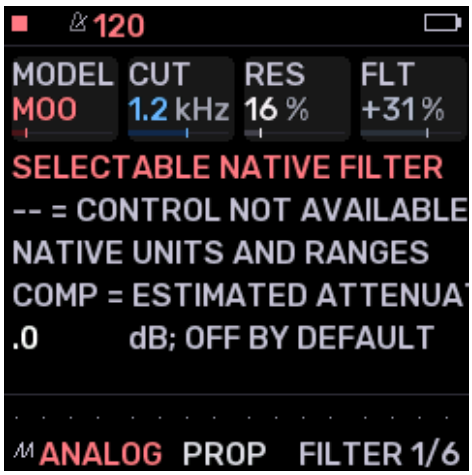
Factory recall in the editor loads ROM data into the working program only. Your saved program at the same numbered slot stays intact until you explicitly save over it.

Shape the sound with EDIT

EDIT focuses on engine and voice controls. Filters appear only where the engine exposes them.



edit engine



edit filter

01 From HOME, SELECT the part to edit and tap EDIT. Tap EDIT again to advance its available pages.

02 Adjust the engine controls on EDIT 1/2. For a useful filter, use the grouped FILTER page: model, cutoff, resonance and envelope amount.

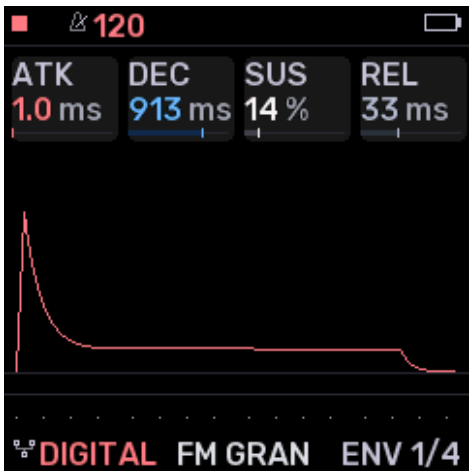
03 If a control is disabled or the page is absent, it does not affect this engine. HYBRID has its own filter architecture; the shared page addresses filter 1.

04 FILTER GAIN groups drive, optional output compensation and trim. Use compensation when resonance/drive makes the sound jump in level, then compare at similar listening volume.

> Compensation is estimated attenuation, not a loudness meter or automatic mastering. The factory sound calibration and curated program mixes are separate from this switch.

Envelopes, LFOs and templates

The envelope shapes articulation; the LFO and matrix provide movement and expression.



envelope



mod templates

01 Tap ENV for attack, decay, sustain and release. Short attack gives definition; a longer attack softens the front. Release controls the tail after key-up.

02 Tap LFO to reach its rate, waveform and destination pages. Add a small amount first and test the full keyboard range.

03 Use LFO > MOD PRESET for velocity-to-brightness, pressure-to-vibrato, wheel-to-vibrato or an envelope swell.

04 Use the editor for the full matrix and graphical envelope controls. HYBRID templates need a free modulation slot; a full matrix is left unchanged.

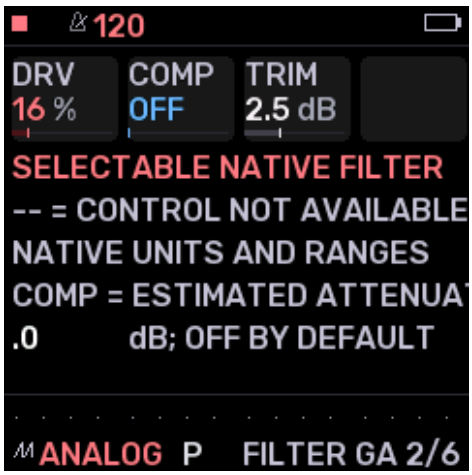
Long releases and sustain can keep voices occupied. Use shorter tails if new chord notes steal old ones. Holding ENV for 3 seconds opens CHORDS instead of the envelope pages.

Use effects without losing the instrument

Start with the dry voice, then add only the space or movement it needs.



effects



filter gain

01 SELECT the part, tap FX, then adjust the displayed distortion, chorus, delay and reverb sends.

02 Tap FX to reach additional effect pages. The insert section offers its own type and amount controls; engine-specific chorus can be separate.

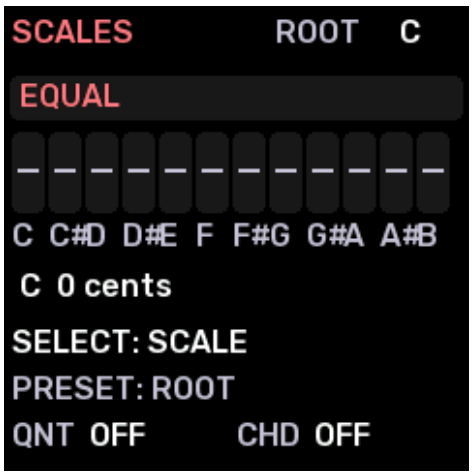
03 Keep bass near centre and use less wet signal for it. On leads, check that delay repeats do not hide the next phrase.

04 Compare the MAIN and DRY factory scenes. Bring effects back gradually, then save your preferred version.

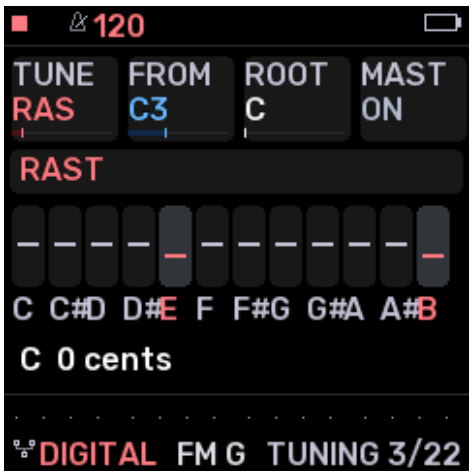
Some effect settings are global. A saved program stores each part's supported sends and insert settings, but changing shared effect configuration can alter how the same mix sounds later.

Select a scale, root and octave range

Microtuning adjusts pitches; quantization chooses notes. They are separate controls.



scale browser



tuning from c3

01 Tap SEL to open the scale browser. Turn SELECT to cycle scales; PRESET or KNOB 1 changes the root. The twelve cells visualize chromatic offsets.

02 For an octave boundary, open GLO > TUNING. KNOB 1 selects system, KNOB 2 FROM cycles C0-C8 by octaves only, KNOB 3 selects root, KNOB 4 controls master tuning for the focused part.

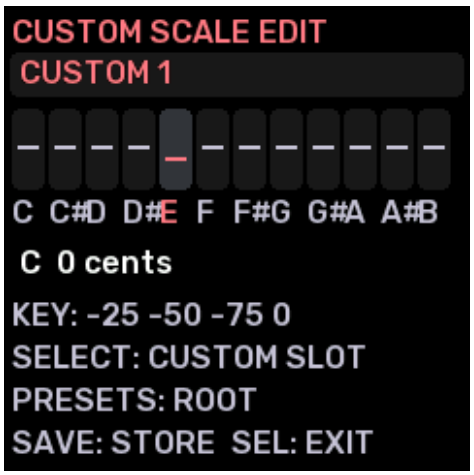
03 FROM C3 means MIDI 48 and higher use the selected tuning. Lower notes keep equal-tempered intervals. The A4 reference remains global; find it on TUNING 2.

04 Use QNT ALL to apply the focused ROOT/SCL note selection across all four parts and incoming MIDI. Use Apply master tuning to let an individual part follow or bypass master microtuning.

Bypassing master tuning does not turn off a separately selected quantizer or chord generator. Factory programs start in equal temperament; loading P4 does not secretly select a maqam.

Create a custom chromatic scale

Use the physical key for the pitch class that should change. Other chromatic keys stay unaltered.



custom scale

01 Hold SEL for 3 seconds, then release. CUSTOM SCALE EDIT remains open. SELECT chooses CUSTOM 1-4 and PRESET chooses ROOT.

02 Press a key once for -25 cents, twice for -50, three times for -75 and four times to restore 0. Repeat for each pitch class you want to detune.

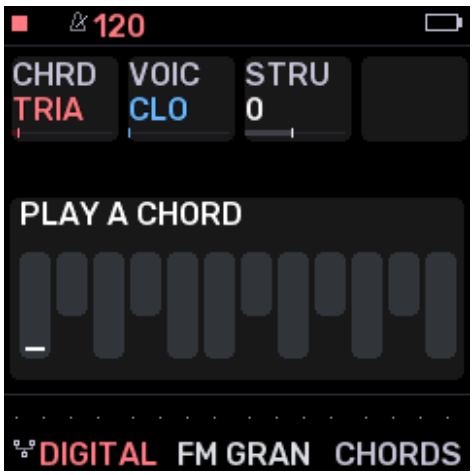
03 For example, to lower E by 50 cents, press an E key twice. E in other octaves receives the same pitch-class adjustment, subject to FROM.

04 Tap SAVE to store the custom scale, then SEL to exit. Save the program if you also want its complete four-part setup preserved.

ROM factory recall preserves your four custom-scale definitions. Loading a saved user program restores the definitions stored in that program. Export a backup before replacing an important setup.

Chords, voicings and strum

Use the three-second ENV shortcut. Normal ENV taps still open envelope editing.



chords

01 Hold ENV for 3 seconds and release. CHORDS stays open. Use KNOB 1 for CHORD, KNOB 2 for VOIC and KNOB 3 for STRUM.

02 Play any key, including black keys. Chords follow the selected scale/root; out-of-scale trigger roots map to an available scale degree.

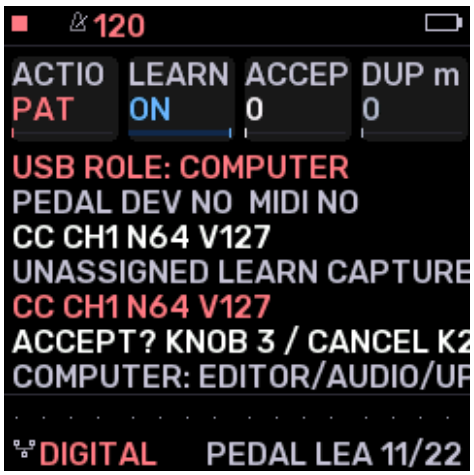
03 Choose a voicing such as CLOSE, SMOOTH, OPEN or an inversion. STRUM offers signed millisecond direction and tempo-related divisions.

04 With nonzero microtonal tuning, the selector offers OFF, OPEN 5 and NEUTRAL. Incompatible stored shapes sound as OPEN 5 until changed; they do not overwrite the saved shape.

Neutral harmony chooses existing tuned scale notes near its target intervals; it does not add arbitrary detuned keys. Mono main voices remain monophonic, and generated chord tones consume the shared voice budget.

Teach a USB pedal an action

Use Learn instead of manually typing message type, number, value and channel.



pedal learn

01 Choose PEDAL/HOST and attach a class-compliant pedal through powered host hardware. Enable program/patch MIDI as needed.

02 Open GLO > PEDAL LEARN. Select Patch Up, Patch Down, Program Next or the required supported action, then arm LEARN.

03 Press the pedal. Read the captured CC/PC message and channel. ACCEPT it to store the assignment, or cancel and try again.

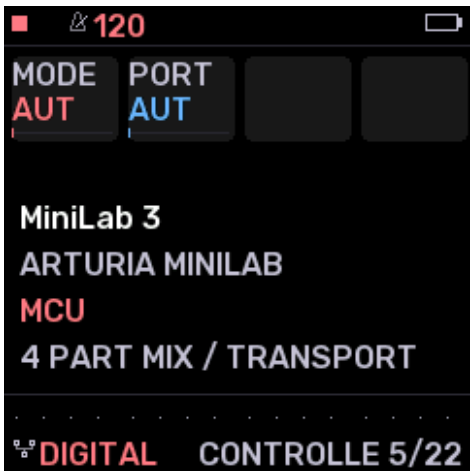
04 Press again to test. Check the last-message, matched-action and trigger/suppression indication. Conflicting duplicate assignments are refused.

05 Set REPEAT only when the pedal sends unwanted duplicates. The 0/OFF default preserves repeated taps. CC release re-arms; identical PC taps inside the window may also be suppressed.

Return to COMPUTER for the editor or installer. Bluetooth is removed. A USB device being present does not by itself mean a MIDI interface was detected.

Use keyboard and surface controllers

Identification is automatic in active host mode; role switching remains manual.



controller

01 Select PEDAL/HOST, connect powered class-compliant hardware and inspect GLO > CONTROLLER for product, family and protocol.

02 Supported Alesis, Novation, Arturia, Akai and Mackie families receive the documented profiles. Unknown controllers keep standard notes and MIDI controllers.

03 MCU/HUI uses the first four strips for part volume, pan, mute, solo, select and record arm, plus Play/Stop/Record. Music and DAW ports are kept separate.

04 APC Mini and supported modern Launchpads have chromatic pad mappings. Named control jacks take precedence; MiniLab 3 skips its separate DIN Thru port.

05 Use the advanced protocol/port override only when the controller's layout differs. Check the compatibility guide for family and generation limits.

Scribble strips, a full DAW, proprietary drivers and every custom controller template are not implemented. The pictured identity is simulated; physical interoperability is still subject to testing.

Move between scenes and programs

A scene edits the current program. A setlist chooses saved programs in an order.



scenes



setlist

01 Open GLO > SCENES. Factory scenes 1-4 are MAIN, INTIMATE, AIR and DRY. Select a scene and RECALL it.

02 To make your own scene, adjust levels, mutes, effect sends, macros and tuning, then CAPTURE. Save the program to preserve the scene.

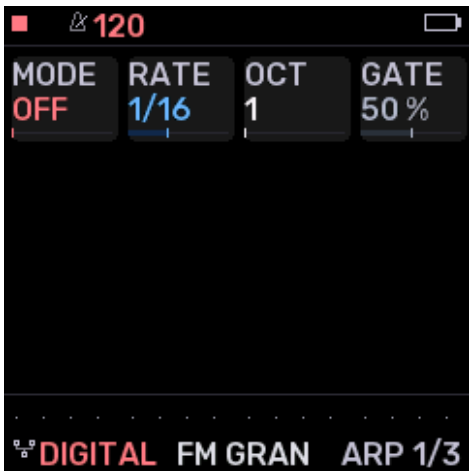
03 Open GLO > SETLIST to choose ENTRY, PROGRAM and LENGTH, or use the editor to reorder up to 32 references and Save order.

04 Assign Scene Next/Prev, Program Next/Prev or Setlist Next/Prev with pedal Learn. Program loads stop transport and wait for quiet; they do not provide gapless sample playback.

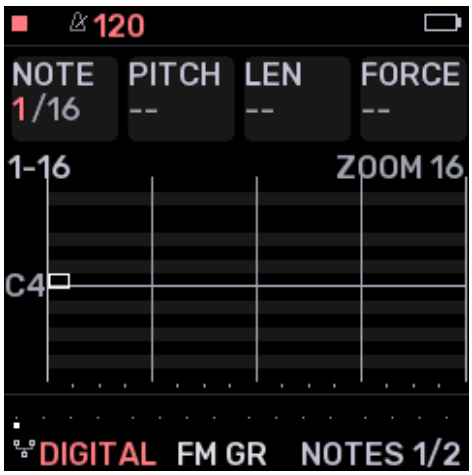
Scenes do not replace engines, pan, split routing or complete modulation matrices. The setlist is global; its entries reference saved slots, so saving a different program in a slot changes what that entry loads.

Arpeggiate and record a pattern

The factory programs contain sounds and scenes, not pre-recorded songs.



arpeggiator



sequencer

01 Tap ARP and choose a mode, rate, octave range and gate. Hold notes to hear the pattern; use HOLD only when you want notes latched.

02 Tap SEQ for the selected part's pattern. Set length/division as needed and use PLAY to start or stop.

03 Use REC to arm/record the focused part, then play the keyboard or send MIDI. Press REC again to stop recording.

04 Use the editor for detailed note, velocity and performance-lane editing. If external sync is selected, the sequencer waits for the required clock/start messages.

REC hold is a destructive pattern-clear gesture with its own countdown: releasing before completion cancels it. It is an exception to the normal function-layer timing. Back up important patterns.

Use the editor and back up your work

Use COMPUTER mode and the matching browser editor. A mock preview is not a physical-device connection.

A reliable editing session

01 Connect, select the FM-1 and read its current state. Choose a part before editing its engine, envelope, filter or modulation panel.

02 Use Sound > Layers for the four levels, pan, mute/solo and routing. Use the graphical envelopes and available engine visualizations to understand edits.

03 Programs > Factory programs recalls any P1-P8 into the working program without replacing a saved slot. Use the saved-program name field and Save when ready.

04 Export a full device backup and an editor-library export separately. A browser library entry is not a saved device program. Keep imported source files as well.

Import limits worth knowing

Source	What to expect
DX7 / .syx	FM6 can read supported voices and banks. Operator-bank storage and a playable user preset are separate actions.
Vital / .vital	Imports are HYBRID approximations, not Vital playback. Unsupported effects and modulation can change the sound. Save custom patch bytes in a program and export your library.
Wavetables / WAV	Supported wavetable conversion can fill user slots. USER WT stays silent until user data exists.
Scala / .scl	Imports tuning data, not new synthesis sounds. Select the root, FROM boundary and master-tuning behavior separately.

Stop transport and release notes before storage/restore operations when requested. Loading a program replaces working edits. Restore only compatible backups and keep an untouched original copy.

Shortcuts and secondary functions

Short taps and holds have different jobs. A brief tap does not show a held-layer screen.

Gesture / context	Result
ALGORITHM on HOME	Engine for the focused part
PRESET on HOME	Preset within that engine; wrap at ends
SELECT on HOME	Focus part 1-4
SEL tap	Scale browser; SELECT cycles tuning scales
SEL hold 3 seconds	Latched custom-scale editor; repeated key presses cycle -25/-50/-75/0 cents
ENV tap / hold 3 seconds	Envelope pages / CHORDS menu
EDIT tap / hold 2 / hold 3 seconds	Edit pages / mixer preview / latched mixer
Mixer KNOBS 1-4	Four part volumes
Mixer SELECT / PRESET / ALGORITHM	Part focus / pan / split point A1-A6
GLO tap / hold 2 seconds	Global pages / toggle layering once; never latch
SAVE tap	Program load/save and naming
SAVE page SELECT / ALGORITHM / PRESET	Slot / name position / name character
HOME tap away from HOME	Return to the four-part HOME display; full-screen visualizer is disabled
HOME hold 3 seconds	System menu
Other supported function layers	Preview after 2 seconds; latch after 3; tap the same function button or HOME to leave
REC tap / hold	Record action / separate clear countdown; release early to cancel

Troubleshoot the sound

Check the performance setup before assuming an engine or preset is broken.

Symptom	Try this
Only one part sounds	Check layering, enable/mute/solo, focus, split zones and input routing.
New chord notes cut old notes	Reduce chord density, unison and release tails. Eight voices are shared across all parts.
Piano is not an acoustic grand	P1 is a synthesized stage-piano design. It is not a multisampled acoustic instrument.
Mono support does not play every chord tone	This is intentional. The polyphonic main carries the chord; mono followers supply quieter colour.
Unexpected detuning	Check tuning system, root, FROM, master-tuning bypass, A4, pitch bend and oscillator detune.
Pedal/controller detected but silent	Check that a MIDI interface was detected, KEY IN/channel settings and the controller's own mode.
Editor cannot connect	Use COMPUTER, a data cable, Chrome/Edge and MIDI/SysEx permission; close competing MIDI applications.
Notes keep sounding	Release sustain, check HOLD/release/loops and send an all-notes/all-sound-off safety command if needed.
Old sound returns after upgrade	Existing autosave is preserved. Recall a factory program explicitly to try the new bank.

Voice examples: four fully polyphonic layers usually allow only two simultaneous keys. One polyphonic main plus three mono followers leaves more notes for the main voice. Two polyphonic parts plus two mono supports can fill eight voices with three-note chords, before release tails.

Physical USB/controller validation remains pending. Host tests and rendered audio do not establish noise, latency, speaker response or stability on every FM-1 and accessory combination.

Research, validation and credits

The bank is designed for recognizable foreground instruments, complementary supports and reviewable audio.

Firecrawl research informed layer roles, envelope relationships, expressive controls, masking, mono compatibility and performance organization. These are design principles, not a guarantee of individual taste. The project includes recipes, source references and audio previews for review.

Sources used

Roland: The Art of Layering

Sound On Sound: Envelopes, Gates & Triggers

Sound On Sound: Synth Secrets conclusion

iZotope: Frequency Masking

iZotope: Mono vs. Stereo

Ableton: Instrument, Drum and Effect Racks

Sequential: Prophet Rev2

Spectrasonics: Live and Stack Modes

What was checked

Solo playback retains calibrated levels and stored solo compensation. When multiple program parts are enabled, 12 dB of headroom is applied before part summing and effect sends. Solo and Layer OFF playback bypass that attenuation. Layered mixes use fixed limiter timing and sample look-ahead; solo playback retains its prior limiter behavior. Audits report loudness and verify clipping, peaks, audible parts and mono compatibility. Flashing firmware clears user settings, programs, presets, samples and autosave after update verification; back up before installation. Hardware testing reported 135% CPU with layered playback and crackling at low volume. Layer OFF avoids that workload; the CPU overload remains under investigation. Host renders do not validate device deadlines. The loader correction is validated with the hardware flash-range guard and watchdog timing model.

Code and image provenance

SynthMaven is based on Clipper by 3dSam. Upstream components, artwork and licences are recorded in the repository NOTICE and source files; the engine implementations are not presented as wholly original SynthMaven code. Screens in this guide use the current firmware renderer with controlled simulated input. The supplied SynthMaven logo is used as provided.

Full design record | Controller coverage

Buttons: primary and secondary functions

Button	Tap / primary action	Hold / secondary action
HOME	Return to four-part HOME	3 seconds: system sections
EDIT	Engine/filter/voice pages	2 seconds: mixer preview; 3 seconds: latch
ENV	Main/second/operator envelopes	3 seconds: CHORDS
SEL	Scale browser; SELECT chooses scale	3 seconds: custom-scale editor
LFO	LFO/modulation/matrix pages	No separate held screen
FX	Sends, insert and shared processors	No alternate SynthMaven punch screen
GLO	Global/tuning/MIDI/layer pages	2 seconds: toggle layers once; release returns
SAVE	PROGRAM/TOOLS pages	No alternate song/section layer
ARP	Arpeggiator pages	2 seconds: repeat preview; 3 seconds: latch
SEQ	NOTES/PATTERN pages	2 seconds: step layer; 3 seconds: latch
PLAY	Start/stop transport; free take: discard	No menu latch
REC	Arm/start/finish recording	0.7 seconds + 1.3-second clear ring
OCT-	Lower keyboard octave; settings: close	With OCT+ at startup: panel calibration
OCT+	Raise keyboard octave; settings: confirm	On NOTES: 0.4 seconds opens note sheet

Both octave buttons together reset the playing octave. OCT- alone during power-up enters USB rescue. Inside a menu or held function, octave buttons can have the on-screen contextual actions instead of changing pitch.

Knobs: know the current context

Context	SELECT	ALGORITHM	PRESET
HOME	Part 1-4	Engine	Preset inside engine
Ordinary menu page	Previous/next page in family	No sound browsing	No sound browsing
SEL scale browser	Master tuning scale	No sound browsing	Tuning root
Held EDIT mixer	Part 1-4	Split note A1-A6	Selected part pan
Held SEL editor	CUSTOM 1-4	Unused	Root
SAVE > PROGRAM	Saved slot 1-8	Name character position	Name character value
SEQ > NOTES	Family pages	No sound browsing	Zoom 8-64 steps
Held SEQ + step	Tempo	Lock parameter	Lock value
HOME-hold settings	Section	Unused	Row cursor

KNOBS 1-4 normally edit the four columns in left-to-right order. HOME uses the focused engine controls; the held mixer uses the four part volumes. A blank column has no parameter. On a settings section, knob 1 edits row 1, knob 2 row 2, and so on. Inside note sheets, knob 1 chooses the row and knob 2 changes its value.

A latched function can temporarily own the controls. Tap its button or HOME to leave before trying normal sound browsing. This prevents an accidental turn from replacing a sound while you are editing another parameter.

Build your own four-part program

01 Load a starting program and save a backup if it contains wanted edits. On HOME, choose part 1 with SELECT. Use ALGORITHM for its engine and PRESET for a suitable main sound.

02 Turn layering OFF with a two-second GLO hold while shaping the main sound. Set its source in EDIT, articulation in ENV, movement in LFO and processing in FX.

03 SELECT part 2 on HOME, load a contrasting supporting sound, and repeat for parts 3 and 4. Avoid making every part equally bright or equally loud.

04 Turn layering ON. Hold EDIT for 3 seconds and release. Set levels with KNOBS 1-4. SELECT a part and use PRESET for pan. Keep bass/foreground near center and compare in mono.

05 For a split, set the boundary with ALGORITHM, then use GLO > LAYER ROUTE for Lower or Upper. Lower includes the boundary; Upper begins one note higher. Disable SPLIT for a full-range layer.

06 Choose scales/root, FROM octave and each part master-tuning switch. Set chord/arp modes deliberately; turn them OFF when you want plain keyboard playing.

07 Test soft/hard playing, low/high notes, sustain and release tails. All parts share eight voices, and some engines have lower per-part limits. Reduce unison, long tails or the number of polyphonic supports if notes disappear.

08 Tap SAVE. SELECT the slot, choose name positions with ALGORITHM and characters with PRESET/KNOB 2. Turn SAVE twice within its confirmation window. Reload the slot to check it, then export a backup in the editor.

Load, save and back up: separate destinations

Item	Meaning and safe workflow
Sound preset	One engine sound. HOME PRESET replaces only the focused part. Import/play a user sound from the editor, then save a program if it uses custom patch data.
Four-part program	Four sounds plus mix/routing/performance data. SAVE > PROGRAM chooses slots 1-8. Loading replaces the working program; saving overwrites that slot only after confirmation.
Factory program	Read-only ROM recipe. An unused saved slot falls back to it; the editor can recall the ROM recipe even when a user save occupies the slot.
Scene	Partial changes inside the program. CAPTURE stores current supported values, RECALL applies them; save the containing program afterward.
Setlist	Ordered references to program slots. Reorder the references and save the list; it does not duplicate the programs.
Editor library entry	Browser-side record/file until sent to the device. Export the library separately so browser-storage clearing cannot remove your only copy.
Full backup	Device storage export. Stop transport, read/export, keep an untouched original, and restore only a compatible backup. Imported source files should also be kept.
Working autosave	Resume state across power cycles. This is separate from named saves and is not a replacement for an external backup.

To restore a named program on the device: stop and release sounding notes, tap SAVE, choose its slot with SELECT, and turn LOAD twice. To store a new version without losing the old one, choose a different slot before confirming SAVE. Name changes do not write the slot until SAVE completes.

Complete screen reference

Every reachable device page, including engine and insert-effect variants.

The following pages are generated from this build's native menu descriptors and captured using its production renderer. Parameter limits and choices come from those descriptors. Defaults mean the generic control default, not the value in every factory preset or saved program.

Open a family with its printed button. Repeated taps advance through the available pages. SELECT normally moves backward/forward within a family, stopping at the ends; use repeated button taps for later pages in a long family. Tap HOME to return to the four-part display. A blank or "-" control is unused.

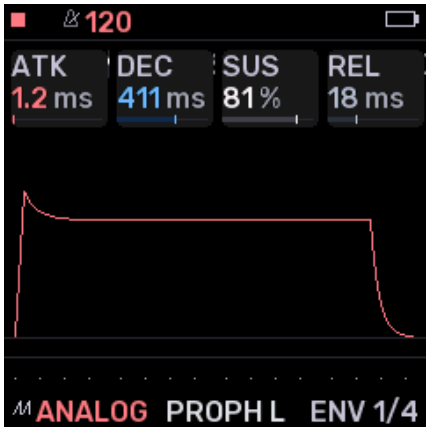
Engine and effect pages may disappear or relabel controls when the engine/model changes. Choose the required engine on HOME before entering EDIT. Ordinary page turns do not load another sound. ALGORITHM/PRESET browse sounds only on HOME; their specialized SAVE, scale-browser, mixer and sequencer functions are described separately.

Storage: sound controls belong to their program part; mix/routing/scenes belong to the four-part program. Explicitly save a named program to retain a deliberate version. Global USB/MIDI/display settings use device settings and may be saved when playback stops. A scene is a partial recall and does not replace all sound settings.

Button / family	Pages shown in this build
ENV	ENV, ENV DEST, ENV 2, ENV 2 LOOP, OP ENV
EDIT	FILTER, FILTER GAIN, EDIT 1, EDIT 2, OP, OP 2, ROUTE, FMX, VOICE, CHARACTER
LFO	LFO, LFO DEST, MOD PRESET, MOD, LFO 2, LFO 3, MATRIX, MOD SEQ, MOD SEQ 2
FX	FX, PART FILTER, INSERT, INSERT 2, DLY, DLY 2, REVERB, CHORUS, EQ+COMP
SEL	SEL, SEL 2
CHORDS	CHORDS
GLO	GLOBAL, MASTER, TUNING, TUNING 2, CONTROLLER, SYSTEM, MIDI, MIDI 2, PATCH MIDI, KEY MIDI, PEDAL LEARN, SCENES, SETLIST, PEDAL MAP, PATCH UP, PATCH DOWN, BANK UP, BANK DOWN, USB ROLE, LAYER ROUTE, LAYER MIX, LAYER PITCH
SAVE	PROGRAM, TOOLS
ARP	ARP, ARP 2, SPICE
SEQ	NOTES, PATTERN

ENV

Shape the amplitude contour of the focused sound.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap ENV > tap ENV again until ENV appears.

KNOB 1 - ATK
Attack time: how quickly the envelope rises after a note starts.
Range: 1.0 ms to 10.0 s. Generic default: 2.1 ms. Stored steps: 0 to 127.

KNOB 2 - DEC
Decay time from the attack peak to the sustain level.
Range: 1.0 ms to 10.0 s. Generic default: 160 ms. Stored steps: 0 to 127.

KNOB 3 - SUS
Sustain level while the key or gate remains held; this is a level, not a time.
Range: 0 % to 100 %. Generic default: 71 %. Stored steps: 0 to 127.

KNOB 4 - REL
Release time after key-up; longer tails continue to consume voices.
Range: 1.0 ms to 10.0 s. Generic default: 78 ms. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

ENV DEST

Route the main envelope into filter, pitch and waveform shaping.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap ENV > tap ENV again until ENV DEST appears.

KNOB 1 - FLT
Filter-envelope amount; positive values open the filter with the envelope, negative values invert that movement where supported.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

KNOB 2 - PIT
Pitch or pitch modulation depth. In an envelope/LFO destination page this is an amount, not a tuning reference.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

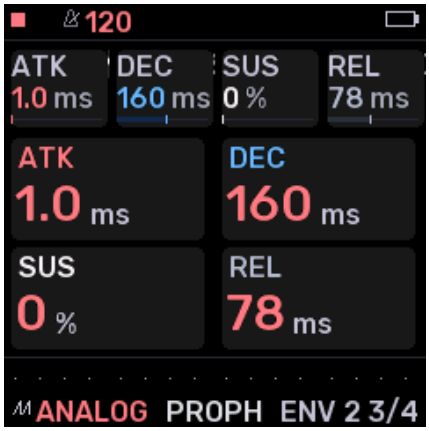
KNOB 3 - SHP
Waveform/timbre shaping or the amount of envelope/LFO shaping, according to this page.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

ENV 2

Set the second envelope used as a modulation source.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap ENV > tap ENV again until ENV 2 appears.

KNOB 1 - ATK
Attack time: how quickly the envelope rises after a note starts.
Range: 1.0 ms to 10.0 s. Generic default: 1.0 ms. Stored steps: 0 to 127.

KNOB 2 - DEC
Decay time from the attack peak to the sustain level.
Range: 1.0 ms to 10.0 s. Generic default: 160 ms. Stored steps: 0 to 127.

KNOB 3 - SUS
Sustain level while the key or gate remains held; this is a level, not a time.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 4 - REL
Release time after key-up; longer tails continue to consume voices.
Range: 1.0 ms to 10.0 s. Generic default: 78 ms. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

ENV 2 LOOP

Choose one-shot or repeating operation for envelope 2.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap ENV > tap ENV again until ENV 2 LOOP appears.

KNOB 1 - LOOP
Repeat/loop behavior. On ENV 2, ONE is a single contour and LOOP repeats it; on SPICE/TURNG it repeats generated variation.
Range: ONE to LOOP. Generic default: ONE. Stored steps: 0 to 1.

Choices: ONE / LOOP.

KNOB 2 - Unused
No assigned parameter on this page.

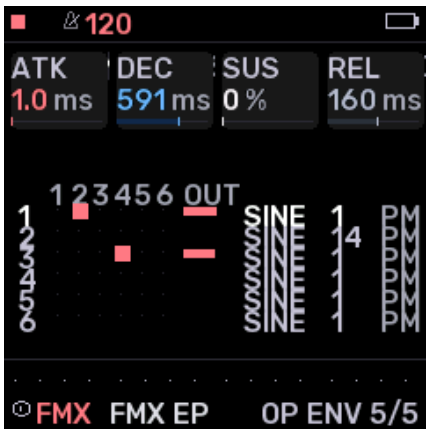
KNOB 3 - Unused
No assigned parameter on this page.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

OP ENV / FMX

Edit the envelope of the operator selected on FMX OP.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap ENV > tap ENV again until OP ENV appears.

KNOB 1 - ATK
Attack time: how quickly the envelope rises after a note starts.
Range: 1.0 ms to 10.0 s. Generic default: 1.0 ms. Stored steps: 0 to 127.

KNOB 2 - DEC
Decay time from the attack peak to the sustain level.
Range: 1.0 ms to 10.0 s. Generic default: 160 ms. Stored steps: 0 to 127.

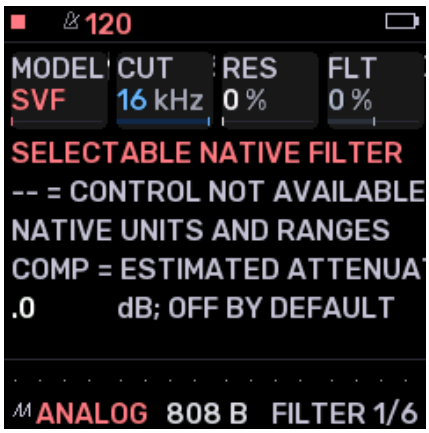
KNOB 3 - SUS
Sustain level while the key or gate remains held; this is a level, not a time.
Range: 0 % to 100 %. Generic default: 100 %. Stored steps: 0 to 127.

KNOB 4 - REL
Release time after key-up; longer tails continue to consume voices.
Range: 1.0 ms to 10.0 s. Generic default: 18 ms. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER / ANALOG

Grouped engine filter controls; absent on engines without exposed filter controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER appears.

KNOB 1 - MODEL

Select the filter model exposed by this engine. The capability map determines which controls actually operate.
Range: SVF to HP. Generic default: SVF. Stored steps: 1 to 8.

Choices: SVF / MOOG / MS20 / DIODE / SEM / NOTCH / BP / HP.

KNOB 2 - CUT

Low-pass/filter cutoff: lower values remove more upper harmonics.
Range: 30 Hz to 16 kHz. Generic default: 2.5 kHz. Stored steps: 0 to 127.

KNOB 3 - RES

Filter resonance emphasizes frequencies around cutoff. High values may change level or approach self-oscillation.
Range: 0 % to 100 %. Generic default: 24 %. Stored steps: 0 to 127.

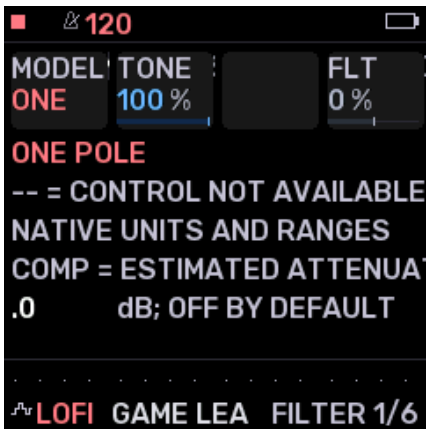
KNOB 4 - FLT

Filter-envelope amount; positive values open the filter with the envelope, negative values invert that movement where supported.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER / LOFI

Grouped engine filter controls; absent on engines without exposed filter controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER appears.

KNOB 1 - MODEL

Select the filter model exposed by this engine. The capability map determines which controls actually operate.
Range: ONE P to ONE P. Generic default: ONE P. Stored steps: 0 to 0.

Choices: ONE POLE.

KNOB 2 - TONE

Tone or spectral balance of this sound/processor.
Range: 0 % to 100 %. Generic default: 100 %. Stored steps: 0 to 127.

KNOB 3 - --

Unused control position; turning this knob has no assigned sound parameter on this layout.
Range: -- to --. Generic default: --. Stored steps: 0 to 0.

Choices: --.

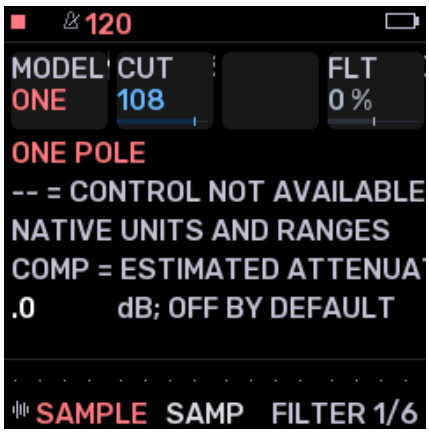
KNOB 4 - FLT

Filter-envelope amount; positive values open the filter with the envelope, negative values invert that movement where supported.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER / SAMPLE

Grouped engine filter controls; absent on engines without exposed filter controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER appears.

KNOB 1 - MODEL
Select the filter model exposed by this engine. The capability map determines which controls actually operate.
Range: ONE P to ONE P. Generic default: ONE P. Stored steps: 0 to 0.

Choices: ONE POLE.

KNOB 2 - CUT
Low-pass/filter cutoff: lower values remove more upper harmonics.
Range: 0 to 127. Generic default: 127. Stored steps: 0 to 127.

KNOB 3 - --
Unused control position; turning this knob has no assigned sound parameter on this layout.
Range: -- to --. Generic default: --. Stored steps: 0 to 0.

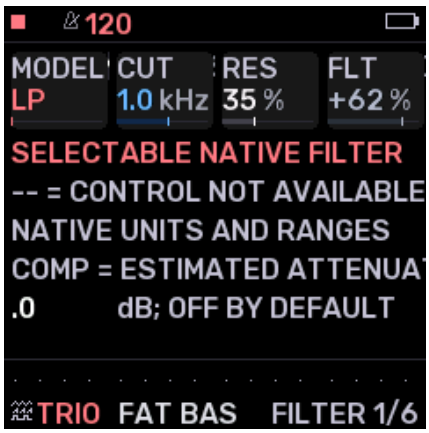
Choices: --.

KNOB 4 - FLT
Filter-envelope amount; positive values open the filter with the envelope, negative values invert that movement where supported.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER / TRIO

Grouped engine filter controls; absent on engines without exposed filter controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER appears.

KNOB 1 - MODEL
Select the filter model exposed by this engine. The capability map determines which controls actually operate.
Range: LP to NOT. Generic default: LP. Stored steps: 0 to 3.

Choices: LP / BP / HP / NOT.

KNOB 2 - CUT
Low-pass/filter cutoff: lower values remove more upper harmonics.
Range: 30 Hz to 16 kHz. Generic default: 1.5 kHz. Stored steps: 0 to 127.

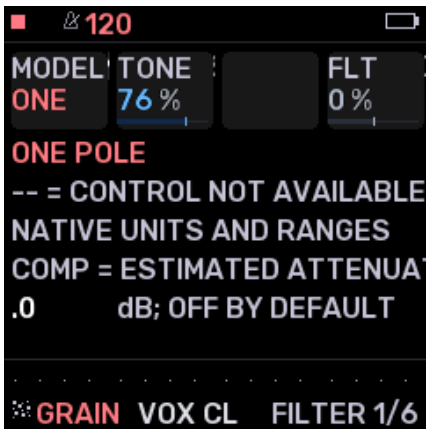
KNOB 3 - RES
Filter resonance emphasizes frequencies around cutoff. High values may change level or approach self-oscillation.
Range: 0 % to 100 %. Generic default: 31 %. Stored steps: 0 to 127.

KNOB 4 - FLT
Filter-envelope amount; positive values open the filter with the envelope, negative values invert that movement where supported.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER / GRAIN

Grouped engine filter controls; absent on engines without exposed filter controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER appears.

KNOB 1 - MODEL
Select the filter model exposed by this engine. The capability map determines which controls actually operate.
Range: ONE P to ONE P. Generic default: ONE P. Stored steps: 0 to 0.

Choices: ONE POLE.

KNOB 2 - TONE
Tone or spectral balance of this sound/processor.
Range: 0 % to 100 %. Generic default: 100 %. Stored steps: 0 to 127.

KNOB 3 - --
Unused control position; turning this knob has no assigned sound parameter on this layout.
Range: -- to --. Generic default: --. Stored steps: 0 to 0.

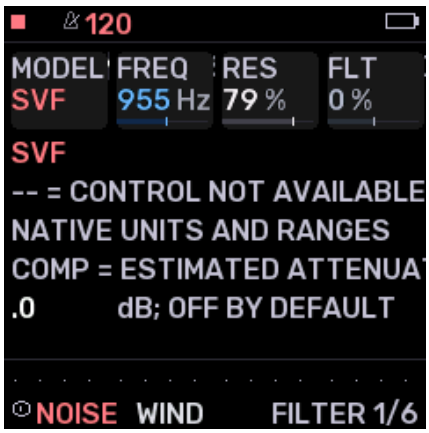
Choices: --.

KNOB 4 - FLT
Filter-envelope amount; positive values open the filter with the envelope, negative values invert that movement where supported.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER / NOISE

Grouped engine filter controls; absent on engines without exposed filter controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER appears.

KNOB 1 - MODEL
Select the filter model exposed by this engine. The capability map determines which controls actually operate.
Range: SVF to SVF. Generic default: SVF. Stored steps: 0 to 0.

Choices: SVF.

KNOB 2 - FREQ
Fixed-versus-ratio frequency behavior for the selected FMX operator.
Range: 30 Hz to 16 kHz. Generic default: 1.5 kHz. Stored steps: 0 to 127.

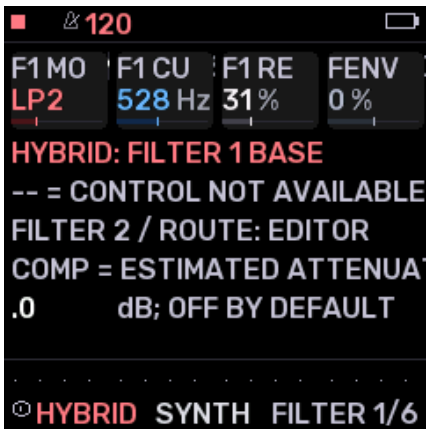
KNOB 3 - RES
Filter resonance emphasizes frequencies around cutoff. High values may change level or approach self-oscillation.
Range: 0 % to 100 %. Generic default: 24 %. Stored steps: 0 to 127.

KNOB 4 - FLT
Filter-envelope amount; positive values open the filter with the envelope, negative values invert that movement where supported.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER / HYBRID

Grouped engine filter controls; absent on engines without exposed filter controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER appears.

KNOB 1 - F1 MODEL

Select the filter model exposed by this engine. The capability map determines which controls actually operate.
Range: OFF to MOOG. Generic default: OFF. Stored steps: 0 to 7.

Choices: OFF / LP12 / LP24 / BP / HP / NOTCH / PEAK / MOOG.

KNOB 2 - F1 CUT

Low-pass/filter cutoff: lower values remove more upper harmonics.
Range: 30 Hz to 16 kHz. Generic default: 30 Hz. Stored steps: 0 to 127.

KNOB 3 - F1 RES

Filter resonance emphasizes frequencies around cutoff. High values may change level or approach self-oscillation.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

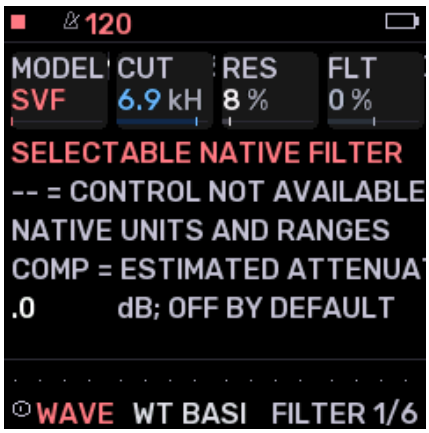
KNOB 4 - FENV

Filter-envelope amount; positive values open the filter with the envelope, negative values invert that movement where supported.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER / WAVE

Grouped engine filter controls; absent on engines without exposed filter controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER appears.

KNOB 1 - MODEL
Select the filter model exposed by this engine. The capability map determines which controls actually operate.
Range: SVF to HP. Generic default: SVF. Stored steps: 1 to 8.

Choices: SVF / MOOG / MS20 / DIODE / SEM / NOTCH / BP / HP.

KNOB 2 - CUT
Low-pass/filter cutoff: lower values remove more upper harmonics.
Range: 30 Hz to 16 kHz. Generic default: 6.9 kHz. Stored steps: 0 to 127.

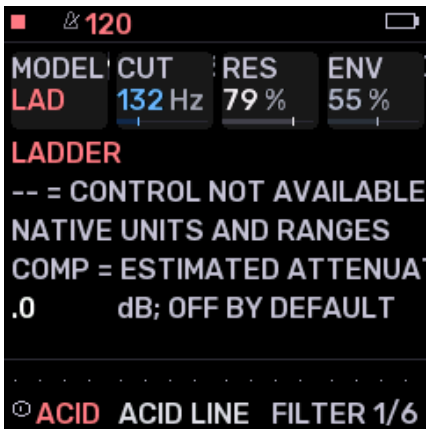
KNOB 3 - RES
Filter resonance emphasizes frequencies around cutoff. High values may change level or approach self-oscillation.
Range: 0 % to 100 %. Generic default: 8 %. Stored steps: 0 to 127.

KNOB 4 - FLT
Filter-envelope amount; positive values open the filter with the envelope, negative values invert that movement where supported.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER / ACID

Grouped engine filter controls; absent on engines without exposed filter controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER appears.

KNOB 1 - MODEL
Select the filter model exposed by this engine. The capability map determines which controls actually operate.
Range: LADDE to LADDE. Generic default: LADDE. Stored steps: 0 to 0.

Choices: LADDER.

KNOB 2 - CUT
Low-pass/filter cutoff: lower values remove more upper harmonics.
Range: 30 Hz to 16 kHz. Generic default: 132 Hz. Stored steps: 0 to 127.

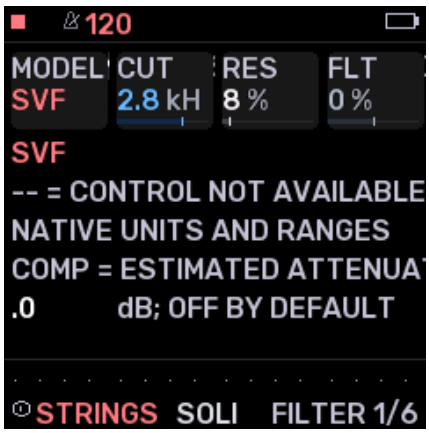
KNOB 3 - RES
Filter resonance emphasizes frequencies around cutoff. High values may change level or approach self-oscillation.
Range: 0 % to 100 %. Generic default: 79 %. Stored steps: 0 to 127.

KNOB 4 - ENV
Filter-envelope amount; positive values open the filter with the envelope, negative values invert that movement where supported.
Range: 0 % to 100 %. Generic default: 55 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER / STRINGS

Grouped engine filter controls; absent on engines without exposed filter controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER appears.

KNOB 1 - MODEL

Select the filter model exposed by this engine. The capability map determines which controls actually operate.
Range: SVF to SVF. Generic default: SVF. Stored steps: 0 to 0.

Choices: SVF.

KNOB 2 - CUT

Low-pass/filter cutoff: lower values remove more upper harmonics.
Range: 30 Hz to 16 kHz. Generic default: 2.8 kHz. Stored steps: 0 to 127.

KNOB 3 - RES

Filter resonance emphasizes frequencies around cutoff. High values may change level or approach self-oscillation.
Range: 0 % to 100 %. Generic default: 8 %. Stored steps: 0 to 127.

KNOB 4 - FLT

Filter-envelope amount; positive values open the filter with the envelope, negative values invert that movement where supported.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER / MACRO

Grouped engine filter controls; absent on engines without exposed filter controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER appears.

KNOB 1 - MODEL
Select the filter model exposed by this engine. The capability map determines which controls actually operate.
Range: LPG to LPG. Generic default: LPG. Stored steps: 0 to 0.

Choices: LPG.

KNOB 2 - CUT
Low-pass/filter cutoff: lower values remove more upper harmonics.
Range: 30 Hz to 16 kHz. Generic default: 6.9 kHz. Stored steps: 0 to 127.

KNOB 3 - --
Unused control position; turning this knob has no assigned sound parameter on this layout.
Range: -- to --. Generic default: --. Stored steps: 0 to 0.

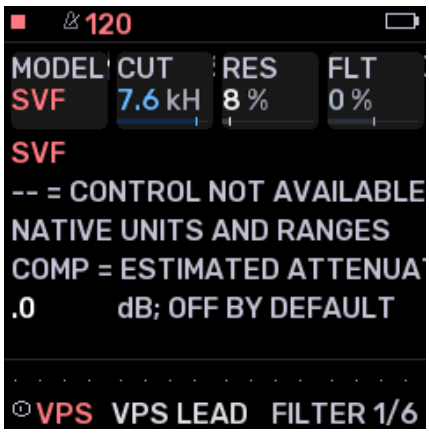
Choices: --.

KNOB 4 - FLT
Filter-envelope amount; positive values open the filter with the envelope, negative values invert that movement where supported.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER / VPS

Grouped engine filter controls; absent on engines without exposed filter controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER appears.

KNOB 1 - MODEL
Select the filter model exposed by this engine. The capability map determines which controls actually operate.
Range: SVF to SVF. Generic default: SVF. Stored steps: 0 to 0.

Choices: SVF.

KNOB 2 - CUT
Low-pass/filter cutoff: lower values remove more upper harmonics.
Range: 30 Hz to 16 kHz. Generic default: 6.9 kHz. Stored steps: 0 to 127.

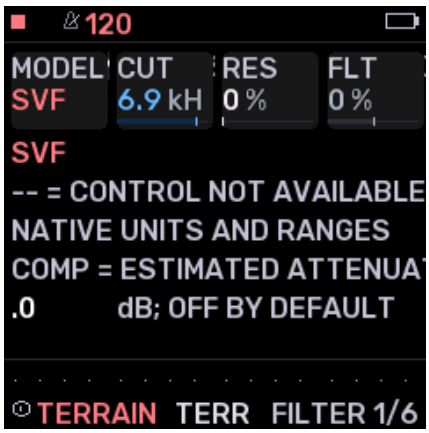
KNOB 3 - RES
Filter resonance emphasizes frequencies around cutoff. High values may change level or approach self-oscillation.
Range: 0 % to 100 %. Generic default: 8 %. Stored steps: 0 to 127.

KNOB 4 - FLT
Filter-envelope amount; positive values open the filter with the envelope, negative values invert that movement where supported.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER / TERRAIN

Grouped engine filter controls; absent on engines without exposed filter controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER appears.

KNOB 1 - MODEL
Select the filter model exposed by this engine. The capability map determines which controls actually operate.
Range: SVF to SVF. Generic default: SVF. Stored steps: 0 to 0.

Choices: SVF.

KNOB 2 - CUT
Low-pass/filter cutoff: lower values remove more upper harmonics.
Range: 30 Hz to 16 kHz. Generic default: 6.9 kHz. Stored steps: 0 to 127.

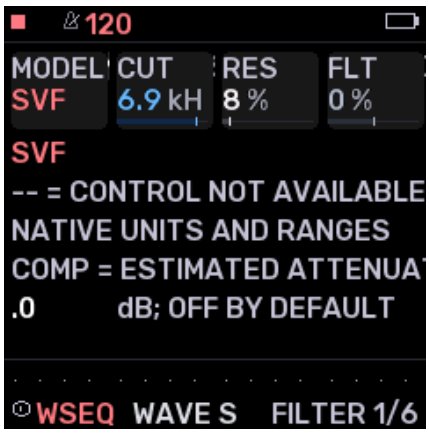
KNOB 3 - RES
Filter resonance emphasizes frequencies around cutoff. High values may change level or approach self-oscillation.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 4 - FLT
Filter-envelope amount; positive values open the filter with the envelope, negative values invert that movement where supported.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER / WSEQ

Grouped engine filter controls; absent on engines without exposed filter controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER appears.

KNOB 1 - MODEL
Select the filter model exposed by this engine. The capability map determines which controls actually operate.
Range: SVF to SVF. Generic default: SVF. Stored steps: 0 to 0.

Choices: SVF.

KNOB 2 - CUT
Low-pass/filter cutoff: lower values remove more upper harmonics.
Range: 30 Hz to 16 kHz. Generic default: 6.9 kHz. Stored steps: 0 to 127.

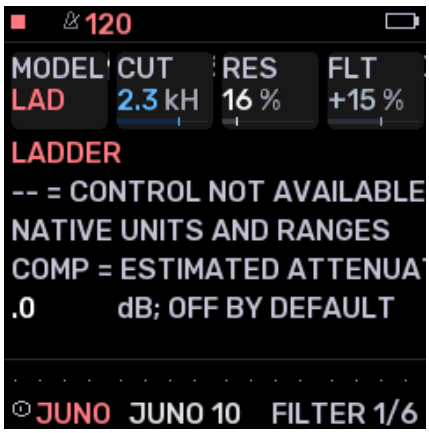
KNOB 3 - RES
Filter resonance emphasizes frequencies around cutoff. High values may change level or approach self-oscillation.
Range: 0 % to 100 %. Generic default: 8 %. Stored steps: 0 to 127.

KNOB 4 - FLT
Filter-envelope amount; positive values open the filter with the envelope, negative values invert that movement where supported.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER / JUNO

Grouped engine filter controls; absent on engines without exposed filter controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER appears.

KNOB 1 - MODEL
Select the filter model exposed by this engine. The capability map determines which controls actually operate.
Range: LADDE to LADDE. Generic default: LADDE. Stored steps: 0 to 0.

Choices: LADDER.

KNOB 2 - CUT
Low-pass/filter cutoff: lower values remove more upper harmonics.
Range: 30 Hz to 16 kHz. Generic default: 2.5 kHz. Stored steps: 0 to 127.

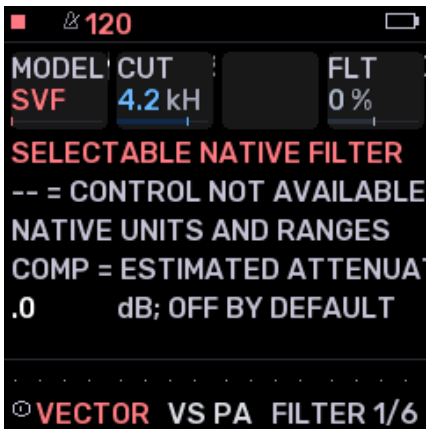
KNOB 3 - RES
Filter resonance emphasizes frequencies around cutoff. High values may change level or approach self-oscillation.
Range: 0 % to 100 %. Generic default: 16 %. Stored steps: 0 to 127.

KNOB 4 - FLT
Filter-envelope amount; positive values open the filter with the envelope, negative values invert that movement where supported.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER / VECTOR

Grouped engine filter controls; absent on engines without exposed filter controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER appears.

KNOB 1 - MODEL

Select the filter model exposed by this engine. The capability map determines which controls actually operate.
Range: SVF to HP. Generic default: SVF. Stored steps: 1 to 8.

Choices: SVF / MOOG / MS20 / DIODE / SEM / NOTCH / BP / HP.

KNOB 2 - CUT

Low-pass/filter cutoff: lower values remove more upper harmonics.
Range: 30 Hz to 16 kHz. Generic default: 4.2 kHz. Stored steps: 0 to 127.

KNOB 3 - --

Unused control position; turning this knob has no assigned sound parameter on this layout.
Range: -- to --. Generic default: --. Stored steps: 0 to 0.

Choices: --.

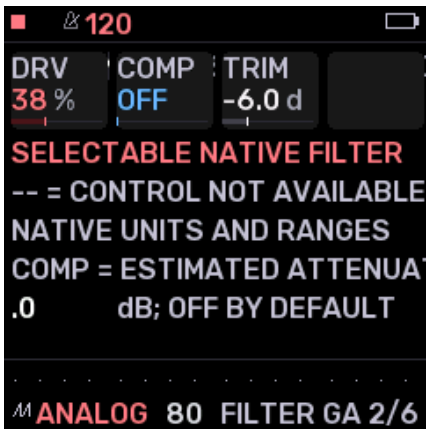
KNOB 4 - FLT

Filter-envelope amount; positive values open the filter with the envelope, negative values invert that movement where supported.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER GAIN / ANALOG

Balance filter drive, optional output compensation and final trim.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER GAIN appears.

KNOB 1 - DRV

Drive into a saturation/filter stage. More drive can add harmonics and change level.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 2 - COMP

Optional estimated filter-output compensation. ON attenuates large resonance/drive level changes; this is not a measured loudness-normalization control.
Range: OFF to ON. Generic default: OFF. Stored steps: 0 to 1.

KNOB 3 - TRIM

Sound output calibration in half-dB steps. This changes the sound gain independently of the part mixer level.
Range: -32.0 dB to 63.5 dB. Generic default: .0 dB. Stored steps: -64 to 127.

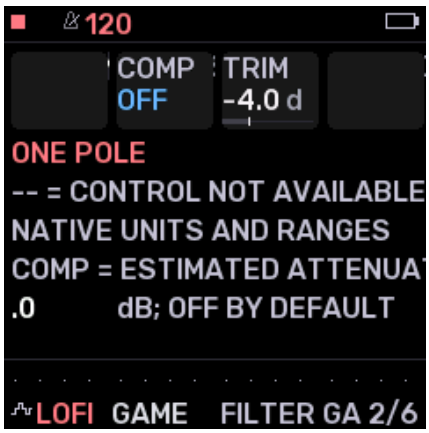
KNOB 4 - Unused

No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER GAIN / LOFI

Balance filter drive, optional output compensation and final trim.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER GAIN appears.

KNOB 1 - --
Unused control position; turning this knob has no assigned sound parameter on this layout.
Range: -- to --. Generic default: --. Stored steps: 0 to 0.

Choices: --.

KNOB 2 - COMP
Optional estimated filter-output compensation. ON attenuates large resonance/drive level changes; this is not a measured loudness-normalization control.
Range: OFF to OFF. Generic default: OFF. Stored steps: 0 to 0.

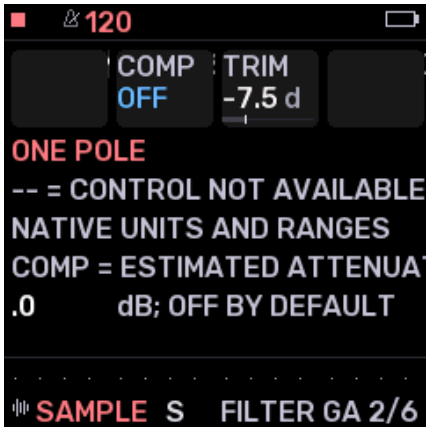
KNOB 3 - TRIM
Sound output calibration in half-dB steps. This changes the sound gain independently of the part mixer level.
Range: -32.0 dB to 63.5 dB. Generic default: .0 dB. Stored steps: -64 to 127.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER GAIN / SAMPLE

Balance filter drive, optional output compensation and final trim.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER GAIN appears.

KNOB 1 - --
Unused control position; turning this knob has no assigned sound parameter on this layout.
Range: -- to --. Generic default: --. Stored steps: 0 to 0.

Choices: --.

KNOB 2 - COMP
Optional estimated filter-output compensation. ON attenuates large resonance/drive level changes; this is not a measured loudness-normalization control.
Range: OFF to OFF. Generic default: OFF. Stored steps: 0 to 0.

KNOB 3 - TRIM
Sound output calibration in half-dB steps. This changes the sound gain independently of the part mixer level.
Range: -32.0 dB to 63.5 dB. Generic default: .0 dB. Stored steps: -64 to 127.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER GAIN / TRIO

Balance filter drive, optional output compensation and final trim.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER GAIN appears.

KNOB 1 - --
Unused control position; turning this knob has no assigned sound parameter on this layout.
Range: -- to --. Generic default: --. Stored steps: 0 to 0.

Choices: --.

KNOB 2 - COMP
Optional estimated filter-output compensation. ON attenuates large resonance/drive level changes; this is not a measured loudness-normalization control.
Range: OFF to ON. Generic default: OFF. Stored steps: 0 to 1.

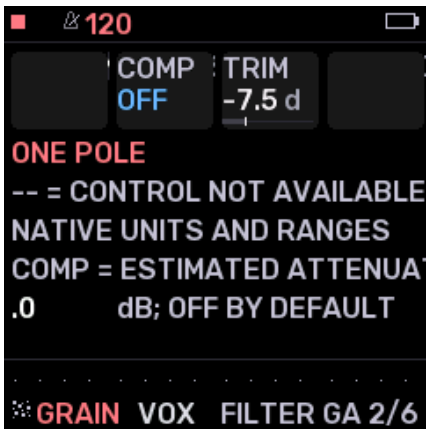
KNOB 3 - TRIM
Sound output calibration in half-dB steps. This changes the sound gain independently of the part mixer level.
Range: -32.0 dB to 63.5 dB. Generic default: .0 dB. Stored steps: -64 to 127.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER GAIN / GRAIN

Balance filter drive, optional output compensation and final trim.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER GAIN appears.

KNOB 1 - --
Unused control position; turning this knob has no assigned sound parameter on this layout.
Range: -- to --. Generic default: --. Stored steps: 0 to 0.

Choices: --.

KNOB 2 - COMP
Optional estimated filter-output compensation. ON attenuates large resonance/drive level changes; this is not a measured loudness-normalization control.
Range: OFF to OFF. Generic default: OFF. Stored steps: 0 to 0.

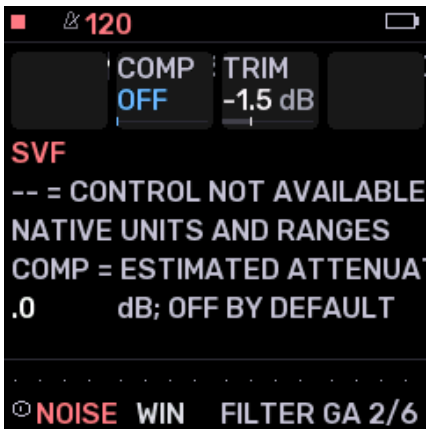
KNOB 3 - TRIM
Sound output calibration in half-dB steps. This changes the sound gain independently of the part mixer level.
Range: -32.0 dB to 63.5 dB. Generic default: .0 dB. Stored steps: -64 to 127.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER GAIN / NOISE

Balance filter drive, optional output compensation and final trim.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER GAIN appears.

KNOB 1 - --
Unused control position; turning this knob has no assigned sound parameter on this layout.
Range: -- to --. Generic default: --. Stored steps: 0 to 0.

Choices: --.

KNOB 2 - COMP
Optional estimated filter-output compensation. ON attenuates large resonance/drive level changes; this is not a measured loudness-normalization control.
Range: OFF to ON. Generic default: OFF. Stored steps: 0 to 1.

KNOB 3 - TRIM
Sound output calibration in half-dB steps. This changes the sound gain independently of the part mixer level.
Range: -32.0 dB to 63.5 dB. Generic default: .0 dB. Stored steps: -64 to 127.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER GAIN / HYBRID

Balance filter drive, optional output compensation and final trim.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER GAIN appears.

KNOB 1 - DRIVE

Drive into a saturation/filter stage. More drive can add harmonics and change level.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 2 - COMP

Optional estimated filter-output compensation. ON attenuates large resonance/drive level changes; this is not a measured loudness-normalization control.
Range: OFF to ON. Generic default: OFF. Stored steps: 0 to 1.

KNOB 3 - TRIM

Sound output calibration in half-dB steps. This changes the sound gain independently of the part mixer level.
Range: -32.0 dB to 63.5 dB. Generic default: .0 dB. Stored steps: -64 to 127.

KNOB 4 - Unused

No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER GAIN / WAVE

Balance filter drive, optional output compensation and final trim.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER GAIN appears.

KNOB 1 - DRV

Drive into a saturation/filter stage. More drive can add harmonics and change level.

Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 2 - COMP

Optional estimated filter-output compensation. ON attenuates large resonance/drive level changes; this is not a measured loudness-normalization control.

Range: OFF to ON. Generic default: OFF. Stored steps: 0 to 1.

KNOB 3 - TRIM

Sound output calibration in half-dB steps. This changes the sound gain independently of the part mixer level.

Range: -32.0 dB to 63.5 dB. Generic default: .0 dB. Stored steps: -64 to 127.

KNOB 4 - Unused

No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER GAIN / ACID

Balance filter drive, optional output compensation and final trim.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER GAIN appears.

KNOB 1 - DIST
Amount sent to distortion.
Range: 0 % to 100 %. Generic default: 16 %. Stored steps: 0 to 127.

KNOB 2 - COMP
Optional estimated filter-output compensation. ON attenuates large resonance/drive level changes; this is not a measured loudness-normalization control.
Range: OFF to ON. Generic default: OFF. Stored steps: 0 to 1.

KNOB 3 - TRIM
Sound output calibration in half-dB steps. This changes the sound gain independently of the part mixer level.
Range: -32.0 dB to 63.5 dB. Generic default: .0 dB. Stored steps: -64 to 127.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER GAIN / STRINGS

Balance filter drive, optional output compensation and final trim.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER GAIN appears.

KNOB 1 - --
Unused control position; turning this knob has no assigned sound parameter on this layout.
Range: -- to --. Generic default: --. Stored steps: 0 to 0.

Choices: --.

KNOB 2 - COMP
Optional estimated filter-output compensation. ON attenuates large resonance/drive level changes; this is not a measured loudness-normalization control.
Range: OFF to ON. Generic default: OFF. Stored steps: 0 to 1.

KNOB 3 - TRIM
Sound output calibration in half-dB steps. This changes the sound gain independently of the part mixer level.
Range: -32.0 dB to 63.5 dB. Generic default: .0 dB. Stored steps: -64 to 127.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER GAIN / MACRO

Balance filter drive, optional output compensation and final trim.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER GAIN appears.

KNOB 1 - --
Unused control position; turning this knob has no assigned sound parameter on this layout.
Range: -- to --. Generic default: --. Stored steps: 0 to 0.

Choices: --.

KNOB 2 - COMP
Optional estimated filter-output compensation. ON attenuates large resonance/drive level changes; this is not a measured loudness-normalization control.
Range: OFF to OFF. Generic default: OFF. Stored steps: 0 to 0.

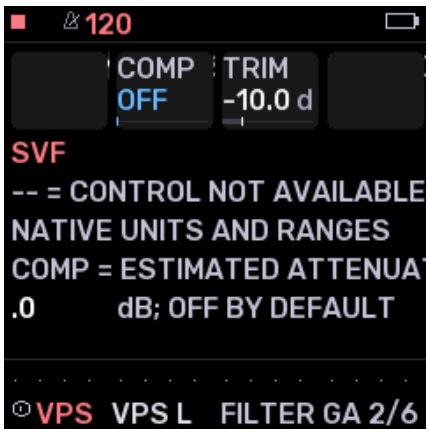
KNOB 3 - TRIM
Sound output calibration in half-dB steps. This changes the sound gain independently of the part mixer level.
Range: -32.0 dB to 63.5 dB. Generic default: .0 dB. Stored steps: -64 to 127.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER GAIN / VPS

Balance filter drive, optional output compensation and final trim.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER GAIN appears.

KNOB 1 - --
Unused control position; turning this knob has no assigned sound parameter on this layout.
Range: -- to --. Generic default: --. Stored steps: 0 to 0.

Choices: --.

KNOB 2 - COMP
Optional estimated filter-output compensation. ON attenuates large resonance/drive level changes; this is not a measured loudness-normalization control.
Range: OFF to ON. Generic default: OFF. Stored steps: 0 to 1.

KNOB 3 - TRIM
Sound output calibration in half-dB steps. This changes the sound gain independently of the part mixer level.
Range: -32.0 dB to 63.5 dB. Generic default: .0 dB. Stored steps: -64 to 127.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER GAIN / TERRAIN

Balance filter drive, optional output compensation and final trim.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER GAIN appears.

KNOB 1 - --
Unused control position; turning this knob has no assigned sound parameter on this layout.
Range: -- to --. Generic default: --. Stored steps: 0 to 0.

Choices: --.

KNOB 2 - COMP
Optional estimated filter-output compensation. ON attenuates large resonance/drive level changes; this is not a measured loudness-normalization control.
Range: OFF to ON. Generic default: OFF. Stored steps: 0 to 1.

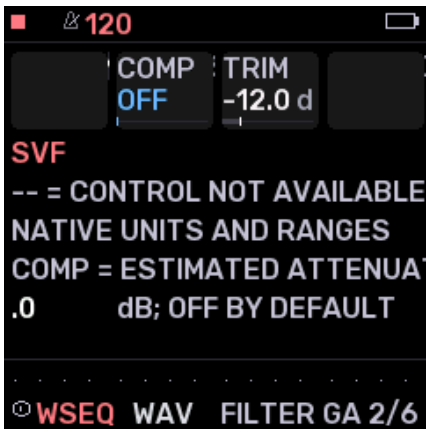
KNOB 3 - TRIM
Sound output calibration in half-dB steps. This changes the sound gain independently of the part mixer level.
Range: -32.0 dB to 63.5 dB. Generic default: .0 dB. Stored steps: -64 to 127.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER GAIN / WSEQ

Balance filter drive, optional output compensation and final trim.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER GAIN appears.

KNOB 1 - --
Unused control position; turning this knob has no assigned sound parameter on this layout.
Range: -- to --. Generic default: --. Stored steps: 0 to 0.

Choices: --.

KNOB 2 - COMP
Optional estimated filter-output compensation. ON attenuates large resonance/drive level changes; this is not a measured loudness-normalization control.
Range: OFF to ON. Generic default: OFF. Stored steps: 0 to 1.

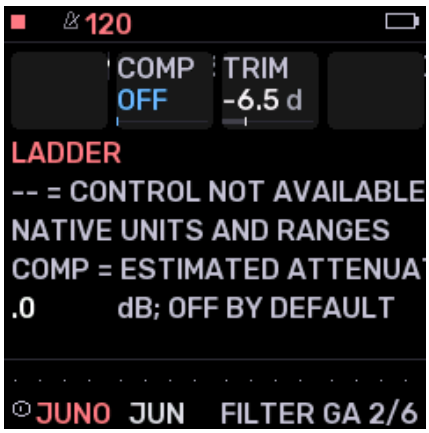
KNOB 3 - TRIM
Sound output calibration in half-dB steps. This changes the sound gain independently of the part mixer level.
Range: -32.0 dB to 63.5 dB. Generic default: .0 dB. Stored steps: -64 to 127.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER GAIN / JUNO

Balance filter drive, optional output compensation and final trim.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER GAIN appears.

KNOB 1 - --
Unused control position; turning this knob has no assigned sound parameter on this layout.
Range: -- to --. Generic default: --. Stored steps: 0 to 0.

Choices: --.

KNOB 2 - COMP
Optional estimated filter-output compensation. ON attenuates large resonance/drive level changes; this is not a measured loudness-normalization control.
Range: OFF to ON. Generic default: OFF. Stored steps: 0 to 1.

KNOB 3 - TRIM
Sound output calibration in half-dB steps. This changes the sound gain independently of the part mixer level.
Range: -32.0 dB to 63.5 dB. Generic default: .0 dB. Stored steps: -64 to 127.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FILTER GAIN / VECTOR

Balance filter drive, optional output compensation and final trim.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FILTER GAIN appears.

KNOB 1 - --
Unused control position; turning this knob has no assigned sound parameter on this layout.
Range: -- to --. Generic default: --. Stored steps: 0 to 0.

Choices: --.

KNOB 2 - COMP
Optional estimated filter-output compensation. ON attenuates large resonance/drive level changes; this is not a measured loudness-normalization control.
Range: OFF to OFF. Generic default: OFF. Stored steps: 0 to 0.

KNOB 3 - TRIM
Sound output calibration in half-dB steps. This changes the sound gain independently of the part mixer level.
Range: -32.0 dB to 63.5 dB. Generic default: .0 dB. Stored steps: -64 to 127.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 1 / ANALOG

Shape the engine source/model. These controls change with the selected engine.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 1 appears.

KNOB 1 - WAVE
Select the oscillator or modulation waveform. Changing the shape changes its harmonic content or movement.
Range: SAW to PWM. Generic default: SAW. Stored steps: 0 to 4.

Choices: SAW / SQR / TRI / SIN / PWM.

KNOB 2 - DTN
Detuning amount between oscillators/voices or FM operators, according to the selected engine.
Range: 0 ct to 127 ct. Generic default: 10 ct. Stored steps: 0 to 127.

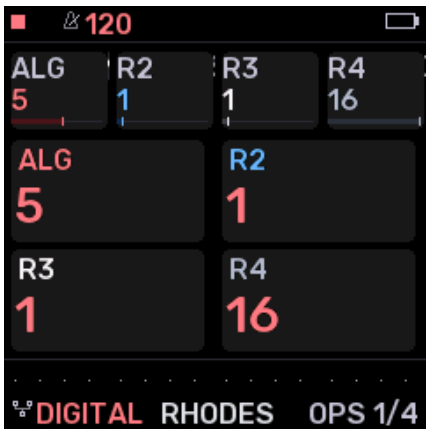
KNOB 3 - MIX
Dry/wet balance or shared effect output level, as identified by the page.
Range: 0 % to 100 %. Generic default: 50 %. Stored steps: 0 to 127.

KNOB 4 - NOIS
Noise oscillator/texture level.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 1 / DIGITAL

Shape the engine source/model. These controls change with the selected engine.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 1 appears.

KNOB 1 - ALG
FM/operator connection algorithm. Each choice changes which oscillators modulate or reach the output.
Range: 1 to 8. Generic default: 1. Stored steps: 0 to 7.

Choices: 1 / 2 / 3 / 4 / 5 / 6 / 7 / 8.

KNOB 2 - R2
Additional oscillator/operator frequency ratio.
Range: .5 to 16. Generic default: 1. Stored steps: 0 to 14.

Choices: .5 / 1 / 2 / 3 / 4 / 5 / 6 / 7 / 8 / 9 / 10 / 11 / 12 / 14 / 16.

KNOB 3 - R3
Additional oscillator/operator frequency ratio.
Range: .5 to 16. Generic default: 1. Stored steps: 0 to 14.

Choices: .5 / 1 / 2 / 3 / 4 / 5 / 6 / 7 / 8 / 9 / 10 / 11 / 12 / 14 / 16.

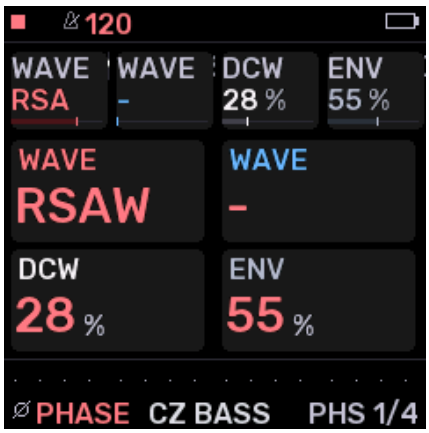
KNOB 4 - R4
Additional oscillator/operator frequency ratio.
Range: .5 to 16. Generic default: 1. Stored steps: 0 to 14.

Choices: .5 / 1 / 2 / 3 / 4 / 5 / 6 / 7 / 8 / 9 / 10 / 11 / 12 / 14 / 16.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 1 / PHASE

Shape the engine source/model. These controls change with the selected engine.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 1 appears.

KNOB 1 - WAVE
Select the oscillator or modulation waveform. Changing the shape changes its harmonic content or movement.
Range: SAW to RTRP. Generic default: SAW. Stored steps: 0 to 7.

Choices: SAW / SQR / PLS / DSIN / SPLS / RSAW / RTRI / RTRP.

KNOB 2 - WAVE2
Select the oscillator or modulation waveform. Changing the shape changes its harmonic content or movement.
Range: - to RTRP. Generic default: -. Stored steps: 0 to 8.

Choices: - / SAW / SQR / PLS / DSIN / SPLS / RSAW / RTRI / RTRP.

KNOB 3 - DCW
Phase-distortion waveform shaping amount.
Range: 0 % to 100 %. Generic default: 47 %. Stored steps: 0 to 127.

KNOB 4 - ENV
Envelope amount applied to phase-distortion waveform shaping (DCW).
Range: 0 % to 100 %. Generic default: 50 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 1 / LOFI

Shape the engine source/model. These controls change with the selected engine.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 1 appears.

KNOB 1 - CHIP
Lo-fi chip waveform/model.
Range: 4BIT to STEP. Generic default: 4BIT. Stored steps: 0 to 4.

Choices: 4BIT / 4B/2 / 8BIT / 1BIT / STEP.

KNOB 2 - WAVE
Select the oscillator or modulation waveform. Changing the shape changes its harmonic content or movement.
Range: PLS to WRAM. Generic default: PLS. Stored steps: 0 to 4.

Choices: PLS / TRI / SAW / NOIS / WRAM.

KNOB 3 - DUTY
Pulse width/duty cycle; changes harmonic balance.
Range: 0 to 127. Generic default: 64. Stored steps: 0 to 127.

KNOB 4 - CRSH
Engine bit-crush/lo-fi intensity.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 1 / LOFI / variant 5

Shape the engine source/model. These controls change with the selected engine.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 1 appears.

This example uses a model-dependent control layout. Choose the model/waveform shown in the screenshot to obtain these labels..

KNOB 1 - CHIP
Lo-fi chip waveform/model.
Range: 4BIT to STEP. Generic default: 4BIT. Stored steps: 0 to 4.

Choices: 4BIT / 4B/2 / 8BIT / 1BIT / STEP.

KNOB 2 - WAVE
Select the oscillator or modulation waveform. Changing the shape changes its harmonic content or movement.
Range: PLS to WRAM. Generic default: PLS. Stored steps: 0 to 4.

Choices: PLS / TRI / SAW / NOIS / WRAM.

KNOB 3 - DUTY
Pulse width/duty cycle; changes harmonic balance.
Range: 0 to 127. Generic default: 64. Stored steps: 0 to 127.

KNOB 4 - DCY
Chip stepped decay: the range selects constant, one-shot decay or looping decay behavior.
Range: CONST to L15. Generic default: CONST. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 1 / LOFI / variant 6

Shape the engine source/model. These controls change with the selected engine.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 1 appears.

This example uses a model-dependent control layout. Choose the model/waveform shown in the screenshot to obtain these labels..

KNOB 1 - CHIP
Lo-fi chip waveform/model.
Range: 4BIT to STEP. Generic default: 4BIT. Stored steps: 0 to 4.

Choices: 4BIT / 4B/2 / 8BIT / 1BIT / STEP.

KNOB 2 - WAVE
Select the oscillator or modulation waveform. Changing the shape changes its harmonic content or movement.
Range: PLS to WRAM. Generic default: PLS. Stored steps: 0 to 4.

Choices: PLS / TRI / SAW / NOIS / WRAM.

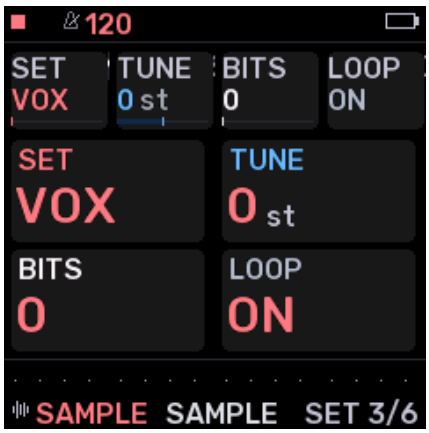
KNOB 3 - WAV#
Index into the LOFI waveform RAM shapes.
Range: SQ50 to STAIR. Generic default: BAS2. Stored steps: 0 to 127.

KNOB 4 - DCY
Chip stepped decay: the range selects constant, one-shot decay or looping decay behavior.
Range: CONST to L15. Generic default: CONST. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 1 / SAMPLE

Shape the engine source/model. These controls change with the selected engine.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 1 appears.

KNOB 1 - SET
Select the source sample/model set.
Range: VOX to AIR. Generic default: VOX. Stored steps: 4 to 7.

Choices: VOX / GLASS / PAD / AIR.

KNOB 2 - TUNE
Sample playback transposition in semitones; this is not the master tuning system.
Range: -24 st to +24 st. Generic default: 0 st. Stored steps: -24 to 24.

KNOB 3 - BITS
Bit depth of the crusher. Lower settings add quantization distortion.
Range: 0 to 127. Generic default: 0. Stored steps: 0 to 127.

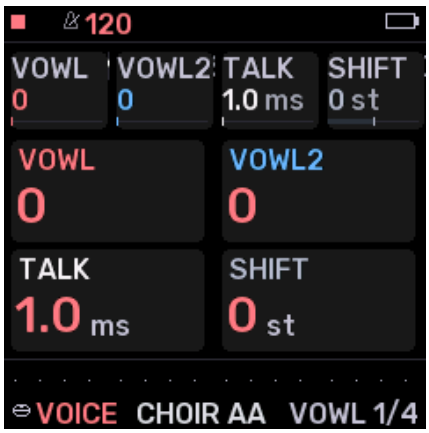
KNOB 4 - LOOP
Repeat/loop behavior. On ENV 2, ONE is a single contour and LOOP repeats it; on SPICE/TURNG it repeats generated variation.
Range: OFF to ON. Generic default: ON. Stored steps: 0 to 1.

Choices: OFF / ON.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 1 / VOICE

Shape the engine source/model. These controls change with the selected engine.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 1 appears.

KNOB 1 - VOWL
Formant/vowel position; moving it changes the vocal resonance.
Range: 0 to 127. Generic default: 0. Stored steps: 0 to 127.

KNOB 2 - VOWL2
Formant/vowel position; moving it changes the vocal resonance.
Range: 0 to 127. Generic default: 64. Stored steps: 0 to 127.

KNOB 3 - TALK
Time taken to move between the two vowel positions.
Range: 1.0 ms to 10.0 s. Generic default: 1.0 ms. Stored steps: 0 to 127.

KNOB 4 - SHIFT
Formant shift in semitones; changes vocal character separately from the note pitch.
Range: -12 st to +12 st. Generic default: 0 st. Stored steps: -12 to 12.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 1 / TRIO

Shape the engine source/model. These controls change with the selected engine.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 1 appears.

KNOB 1 - WAVE

Select the oscillator or modulation waveform. Changing the shape changes its harmonic content or movement.
Range: SAW3 to R+SP. Generic default: SAW3. Stored steps: 0 to 15.

Choices: SAW3 / PLS3 / PPT / SST / TRI3 / S&T / P&S / P+N / NOIS / SYNC / SYNCP / SYNC3 / RING / RING3 / R+S / R+SP.

KNOB 2 - INT2

Pitch interval for the additional oscillator/voice in the trio.
Range: -24 st to +24 st. Generic default: 0 st. Stored steps: -24 to 24.

KNOB 3 - INT3

Pitch interval for the additional oscillator/voice in the trio.
Range: -24 st to +24 st. Generic default: -12 st. Stored steps: -24 to 24.

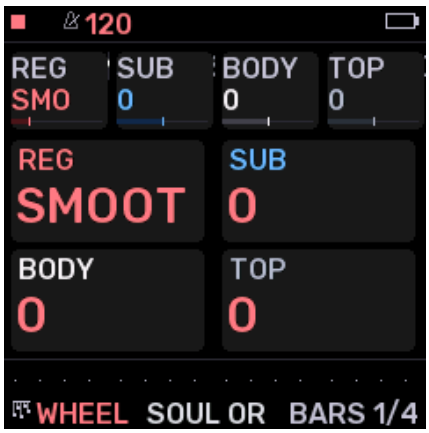
KNOB 4 - DTN

Detuning amount between oscillators/voices or FM operators, according to the selected engine.
Range: 0 ct to 50 ct. Generic default: 6 ct. Stored steps: 0 to 50.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 1 / WHEEL

Shape the engine source/model. These controls change with the selected engine.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 1 appears.

KNOB 1 - REG
Sample/grain register selection.
Range: FLUTE to FULL. Generic default: 3BAR. Stored steps: 0 to 15.

Choices: FLUTE / MELLO / HOLLW / SMOOT / 3BAR / BLUES / GOSPL / ROCK / TOPS / CLARI / REED / STRNG / CHAPL / BRITE / BASS / FULL.

KNOB 2 - SUB
Offset to the low organ drawbar group relative to the selected registration.
Range: -8 to 8. Generic default: 0. Stored steps: -8 to 8.

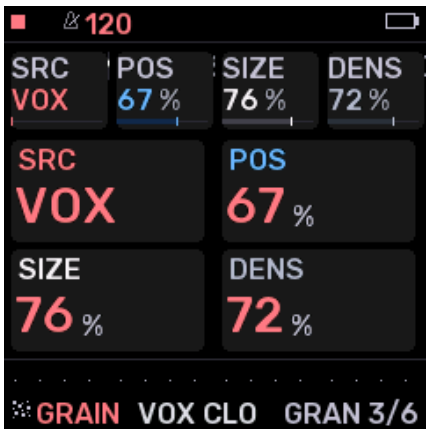
KNOB 3 - BODY
Offset to the middle organ drawbar group.
Range: -8 to 8. Generic default: 0. Stored steps: -8 to 8.

KNOB 4 - TOP
Offset to the upper organ drawbar group.
Range: -8 to 8. Generic default: 0. Stored steps: -8 to 8.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 1 / GRAIN

Shape the engine source/model. These controls change with the selected engine.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 1 appears.

KNOB 1 - SRC
Source sample set used to generate grains.
Range: VOX to AIR. Generic default: VOX. Stored steps: 4 to 7.

Choices: VOX / GLASS / PAD / AIR.

KNOB 2 - POS
Position in a sample, wavetable, wave sequence or excitation, as indicated by the engine.
Range: 0 % to 100 %. Generic default: 25 %. Stored steps: 0 to 127.

KNOB 3 - SIZE
Grain duration; short grains can sound granular, longer grains smoother.
Range: 0 % to 100 %. Generic default: 63 %. Stored steps: 0 to 127.

KNOB 4 - DENS
Grain density; more overlapping grains make a denser texture.
Range: 0 % to 100 %. Generic default: 63 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 1 / FM6

Shape the engine source/model. These controls change with the selected engine.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 1 appears.

KNOB 1 - ALG
PAT retains the patch algorithm; a numbered value overrides it with a DX7-style algorithm.
Range: PAT to 32. Generic default: PAT. Stored steps: 0 to 32.

KNOB 2 - FB
Feedback/recirculation. High values prolong the effect and can greatly increase level.
Range: -7 to 7. Generic default: 0. Stored steps: -7 to 7.

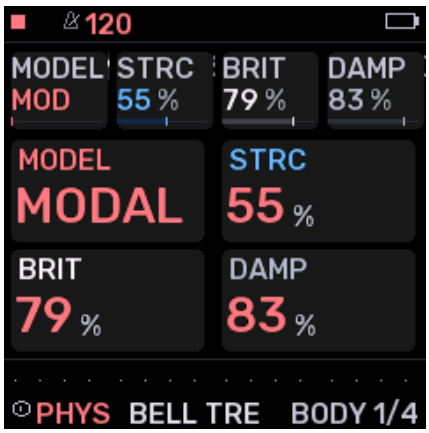
KNOB 3 - MLVL
Modulator level.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

KNOB 4 - MRAT
Modulator frequency ratio.
Range: -16 to 16. Generic default: 0. Stored steps: -16 to 16.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 1 / PHYS

Shape the engine source/model. These controls change with the selected engine.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 1 appears.

KNOB 1 - MODEL
Physical synthesis model: MODAL resonator, STRNG plucked string, MEMB membrane or SYMP sympathetic strings.
Range: MODAL to SYMP. Generic default: MODAL. Stored steps: 0 to 3.

Choices: MODAL / STRNG / MEMB / SYMP.

KNOB 2 - STRC
Resonator structure/material relationship.
Range: 0 % to 100 %. Generic default: 50 %. Stored steps: 0 to 127.

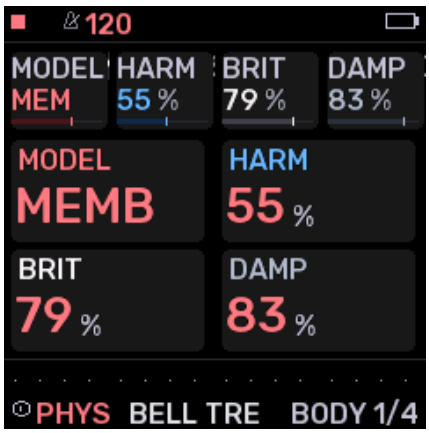
KNOB 3 - BRIT
High-frequency brightness/upper-harmonic balance.
Range: 0 % to 100 %. Generic default: 63 %. Stored steps: 0 to 127.

KNOB 4 - DAMP
High-frequency damping; darker reverberation or faster high-frequency decay as raised.
Range: 0 % to 100 %. Generic default: 63 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 1 / PHYS / variant 3

Shape the engine source/model. These controls change with the selected engine.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 1 appears.

This example uses a model-dependent control layout. Choose the model/waveform shown in the screenshot to obtain these labels..

KNOB 1 - MODEL
Physical synthesis model: MODAL resonator, STRNG plucked string, MEMB membrane or SYMP sympathetic strings.
Range: MODAL to SYMP. Generic default: MODAL. Stored steps: 0 to 3.

Choices: MODAL / STRNG / MEMB / SYMP.

KNOB 2 - HARM
Harmonic content/structure of the selected model.
Range: 0 % to 100 %. Generic default: 50 %. Stored steps: 0 to 127.

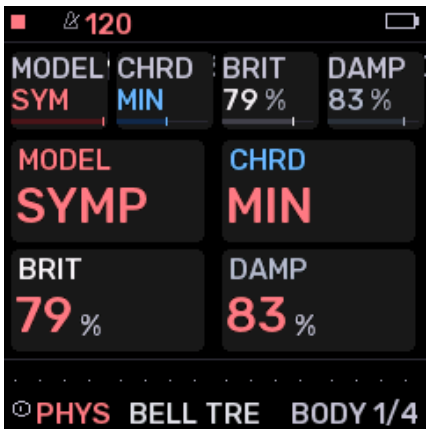
KNOB 3 - BRIT
High-frequency brightness/upper-harmonic balance.
Range: 0 % to 100 %. Generic default: 63 %. Stored steps: 0 to 127.

KNOB 4 - DAMP
High-frequency damping; darker reverberation or faster high-frequency decay as raised.
Range: 0 % to 100 %. Generic default: 63 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 1 / PHYS / variant 4

Shape the engine source/model. These controls change with the selected engine.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 1 appears.

This example uses a model-dependent control layout. Choose the model/waveform shown in the screenshot to obtain these labels..

KNOB 1 - MODEL
Physical synthesis model: MODAL resonator, STRNG plucked string, MEMB membrane or SYMP sympathetic strings.
Range: MODAL to SYMP. Generic default: MODAL. Stored steps: 0 to 3.

Choices: MODAL / STRNG / MEMB / SYMP.

KNOB 2 - CHRD
Sympathetic-string interval set; the named choices select the resonance relationships.
Range: OCT to ROOT. Generic default: MIN. Stored steps: 0 to 127.

KNOB 3 - BRIT
High-frequency brightness/upper-harmonic balance.
Range: 0 % to 100 %. Generic default: 63 %. Stored steps: 0 to 127.

KNOB 4 - DAMP
High-frequency damping; darker reverberation or faster high-frequency decay as raised.
Range: 0 % to 100 %. Generic default: 63 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 1 / NOISE

Shape the engine source/model. These controls change with the selected engine.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 1 appears.

KNOB 1 - MODE
Choose the operating mode for this page: LFO restart, arp direction, delay style, controller protocol or operator frequency mode.
Range: ANLG to META. Generic default: ANLG. Stored steps: 0 to 3.

Choices: ANLG / DUST / LFSR / META.

KNOB 2 - COLR
Tone or spectral balance of this sound/processor.
Range: 0 % to 100 %. Generic default: 50 %. Stored steps: 0 to 127.

KNOB 3 - FREQ
Fixed-versus-ratio frequency behavior for the selected FMX operator.
Range: 30 Hz to 16 kHz. Generic default: 1.5 kHz. Stored steps: 0 to 127.

KNOB 4 - RES
Filter resonance emphasizes frequencies around cutoff. High values may change level or approach self-oscillation.
Range: 0 % to 100 %. Generic default: 24 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 1 / HYBRID

Shape the engine source/model. These controls change with the selected engine.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 1 appears.

KNOB 1 - CUT

Offset to HYBRID filter cutoff relative to its stored patch; use the editor for the complete two-filter architecture.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

KNOB 2 - RES

Offset to HYBRID filter resonance relative to its stored patch.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

KNOB 3 - FENV

Filter-envelope amount; positive values open the filter with the envelope, negative values invert that movement where supported.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

KNOB 4 - PTCH

Select the underlying HYBRID patch, including the Vital-derived factory bank.
Range: SYNTH to V64 A. Generic default: SYNTH. Stored steps: 0 to 99.

Choices: SYNTHMAVEN / POLY PAD / STRINGS / ACID / SYNC LEAD / FM BELL / FM KEYS / WOBBLE / REESE / HOOVER / BRASS / VOX / PLUCK / RISER / METAL / FOLD BASS / CHIP LEAD / GLASS PAD.

Choices (continued): DRONE / 808 SUB / TRANCE / SAW PLUCK / SPACE BEL / INIT / MOOG BASS / MOOG LEAD / LADDER PAD / XD 80 BASS / XD CRYSTAL / XD DARKCHD / XD RHODES / XD GHOST / XD JUNOPAD / XD STRINGS / XD SYNWAVE / XD MOOG LD.

Choices (continued): V01 STROBE / V02 DARK / V03 REESE / V04 PLUCK / V05 SPACE / V06 ROOFTO / V07 ON DAN / V08 NOSTAL / V09 KEYS O / V10 NO TIM / V11 KEYAQU / V12 TOY / V13 CITY L / V14 THE VE / V15 STAY / V16 WOOD / V17 HOLOGR / V18 INSOMN.

Choices (continued): V19 MAINST / V20 HAPPY / V21 XYLOPH / V22 DR WHO / V23 AIRWAV / V24 WOBBLY / V25 ALCOHO / V26 RETROV / V27 STRING / V28 BLUR / V29 ORCHES / V30 VOID / V31 ALIEN / V32 NEGATI / V33 BLACK / V34 GANGNA / V35 YEAH / V36 TYPE A.

EDIT 1 / HYBRID - continued

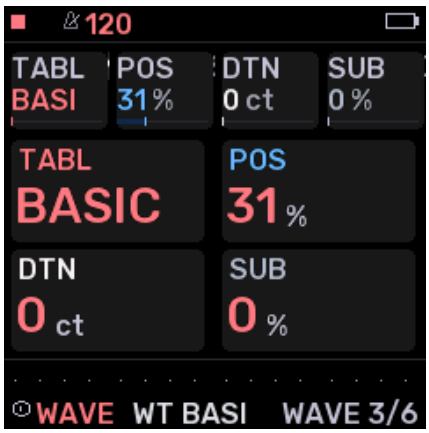
Choices (continued): V37 TAKE 2 / V38 BRASS / V39 JUMP / V40 LIVE A / V41 FALLIN / V42 WIN / V43 HOUSE / V44 SOFT L / V45 COUNTD / V46 LOST / V47 NOW / V48 ORGAN / V49 STRANG / V50 SOLO L / V51 SHOW / V52 BRAMM / V53 BASS G / V54 RUSH.

Choices (continued): V55 POLARI / V56 DEEP R / V57 DEEP R / V58 TUNNEL / V59 SYNTH / V60 STARS / V61 ARP SY / V62 KEYS W / V63 KEYS E / V64 ALIEN.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 1 / WAVE

Shape the engine source/model. These controls change with the selected engine.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 1 appears.

KNOB 1 - TABL

Select the wavetable. USER WT remains silent until a user table is uploaded.

Range: BASIC to USR4. Generic default: BASIC. Stored steps: 0 to 19.

Choices: BASIC / PWM / HARMON / ODD / VOCAL / RESO / SYNC / FM / FM 2:3 / FOLD / PD / DIGITL / BELL / ORGAN / CHIP / COMB / USR1 / USR2.

Choices (continued): USR3 / USR4.

KNOB 2 - POS

Position in a sample, wavetable, wave sequence or excitation, as indicated by the engine.

Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 3 - DTN

Detuning amount between oscillators/voices or FM operators, according to the selected engine.

Range: 0 ct to 50 ct. Generic default: 0 ct. Stored steps: 0 to 50.

KNOB 4 - SUB

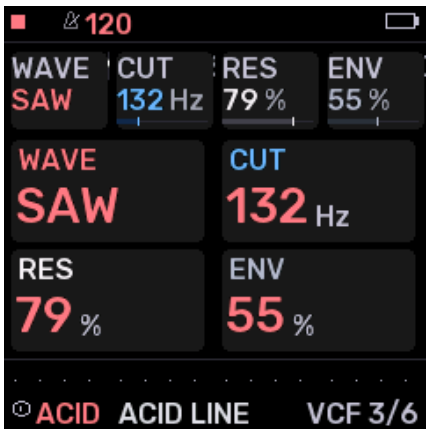
Sub-oscillator level under the main oscillator.

Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 1 / ACID

Shape the engine source/model. These controls change with the selected engine.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 1 appears.

KNOB 1 - WAVE
Select the oscillator or modulation waveform. Changing the shape changes its harmonic content or movement.
Range: SAW to SQR. Generic default: SAW. Stored steps: 0 to 1.

Choices: SAW / SQR.

KNOB 2 - CUT
Low-pass/filter cutoff: lower values remove more upper harmonics.
Range: 30 Hz to 16 kHz. Generic default: 132 Hz. Stored steps: 0 to 127.

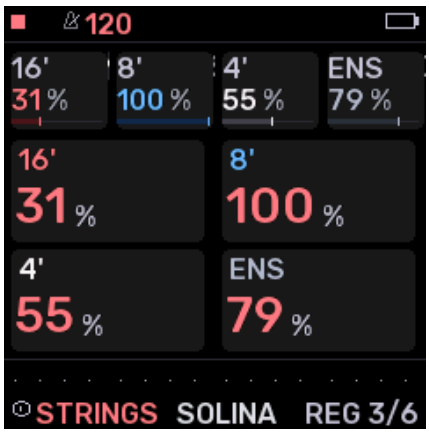
KNOB 3 - RES
Filter resonance emphasizes frequencies around cutoff. High values may change level or approach self-oscillation.
Range: 0 % to 100 %. Generic default: 79 %. Stored steps: 0 to 127.

KNOB 4 - ENV
Filter-envelope amount; positive values open the filter with the envelope, negative values invert that movement where supported.
Range: 0 % to 100 %. Generic default: 55 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 1 / STRINGS

Shape the engine source/model. These controls change with the selected engine.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 1 appears.

KNOB 1 - 16'
Organ drawbar level for the indicated pitch register.
Range: 0 % to 100 %. Generic default: 31 %. Stored steps: 0 to 127.

KNOB 2 - 8'
Organ drawbar level for the indicated pitch register.
Range: 0 % to 100 %. Generic default: 100 %. Stored steps: 0 to 127.

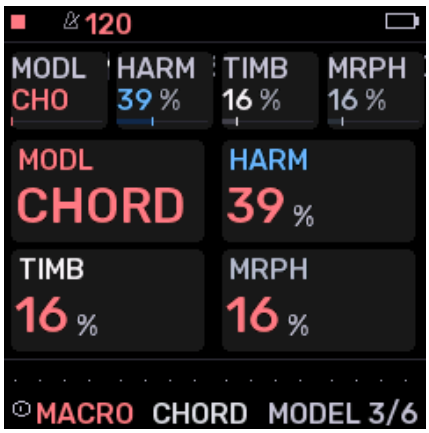
KNOB 3 - 4'
Organ drawbar level for the indicated pitch register.
Range: 0 % to 100 %. Generic default: 55 %. Stored steps: 0 to 127.

KNOB 4 - ENS
String ensemble/chorus amount.
Range: 0 % to 100 %. Generic default: 79 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 1 / MACRO

Shape the engine source/model. These controls change with the selected engine.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 1 appears.

KNOB 1 - MODL
Select the synthesis model inside MACRO; the engine remains MACRO. Each model interprets HARM, TIMB and MRPH differently.
Range: CHORD to SPEEC. Generic default: CHORD. Stored steps: 0 to 3.

Choices: CHORD / SWARM / ADDTV / SPEECH.

KNOB 2 - HARM
Harmonic content/structure of the selected model.
Range: 0 % to 100 %. Generic default: 50 %. Stored steps: 0 to 127.

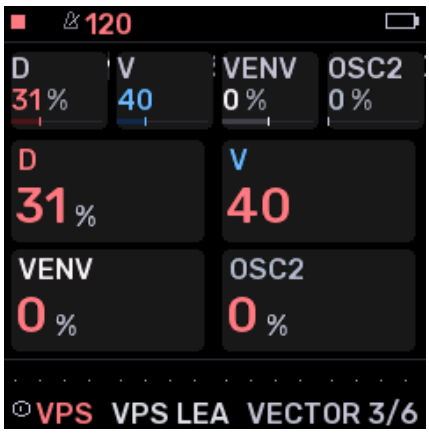
KNOB 3 - TIMB
Model-dependent timbre shaping.
Range: 0 % to 100 %. Generic default: 50 %. Stored steps: 0 to 127.

KNOB 4 - MRPH
Morph amount between the model waveforms.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 1 / VPS

Shape the engine source/model. These controls change with the selected engine.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 1 appears.

KNOB 1 - D
Phase breakpoint/deformation position of the vector phase-shaping oscillator.
Range: 0 % to 100 %. Generic default: 39 %. Stored steps: 0 to 127.

KNOB 2 - V
Phase-shaping slope/ratio parameter; alters the waveform around the breakpoint.
Range: 0 to 127. Generic default: 32. Stored steps: 0 to 127.

KNOB 3 - VENV
Velocity envelope amount.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

KNOB 4 - OSC2
Second oscillator setting/level. Use the engine description to distinguish oscillator type from mix.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 1 / TERRAIN

Shape the engine source/model. These controls change with the selected engine.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 1 appears.

KNOB 1 - TERR
Select the wave-terrain surface.
Range: WAVES to CHAOS. Generic default: WAVES. Stored steps: 0 to 7.

Choices: WAVES / RIPPL / CRYST / BUMPS / SADDL / FOLDS / TWIST / CHAOS.

KNOB 2 - RAD
Wave-terrain radial coordinate.
Range: 0 % to 100 %. Generic default: 39 %. Stored steps: 0 to 127.

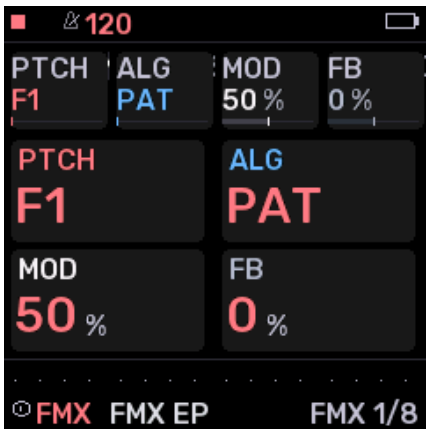
KNOB 3 - X
Coordinate in the vector/terrain mix; moving between sources changes the timbre.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 4 - Y
Coordinate in the vector/terrain mix; moving between sources changes the timbre.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 1 / FMX

Shape the engine source/model. These controls change with the selected engine.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 1 appears.

KNOB 1 - PTCH
Select an internal engine patch/model.
Range: F1 to B27. Generic default: F1. Stored steps: 0 to 42.

KNOB 2 - ALG
FM/operator connection algorithm. Each choice changes which oscillators modulate or reach the output.
Range: PAT to 40. Generic default: PAT. Stored steps: 0 to 40.

KNOB 3 - MOD
On MOD PRESET, choose a ready-made expressive routing; other pages use it for modulation amount.
Range: 0 % to 100 %. Generic default: 50 %. Stored steps: 0 to 127.

KNOB 4 - FB
Feedback/recirculation. High values prolong the effect and can greatly increase level.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 1 / WSEQ

Shape the engine source/model. These controls change with the selected engine.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 1 appears.

KNOB 1 - SEQ
Wave-sequence selection/control.
Range: STEPS to PING. Generic default: STEPS. Stored steps: 0 to 15.

Choices: STEPS / RHYTHM / SWEEP / FIFTHS / MINOR / MAJOR / OCTAVE / GATE / BELLS / 8BIT / MORPH / VOX / NOISY / DUB / DORIAN / PING.

KNOB 2 - RATE
Speed of the selected oscillator/modulator/effect. The displayed unit identifies Hz or a tempo division.
Range: 1/32 to BAR. Generic default: 1/16. Stored steps: 0 to 127.

KNOB 3 - XFADE
Crossfade between wave-sequence stages/sources.
Range: 0 % to 100 %. Generic default: 16 %. Stored steps: 0 to 127.

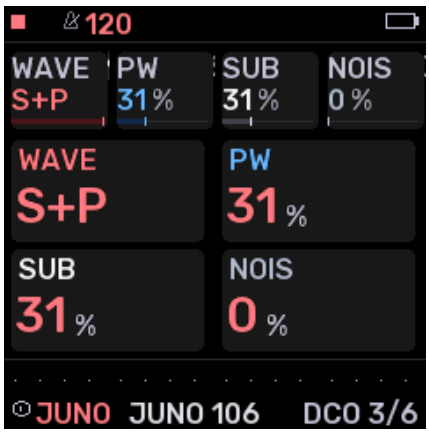
KNOB 4 - LOOP
Repeat/loop behavior. On ENV 2, ONE is a single contour and LOOP repeats it; on SPICE/TURNG it repeats generated variation.
Range: FWD to ONCE. Generic default: FWD. Stored steps: 0 to 3.

Choices: FWD / BACK / PING / ONCE.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 1 / JUNO

Shape the engine source/model. These controls change with the selected engine.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 1 appears.

KNOB 1 - WAVE
Select the oscillator or modulation waveform. Changing the shape changes its harmonic content or movement.
Range: SAW to S+P. Generic default: S+P. Stored steps: 0 to 2.

Choices: SAW / PLS / S+P.

KNOB 2 - PW
Pulse width/duty cycle; changes harmonic balance.
Range: 0 % to 100 %. Generic default: 24 %. Stored steps: 0 to 127.

KNOB 3 - SUB
Sub-oscillator level under the main oscillator.
Range: 0 % to 100 %. Generic default: 31 %. Stored steps: 0 to 127.

KNOB 4 - NOIS
Noise oscillator/texture level.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 1 / VECTOR

Shape the engine source/model. These controls change with the selected engine.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 1 appears.

KNOB 1 - A

Select the source waveform at vector corner A.

Range: SINE to METAL. Generic default: SAW. Stored steps: 0 to 15.

Choices: SINE / TRI / SAW / SQR / PLS25 / PLS10 / ORGAN / VOX A / VOX O / BELL / HOLLW / BRASS / RESO / DIGI / STRNG / METAL.

KNOB 2 - B

Select the source waveform at vector corner B.

Range: SINE to METAL. Generic default: VOX A. Stored steps: 0 to 15.

Choices: SINE / TRI / SAW / SQR / PLS25 / PLS10 / ORGAN / VOX A / VOX O / BELL / HOLLW / BRASS / RESO / DIGI / STRNG / METAL.

KNOB 3 - C

Select the source waveform at vector corner C.

Range: SINE to METAL. Generic default: STRNG. Stored steps: 0 to 15.

Choices: SINE / TRI / SAW / SQR / PLS25 / PLS10 / ORGAN / VOX A / VOX O / BELL / HOLLW / BRASS / RESO / DIGI / STRNG / METAL.

KNOB 4 - D

Select the source waveform at vector corner D.

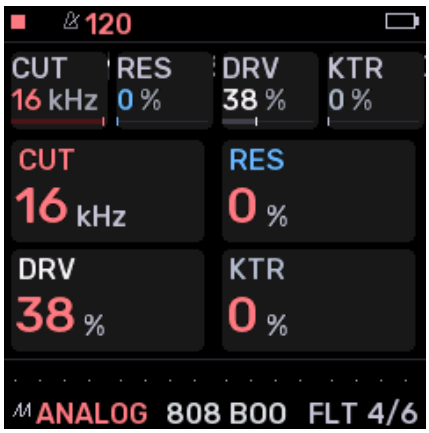
Range: SINE to METAL. Generic default: BELL. Stored steps: 0 to 15.

Choices: SINE / TRI / SAW / SQR / PLS25 / PLS10 / ORGAN / VOX A / VOX O / BELL / HOLLW / BRASS / RESO / DIGI / STRNG / METAL.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 2 / ANALOG

Shape the remaining engine controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 2 appears.

KNOB 1 - CUT
Low-pass/filter cutoff: lower values remove more upper harmonics.
Range: 30 Hz to 16 kHz. Generic default: 2.5 kHz. Stored steps: 0 to 127.

KNOB 2 - RES
Filter resonance emphasizes frequencies around cutoff. High values may change level or approach self-oscillation.
Range: 0 % to 100 %. Generic default: 24 %. Stored steps: 0 to 127.

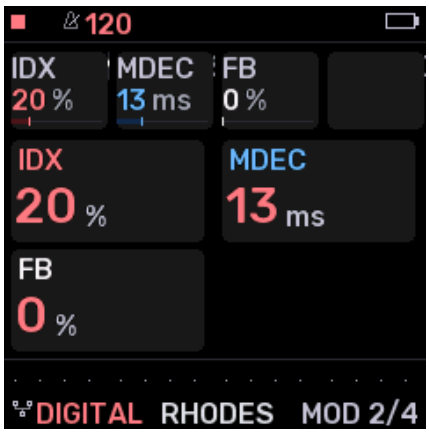
KNOB 3 - DRV
Drive into a saturation/filter stage. More drive can add harmonics and change level.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 4 - KTR
Keyboard tracking: changes the filter/model response with played pitch.
Range: 0 % to 100 %. Generic default: 50 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 2 / DIGITAL

Shape the remaining engine controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 2 appears.

KNOB 1 - IDX
FM/phase modulation index; higher values generally add sidebands.
Range: 0 % to 100 %. Generic default: 47 %. Stored steps: 0 to 127.

KNOB 2 - MDEC
Modulator decay time.
Range: 1.0 ms to 10.0 s. Generic default: 78 ms. Stored steps: 0 to 127.

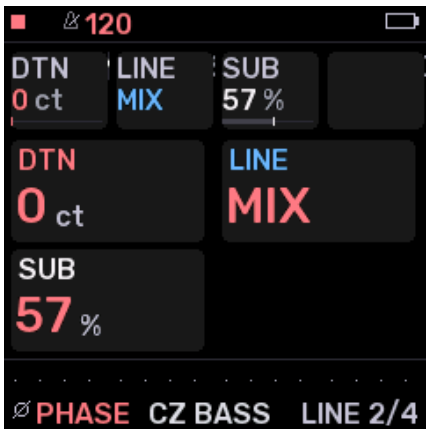
KNOB 3 - FB
Feedback/recirculation. High values prolong the effect and can greatly increase level.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 4 - -
Unused control position; turning this knob has no assigned sound parameter on this layout.
Range: 0 to 0. Generic default: 0. Stored steps: 0 to 0.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 2 / PHASE

Shape the remaining engine controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 2 appears.

KNOB 1 - DTN
Detuning amount between oscillators/voices or FM operators, according to the selected engine.
Range: 0 ct to 127 ct. Generic default: 0 ct. Stored steps: 0 to 127.

KNOB 2 - LINE
Combine the two phase-distortion waveforms by mixing or ring modulation.
Range: MIX to RING. Generic default: MIX. Stored steps: 0 to 1.

Choices: MIX / RING.

KNOB 3 - SUB
Sub-oscillator level under the main oscillator.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 4 - -
Unused control position; turning this knob has no assigned sound parameter on this layout.
Range: 0 to 0. Generic default: 0. Stored steps: 0 to 0.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 2 / LOFI

Shape the remaining engine controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 2 appears.

KNOB 1 - SWP
Sweep amount or rate of the selected engine.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 2 - VIB
Built-in chip vibrato depth.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 3 - ARP
Engine-specific arpeggiation/model control; the dedicated ARP menu remains the performance arpeggiator.
Range: OFF to MIN. Generic default: OFF. Stored steps: 0 to 3.

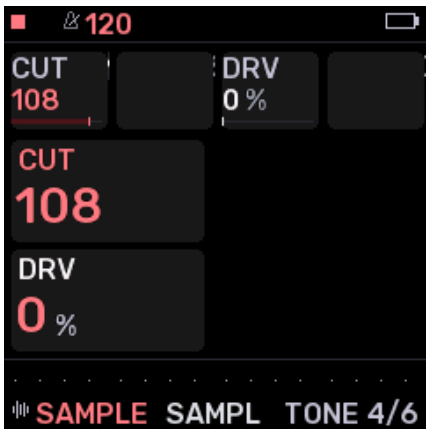
Choices: OFF / OCT / MAJ / MIN.

KNOB 4 - TONE
Tone or spectral balance of this sound/processor.
Range: 0 % to 100 %. Generic default: 100 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 2 / SAMPLE

Shape the remaining engine controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 2 appears.

KNOB 1 - CUT
Low-pass/filter cutoff: lower values remove more upper harmonics.
Range: 0 to 127. Generic default: 127. Stored steps: 0 to 127.

KNOB 2 - -
Unused control position; turning this knob has no assigned sound parameter on this layout.
Range: 0 to 0. Generic default: 0. Stored steps: 0 to 0.

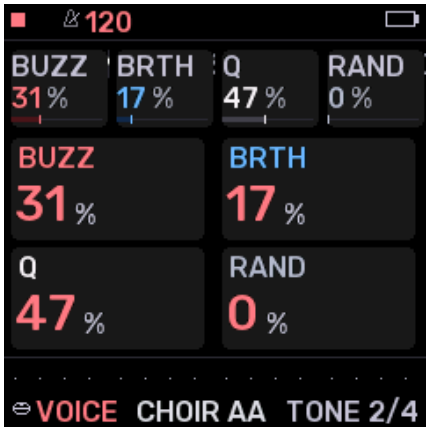
KNOB 3 - DRV
Drive into a saturation/filter stage. More drive can add harmonics and change level.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 4 - -
Unused control position; turning this knob has no assigned sound parameter on this layout.
Range: 0 to 0. Generic default: 0. Stored steps: 0 to 0.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 2 / VOICE

Shape the remaining engine controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 2 appears.

KNOB 1 - BUZZ
Buzz/higher-partial content.
Range: 0 % to 100 %. Generic default: 50 %. Stored steps: 0 to 127.

KNOB 2 - BRTH
Breathy noise contribution.
Range: 0 % to 100 %. Generic default: 8 %. Stored steps: 0 to 127.

KNOB 3 - Q
Filter resonance emphasizes frequencies around cutoff. High values may change level or approach self-oscillation.
Range: 0 % to 100 %. Generic default: 50 %. Stored steps: 0 to 127.

KNOB 4 - RAND
Randomization depth/variation in this engine.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 2 / TRIO

Shape the remaining engine controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 2 appears.

KNOB 1 - MODE
Choose the operating mode for this page: LFO restart, arp direction, delay style, controller protocol or operator frequency mode.
Range: LP to NOT. Generic default: LP. Stored steps: 0 to 3.

Choices: LP / BP / HP / NOT.

KNOB 2 - CUT
Low-pass/filter cutoff: lower values remove more upper harmonics.
Range: 30 Hz to 16 kHz. Generic default: 1.5 kHz. Stored steps: 0 to 127.

KNOB 3 - RES
Filter resonance emphasizes frequencies around cutoff. High values may change level or approach self-oscillation.
Range: 0 % to 100 %. Generic default: 31 %. Stored steps: 0 to 127.

KNOB 4 - PW
Pulse width/duty cycle; changes harmonic balance.
Range: 0 % to 100 %. Generic default: 50 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 2 / WHEEL

Shape the remaining engine controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 2 appears.

KNOB 1 - PERC
Percussive organ/transient component.
Range: OFF to 3SLOW. Generic default: OFF. Stored steps: 0 to 6.

Choices: OFF / 2ND / 3RD / 2SOFT / 3SOFT / 2SLOW / 3SLOW.

KNOB 2 - CLICK
Short attack/click component.
Range: 0 % to 100 %. Generic default: 31 %. Stored steps: 0 to 127.

KNOB 3 - DRV
Drive into a saturation/filter stage. More drive can add harmonics and change level.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

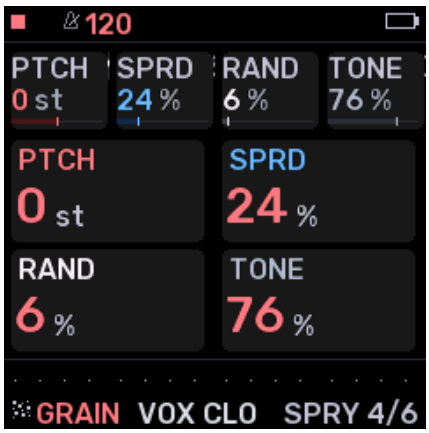
KNOB 4 - ROTR
Rotary speaker movement/speed parameter.
Range: OFF to FAST. Generic default: SLOW. Stored steps: 0 to 2.

Choices: OFF / SLOW / FAST.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 2 / GRAIN

Shape the remaining engine controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 2 appears.

KNOB 1 - PTCH
Grain playback pitch offset in semitones.
Range: -24 st to +24 st. Generic default: 0 st. Stored steps: -24 to 24.

KNOB 2 - SPRD
Stereo spread or oscillator/processor width. Check the result in mono.
Range: 0 % to 100 %. Generic default: 24 %. Stored steps: 0 to 127.

KNOB 3 - RAND
Randomization depth/variation in this engine.
Range: 0 % to 100 %. Generic default: 8 %. Stored steps: 0 to 127.

KNOB 4 - TONE
Tone or spectral balance of this sound/processor.
Range: 0 % to 100 %. Generic default: 100 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 2 / FM6

Shape the remaining engine controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 2 appears.

KNOB 1 - MEG
Offset to the modulator envelope behavior of the underlying FM6 patch.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

KNOB 2 - VMOD
Velocity modulation of model timbre/level.
Range: -7 to 7. Generic default: 0. Stored steps: -7 to 7.

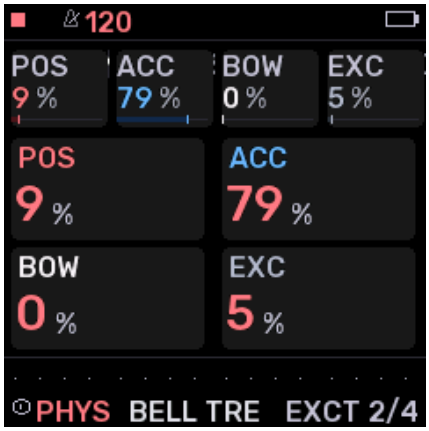
KNOB 3 - DTUN
Detuning amount between oscillators/voices or FM operators, according to the selected engine.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 4 - PTCH
Select an internal engine patch/model.
Range: F1 to ARABE. Generic default: F1. Stored steps: 0 to 67.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 2 / PHYS

Shape the remaining engine controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 2 appears.

KNOB 1 - POS
Position in a sample, wavetable, wave sequence or excitation, as indicated by the engine.
Range: 0 % to 100 %. Generic default: 16 %. Stored steps: 0 to 127.

KNOB 2 - ACC
Accent sensitivity/amount.
Range: 0 % to 100 %. Generic default: 71 %. Stored steps: 0 to 127.

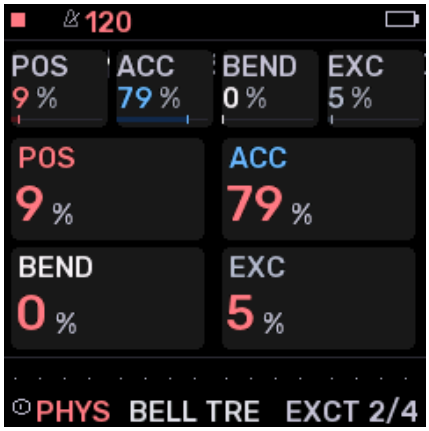
KNOB 3 - BOW
Bowing/excitation behavior of the physical model.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 4 - EXC
Physical-model excitation type or intensity.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 2 / PHYS / variant 3

Shape the remaining engine controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 2 appears.

This example uses a model-dependent control layout. Choose the model/waveform shown in the screenshot to obtain these labels..

KNOB 1 - POS
Position in a sample, wavetable, wave sequence or excitation, as indicated by the engine.
Range: 0 % to 100 %. Generic default: 16 %. Stored steps: 0 to 127.

KNOB 2 - ACC
Accent sensitivity/amount.
Range: 0 % to 100 %. Generic default: 71 %. Stored steps: 0 to 127.

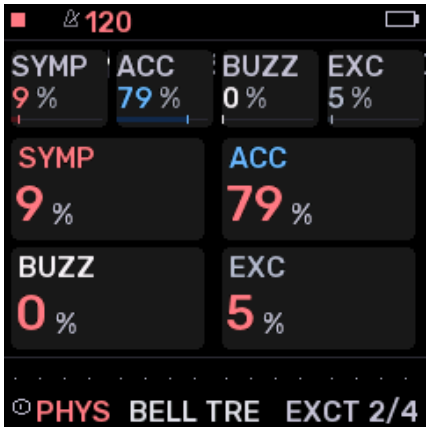
KNOB 3 - BEND
Pitch drop after the membrane strike, up to twelve semitones.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 4 - EXC
Physical-model excitation type or intensity.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 2 / PHYS / variant 4

Shape the remaining engine controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 2 appears.

This example uses a model-dependent control layout. Choose the model/waveform shown in the screenshot to obtain these labels..

KNOB 1 - SYMP
Level of sympathetic strings ringing above the played note.
Range: 0 % to 100 %. Generic default: 16 %. Stored steps: 0 to 127.

KNOB 2 - ACC
Accent sensitivity/amount.
Range: 0 % to 100 %. Generic default: 71 %. Stored steps: 0 to 127.

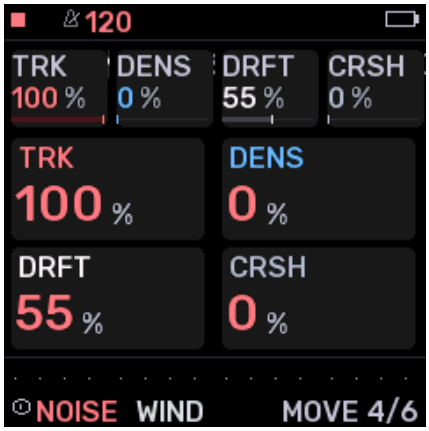
KNOB 3 - BUZZ
Buzz/higher-partial content.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 4 - EXC
Physical-model excitation type or intensity.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 2 / NOISE

Shape the remaining engine controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 2 appears.

KNOB 1 - TRK
Keyboard tracking of the noise resonator frequency.
Range: 0 % to 100 %. Generic default: 50 %. Stored steps: 0 to 127.

KNOB 2 - DENS
Grain density; more overlapping grains make a denser texture.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 3 - DRFT
Slow oscillator pitch variation.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 4 - CRSH
Engine bit-crush/lo-fi intensity.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 2 / NOISE / variant 3

Shape the remaining engine controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 2 appears.

This example uses a model-dependent control layout. Choose the model/waveform shown in the screenshot to obtain these labels..

KNOB 1 - TRK
Keyboard tracking of the noise resonator frequency.
Range: 0 % to 100 %. Generic default: 50 %. Stored steps: 0 to 127.

KNOB 2 - LEN
Length of the current sequence or setlist. Shorter lengths wrap sooner; a setlist length of zero disables the list.
Range: L23 to L7. Generic default: L23. Stored steps: 0 to 127.

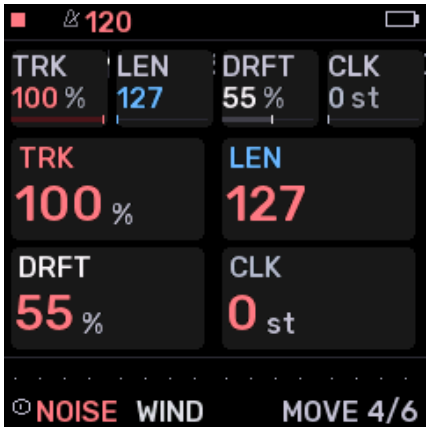
KNOB 3 - DRFT
Slow oscillator pitch variation.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 4 - CLK
Clock/sample-hold rate of the digital noise model.
Range: 0 st to 127 st. Generic default: 0 st. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 2 / NOISE / variant 4

Shape the remaining engine controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 2 appears.

This example uses a model-dependent control layout. Choose the model/waveform shown in the screenshot to obtain these labels..

KNOB 1 - TRK
Keyboard tracking of the noise resonator frequency.
Range: 0 % to 100 %. Generic default: 50 %. Stored steps: 0 to 127.

KNOB 2 - LEN
Length of the current sequence or setlist. Shorter lengths wrap sooner; a setlist length of zero disables the list.
Range: 127 to 7. Generic default: 127. Stored steps: 0 to 127.

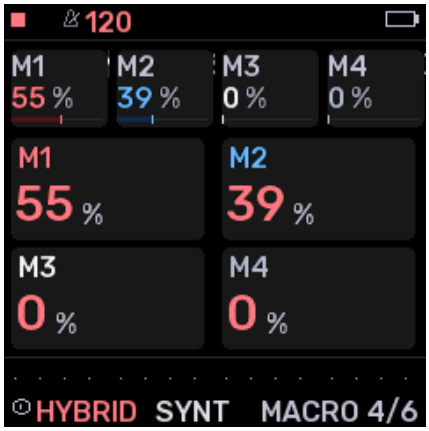
KNOB 3 - DRFT
Slow oscillator pitch variation.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 4 - CLK
Clock/sample-hold rate of the digital noise model.
Range: 0 st to 127 st. Generic default: 0 st. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 2 / HYBRID

Shape the remaining engine controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 2 appears.

KNOB 1 - M1
One of four macro controls. HYBRID uses patch-defined macro assignments; other engines provide brightness, timbre, motion and space behavior.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 2 - M2
One of four macro controls. HYBRID uses patch-defined macro assignments; other engines provide brightness, timbre, motion and space behavior.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

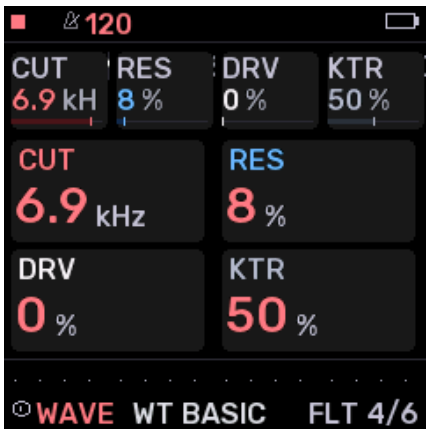
KNOB 3 - M3
One of four macro controls. HYBRID uses patch-defined macro assignments; other engines provide brightness, timbre, motion and space behavior.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 4 - M4
One of four macro controls. HYBRID uses patch-defined macro assignments; other engines provide brightness, timbre, motion and space behavior.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 2 / WAVE

Shape the remaining engine controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 2 appears.

KNOB 1 - CUT
Low-pass/filter cutoff: lower values remove more upper harmonics.
Range: 30 Hz to 16 kHz. Generic default: 6.9 kHz. Stored steps: 0 to 127.

KNOB 2 - RES
Filter resonance emphasizes frequencies around cutoff. High values may change level or approach self-oscillation.
Range: 0 % to 100 %. Generic default: 8 %. Stored steps: 0 to 127.

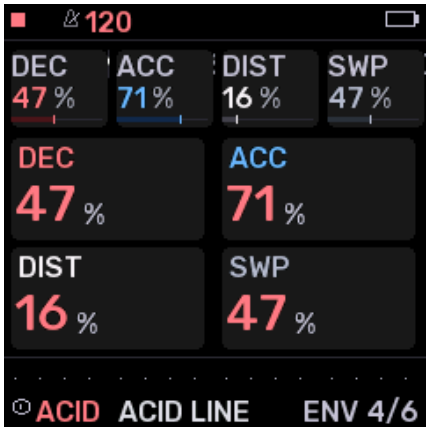
KNOB 3 - DRV
Drive into a saturation/filter stage. More drive can add harmonics and change level.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 4 - KTR
Keyboard tracking: changes the filter/model response with played pitch.
Range: 0 % to 100 %. Generic default: 50 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 2 / ACID

Shape the remaining engine controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 2 appears.

KNOB 1 - DEC
Decay time from the attack peak to the sustain level.
Range: 0 % to 100 %. Generic default: 47 %. Stored steps: 0 to 127.

KNOB 2 - ACC
Accent sensitivity/amount.
Range: 0 % to 100 %. Generic default: 71 %. Stored steps: 0 to 127.

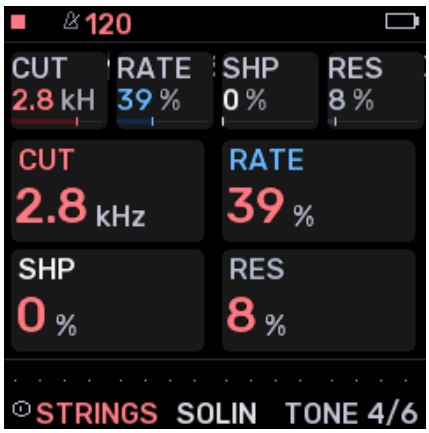
KNOB 3 - DIST
Amount sent to distortion.
Range: 0 % to 100 %. Generic default: 16 %. Stored steps: 0 to 127.

KNOB 4 - SWP
Sweep amount or rate of the selected engine.
Range: 0 % to 100 %. Generic default: 47 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 2 / STRINGS

Shape the remaining engine controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 2 appears.

KNOB 1 - CUT
Low-pass/filter cutoff: lower values remove more upper harmonics.
Range: 30 Hz to 16 kHz. Generic default: 2.8 kHz. Stored steps: 0 to 127.

KNOB 2 - RATE
Speed of the selected oscillator/modulator/effect. The displayed unit identifies Hz or a tempo division.
Range: 0 % to 100 %. Generic default: 39 %. Stored steps: 0 to 127.

KNOB 3 - SHP
Waveform/timbre shaping or the amount of envelope/LFO shaping, according to this page.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 4 - RES
Filter resonance emphasizes frequencies around cutoff. High values may change level or approach self-oscillation.
Range: 0 % to 100 %. Generic default: 8 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 2 / MACRO

Shape the remaining engine controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 2 appears.

KNOB 1 - CUT
Low-pass/filter cutoff: lower values remove more upper harmonics.
Range: 30 Hz to 16 kHz. Generic default: 6.9 kHz. Stored steps: 0 to 127.

KNOB 2 - RES
Filter resonance emphasizes frequencies around cutoff. High values may change level or approach self-oscillation.
Range: 0 % to 100 %. Generic default: 8 %. Stored steps: 0 to 127.

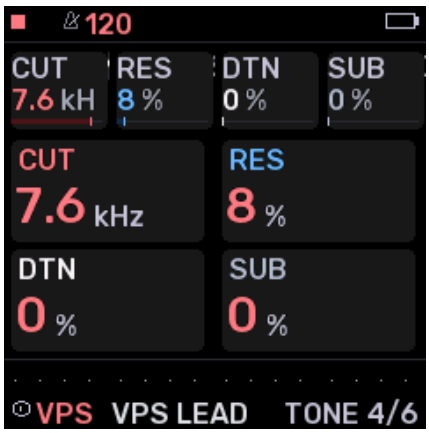
KNOB 3 - DEC
Decay time from the attack peak to the sustain level.
Range: 0 % to 100 %. Generic default: 47 %. Stored steps: 0 to 127.

KNOB 4 - LPG
Low-pass gate behavior of the model.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 2 / VPS

Shape the remaining engine controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 2 appears.

KNOB 1 - CUT
Low-pass/filter cutoff: lower values remove more upper harmonics.
Range: 30 Hz to 16 kHz. Generic default: 6.9 kHz. Stored steps: 0 to 127.

KNOB 2 - RES
Filter resonance emphasizes frequencies around cutoff. High values may change level or approach self-oscillation.
Range: 0 % to 100 %. Generic default: 8 %. Stored steps: 0 to 127.

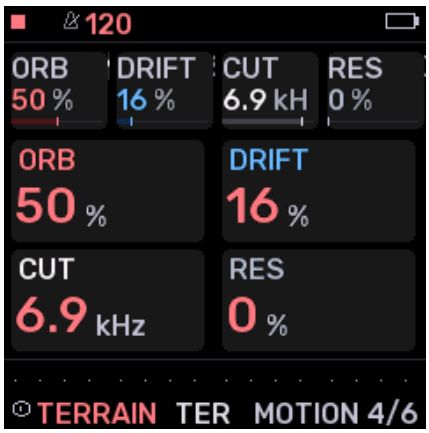
KNOB 3 - DTN
Detuning amount between oscillators/voices or FM operators, according to the selected engine.
Range: 0 % to 100 %. Generic default: 16 %. Stored steps: 0 to 127.

KNOB 4 - SUB
Sub-oscillator level under the main oscillator.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 2 / TERRAIN

Shape the remaining engine controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 2 appears.

KNOB 1 - ORB
Vector/orbital modulation position or movement.
Range: 0 % to 100 %. Generic default: 50 %. Stored steps: 0 to 127.

KNOB 2 - DRIFT
Slow oscillator pitch variation.
Range: 0 % to 100 %. Generic default: 16 %. Stored steps: 0 to 127.

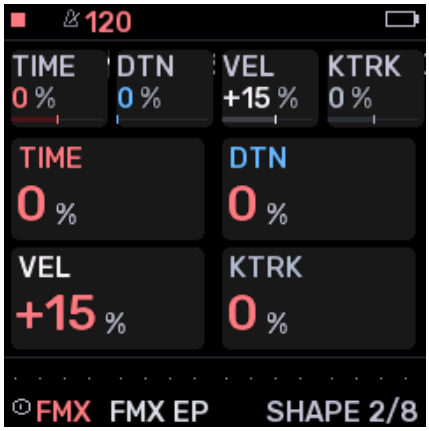
KNOB 3 - CUT
Low-pass/filter cutoff: lower values remove more upper harmonics.
Range: 30 Hz to 16 kHz. Generic default: 6.9 kHz. Stored steps: 0 to 127.

KNOB 4 - RES
Filter resonance emphasizes frequencies around cutoff. High values may change level or approach self-oscillation.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 2 / FMX

Shape the remaining engine controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 2 appears.

KNOB 1 - TIME
Global offset to the FMX operator envelope times; individual operator envelopes remain available under OP ENV.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

KNOB 2 - DTN
Detuning amount between oscillators/voices or FM operators, according to the selected engine.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 3 - VEL
Velocity sensitivity of the operator/modulation target.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

KNOB 4 - KTRK
Keyboard tracking: changes the filter/model response with played pitch.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 2 / WSEQ

Shape the remaining engine controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 2 appears.

KNOB 1 - START
Position in a sample, wavetable, wave sequence or excitation, as indicated by the engine.
Range: 0 to 127. Generic default: 0. Stored steps: 0 to 127.

KNOB 2 - PITCH
Pitch or pitch modulation depth. In an envelope/LFO destination page this is an amount, not a tuning reference.
Range: 0 % to 100 %. Generic default: 50 %. Stored steps: 0 to 127.

KNOB 3 - CUT
Low-pass/filter cutoff: lower values remove more upper harmonics.
Range: 30 Hz to 16 kHz. Generic default: 6.9 kHz. Stored steps: 0 to 127.

KNOB 4 - RES
Filter resonance emphasizes frequencies around cutoff. High values may change level or approach self-oscillation.
Range: 0 % to 100 %. Generic default: 8 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 2 / JUNO

Shape the remaining engine controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 2 appears.

KNOB 1 - HPF
High-pass cutoff; reduce unwanted low-frequency energy.
Range: 0 to 3. Generic default: 1. Stored steps: 0 to 3.

Choices: 0 / 1 / 2 / 3.

KNOB 2 - CUT
Low-pass/filter cutoff: lower values remove more upper harmonics.
Range: 30 Hz to 16 kHz. Generic default: 2.5 kHz. Stored steps: 0 to 127.

KNOB 3 - RES
Filter resonance emphasizes frequencies around cutoff. High values may change level or approach self-oscillation.
Range: 0 % to 100 %. Generic default: 16 %. Stored steps: 0 to 127.

KNOB 4 - CHOR
Chorus amount/send; adds detuned movement and width.
Range: OFF to I+II. Generic default: I. Stored steps: 0 to 3.

Choices: OFF / I / II / I+II.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EDIT 2 / VECTOR

Shape the remaining engine controls.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until EDIT 2 appears.

KNOB 1 - X
Coordinate in the vector/terrain mix; moving between sources changes the timbre.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

KNOB 2 - Y
Coordinate in the vector/terrain mix; moving between sources changes the timbre.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

KNOB 3 - MOVE
Wave/vector movement amount or rate.
Range: OFF to RND+. Generic default: CIRC. Stored steps: 0 to 10.

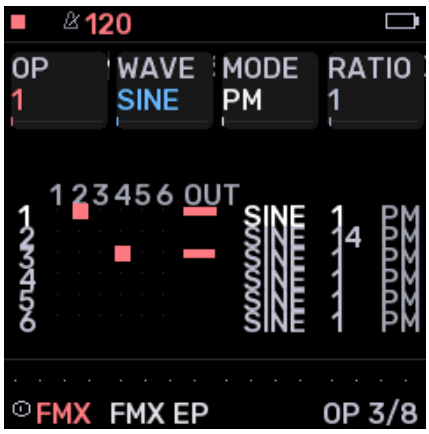
Choices: OFF / CIRC / CIRC+ / 8 / 8+ / SWEEP / SWP+ / LOOP / LOOP+ / RND / RND+.

KNOB 4 - CUT
Low-pass/filter cutoff: lower values remove more upper harmonics.
Range: 30 Hz to 16 kHz. Generic default: 4.2 kHz. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

OP / FMX

Select and tune an FMX operator.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until OP appears.

KNOB 1 - OP

Select one of the six FMX operators to edit.

Range: 1 to 6. Generic default: 1. Stored steps: 1 to 6.

KNOB 2 - WAVE

Select the oscillator or modulation waveform. Changing the shape changes its harmonic content or movement.

Range: SINE to NOISE. Generic default: SINE. Stored steps: 0 to 9.

Choices: SINE / TRI / SAW / SQR / HALF / ABS / QUART / PULSE / SIN8 / NOISE.

KNOB 3 - MODE

Choose the operating mode for this page: LFO restart, arp direction, delay style, controller protocol or operator frequency mode.

Range: PM to RING. Generic default: PM. Stored steps: 0 to 2.

Choices: PM / FM / RING.

KNOB 4 - RATIO

Operator frequency ratio relative to the played note.

Range: 0.5 to 31. Generic default: 1. Stored steps: 0 to 31.

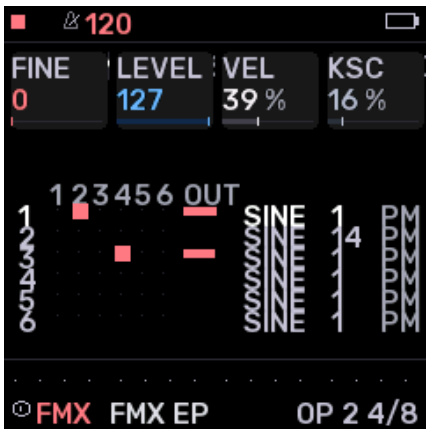
Choices: 0.5 / 1 / 2 / 3 / 4 / 5 / 6 / 7 / 8 / 9 / 10 / 11 / 12 / 13 / 14 / 15 / 16 / 17.

Choices (continued): 18 / 19 / 20 / 21 / 22 / 23 / 24 / 25 / 26 / 27 / 28 / 29 / 30 / 31.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

OP 2 / FMX

Set FMX operator level and performance response.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until OP 2 appears.

KNOB 1 - FINE
Fine operator tuning within the selected ratio.
Range: 0 to 99. Generic default: 0. Stored steps: 0 to 99.

KNOB 2 - LEVEL
Level of the selected part/operator/processor; use the page context to identify which one. A zero part level silences that part.
Range: 0 to 127. Generic default: 127. Stored steps: 0 to 127.

KNOB 3 - VEL
Velocity sensitivity of the operator/modulation target.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 4 - KSC
Keyboard scaling of the selected operator level.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

ROUTE / FMX

Connect FMX operators; FROM modulates TO, with independently controlled output.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until ROUTE appears.

KNOB 1 - TO
Destination FMX operator receiving modulation.
Range: 1 to 6. Generic default: 1. Stored steps: 1 to 6.

KNOB 2 - FROM
On TUNING, select the lowest affected octave C0-C8. Notes below it remain equal temperament. On FMX ROUTE, choose the source operator.
Range: 1 to 6. Generic default: 2. Stored steps: 1 to 6.

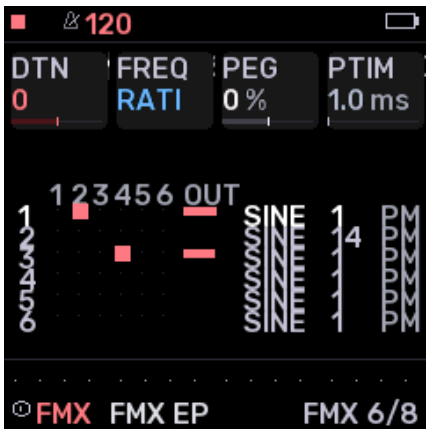
KNOB 3 - AMT
Depth of the selected modulation connection or processor. Bipolar ranges reverse the effect below zero.
Range: 0 to 127. Generic default: 0. Stored steps: 0 to 127.

KNOB 4 - OUT
On MIDI: output channel. On FMX ROUTE: selected operator output level.
Range: 0 to 127. Generic default: 0. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FMX / FMX

Set operator detune/frequency behavior and the FMX pitch envelope.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until FMX appears.

KNOB 1 - DTN
Detuning amount between oscillators/voices or FM operators, according to the selected engine.
Range: -7 to 7. Generic default: 0. Stored steps: -7 to 7.

KNOB 2 - FREQ
Fixed-versus-ratio frequency behavior for the selected FMX operator.
Range: RATIO to FIX. Generic default: RATIO. Stored steps: 0 to 1.

Choices: RATIO / FIX.

KNOB 3 - PEG
FMX pitch-envelope amount.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

KNOB 4 - PTIM
FMX pitch-envelope time.
Range: 1.0 ms to 10.0 s. Generic default: 18 ms. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

VOICE

Choose polyphony/mono articulation and glide.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until VOICE appears.

KNOB 1 - VCE
Part voice mode: POLY, MONO, LEG, UNI, UNI2 or UNI4. Unison consumes additional voices from the shared budget.
Range: POLY to UNI4. Generic default: POLY. Stored steps: 0 to 5.

Choices: POLY / MONO / LEG / UNI / UNI2 / UNI4.

KNOB 2 - GLD
Glide/portamento time between pitches.
Range: 1.0 ms to 10.0 s. Generic default: 1.0 ms. Stored steps: 0 to 127.

KNOB 3 - GLMOD
Choose which note transitions use glide.
Range: RATE to TIME. Generic default: RATE. Stored steps: 0 to 1.

Choices: RATE / TIME.

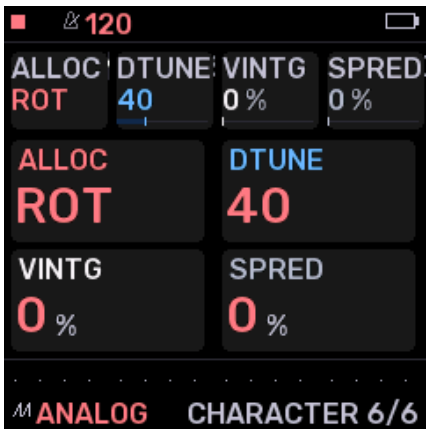
KNOB 4 - PRIO
Priority of held notes in a monophonic part.
Range: LAST to HIGH. Generic default: LAST. Stored steps: 0 to 2.

Choices: LAST / LOW / HIGH.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

CHARACTER

Choose allocation, detuning, vintage variation and stereo character.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap EDIT > tap EDIT again until CHARACTER appears.

KNOB 1 - ALLOC
ROT rotates available voice assignments; REUSE prefers reusing a matching/released voice. Unison is selected by VOIC on the VOICE page.
Range: ROT to REUSE. Generic default: ROT. Stored steps: 0 to 1.

Choices: ROT / REUSE.

KNOB 2 - DTUNE
Detuning amount between oscillators/voices or FM operators, according to the selected engine.
Range: 0 to 127. Generic default: 40. Stored steps: 0 to 127.

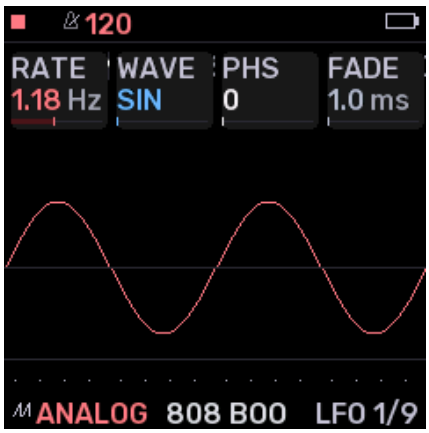
KNOB 3 - VINTG
Vintage variation/drift; adds small imperfections rather than changing the scale.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 4 - SPRED
Stereo spread or oscillator/processor width. Check the result in mono.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

LFO

Set the first low-frequency oscillator.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap LFO > tap LFO again until LFO appears.

KNOB 1 - RATE

Speed of the selected oscillator/modulator/effect. The displayed unit identifies Hz or a tempo division.
Range: .05 Hz to 40.0 Hz. Generic default: 1.18 Hz. Stored steps: 0 to 127.

KNOB 2 - WAVE

Select the oscillator or modulation waveform. Changing the shape changes its harmonic content or movement.
Range: SIN to S&H. Generic default: SIN. Stored steps: 0 to 4.

Choices: SIN / TRI / SAW / SQR / S&H.

KNOB 3 - PHS

Starting phase of LFO 1.
Range: 0 to 127. Generic default: 0. Stored steps: 0 to 127.

KNOB 4 - FADE

Time for LFO modulation to fade in after a note starts.
Range: 1.0 ms to 10.0 s. Generic default: 1.0 ms. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

LFO DEST

Set LFO 1 destination depths.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap LFO > tap LFO again until LFO DEST appears.

KNOB 1 - PIT
Pitch or pitch modulation depth. In an envelope/LFO destination page this is an amount, not a tuning reference.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

KNOB 2 - FLT
Filter-envelope amount; positive values open the filter with the envelope, negative values invert that movement where supported.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

KNOB 3 - SHP
Waveform/timbre shaping or the amount of envelope/LFO shaping, according to this page.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

KNOB 4 - AMP
Amplitude modulation depth. Start with a small amount to avoid large level swings.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

MOD PRESET

Apply an expressive modulation template.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap LFO > tap LFO again until MOD PRESET appears.

KNOB 1 - MOD
On MOD PRESET, choose a ready-made expressive routing; other pages use it for modulation amount.
Range: VEL B to ENV S. Generic default: VEL B. Stored steps: 0 to 3.

Choices: VEL BRIGHT / PRESS VIB / WHEEL VIB / ENV SWELL.

KNOB 2 - APPLY
Apply the selected modulation template. Existing required slots are checked before changing routings.
Range: 0 to 1. Generic default: 0. Stored steps: 0 to 1.

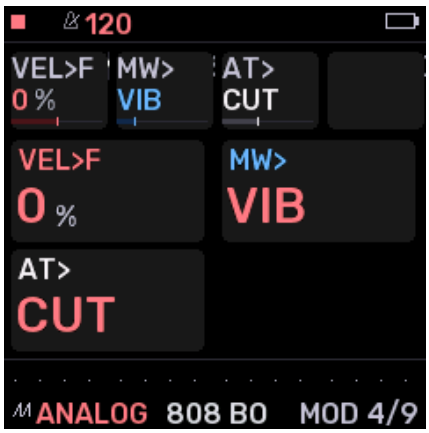
KNOB 3 - Unused
No assigned parameter on this page.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

MOD

Set velocity, wheel and pressure behavior.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap LFO > tap LFO again until MOD appears.

KNOB 1 - VEL>F
Velocity-to-filter sensitivity; stronger key strikes change brightness.
Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

KNOB 2 - MW>
Destination controlled by the MIDI modulation wheel.
Range: OFF to TREM. Generic default: VIB. Stored steps: 0 to 5.

Choices: OFF / VIB / CUT / SHP / WAH / TREM.

KNOB 3 - AT>
Destination controlled by MIDI channel pressure/aftertouch.
Range: OFF to TREM. Generic default: CUT. Stored steps: 0 to 5.

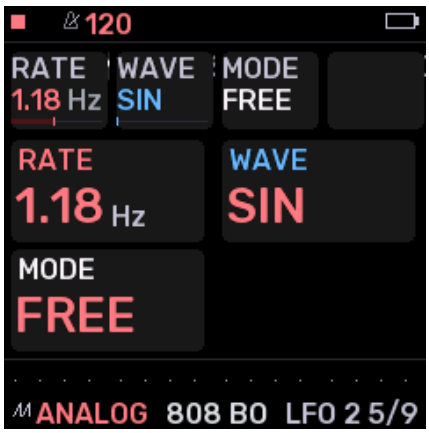
Choices: OFF / VIB / CUT / SHP / WAH / TREM.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

LFO 2

Set the second modulation oscillator.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap LFO > tap LFO again until LFO 2 appears.

KNOB 1 - RATE

Speed of the selected oscillator/modulator/effect. The displayed unit identifies Hz or a tempo division.
Range: .05 Hz to 40.0 Hz. Generic default: 1.18 Hz. Stored steps: 0 to 127.

KNOB 2 - WAVE

Select the oscillator or modulation waveform. Changing the shape changes its harmonic content or movement.
Range: SIN to S&H. Generic default: SIN. Stored steps: 0 to 4.

Choices: SIN / TRI / SAW / SQR / S&H.

KNOB 3 - MODE

Choose the operating mode for this page: LFO restart, arp direction, delay style, controller protocol or operator frequency mode.
Range: FREE to KEY. Generic default: FREE. Stored steps: 0 to 1.

Choices: FREE / KEY.

KNOB 4 - Unused

No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

LFO 3

Set the third modulation oscillator.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap LFO > tap LFO again until LFO 3 appears.

KNOB 1 - RATE
Speed of the selected oscillator/modulator/effect. The displayed unit identifies Hz or a tempo division.
Range: .05 Hz to 40.0 Hz. Generic default: .41 Hz. Stored steps: 0 to 127.

KNOB 2 - WAVE
Select the oscillator or modulation waveform. Changing the shape changes its harmonic content or movement.
Range: SIN to S&H. Generic default: SIN. Stored steps: 0 to 4.

Choices: SIN / TRI / SAW / SQR / S&H.

KNOB 3 - MODE
Choose the operating mode for this page: LFO restart, arp direction, delay style, controller protocol or operator frequency mode.
Range: FREE to KEY. Generic default: FREE. Stored steps: 0 to 1.

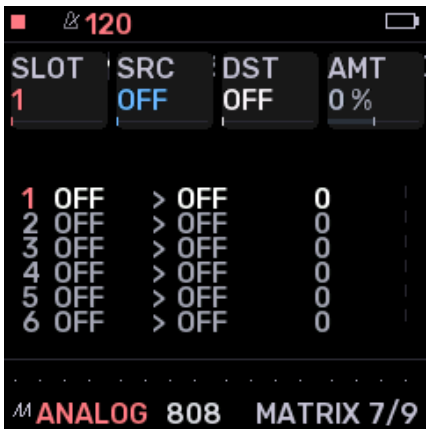
Choices: FREE / KEY.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

MATRIX

Edit one of twelve source/destination/depth connections.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap LFO > tap LFO again until MATRIX appears.

KNOB 1 - SLOT

Select the item being edited; on PROGRAM this is the saved program slot, on MATRIX it is one of the 12 modulation connections.

Range: 1 to 12. Generic default: 1. Stored steps: 1 to 12.

KNOB 2 - SRC

Modulation source for the selected matrix slot.

Range: OFF to MAC4. Generic default: OFF. Stored steps: 0 to 19.

Choices: OFF / LFO1 / LFO2 / ENV1 / ENV2 / VEL / KEY / MW / AT / RAND / SLIDE / SEQ1 / SEQ2 / SEQ3 / SEQ4 / LFO3 / MAC1 / MAC2.

Choices (continued): MAC3 / MAC4.

KNOB 3 - DST

Modulation destination for the selected matrix slot; engine targets follow the current engine.

Range: OFF to KTR. Generic default: OFF. Stored steps: 0 to 14.

Choices: OFF / PITCH / CUT / SHAPE / AMP / PAN / LFO2 / WAVE / DTN / MIX / NOIS / CUT / RES / DRV / KTR.

KNOB 4 - AMT

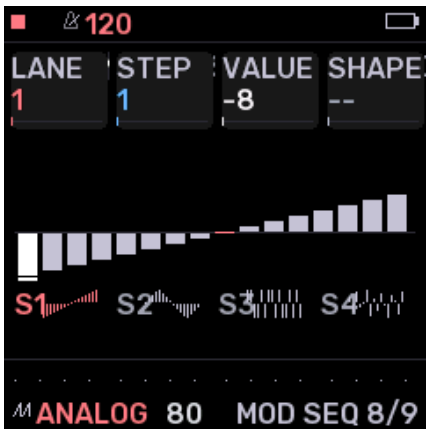
Depth of the selected modulation connection or processor. Bipolar ranges reverse the effect below zero.

Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

MOD SEQ

Draw a sixteen-step modulation lane.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap LFO > tap LFO again until MOD SEQ appears.

KNOB 1 - LANE
Select one of four modulation sequencer lanes.
Range: 1 to 4. Generic default: 1. Stored steps: 1 to 4.

KNOB 2 - STEP
Choose the modulation-sequencer step to edit.
Range: 1 to 16. Generic default: 1. Stored steps: 1 to 16.

KNOB 3 - VALUE
Set the selected step value, or the exact MIDI CC value required by a pedal assignment.
Range: -8 to 7. Generic default: 0. Stored steps: -8 to 7.

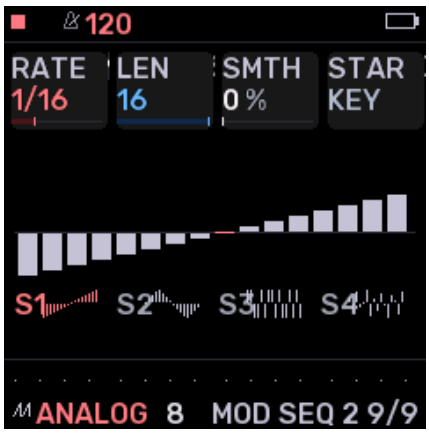
KNOB 4 - SHAPE
Waveform/timbre shaping or the amount of envelope/LFO shaping, according to this page.
Range: -- to SHIFT. Generic default: --. Stored steps: 0 to 9.

Choices: -- / RAMP+ / RAMP- / TRI / SINE / SQR / RND / FLAT / INVRT / SHIFT.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

MOD SEQ 2

Set modulation-lane timing, length, smoothing and restart.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap LFO > tap LFO again until MOD SEQ 2 appears.

KNOB 1 - RATE

Speed of the selected oscillator/modulator/effect. The displayed unit identifies Hz or a tempo division.
Range: 1/32 to 2 BAR. Generic default: 1/16. Stored steps: 0 to 8.

Choices: 1/32 / 16T / 1/16 / 8T / 1/8 / 1/4 / 1/2 / BAR / 2 BAR.

KNOB 2 - LEN

Length of the current sequence or setlist. Shorter lengths wrap sooner; a setlist length of zero disables the list.
Range: 1 to 16. Generic default: 16. Stored steps: 1 to 16.

KNOB 3 - SMTH

Smooth transitions between modulation-sequencer step values.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 4 - START

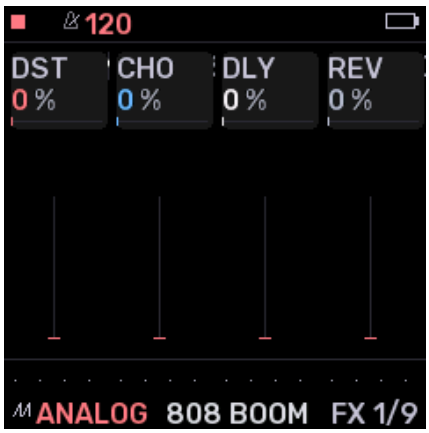
Position in a sample, wavetable, wave sequence or excitation, as indicated by the engine.
Range: SYNC to KEY. Generic default: KEY. Stored steps: 0 to 1.

Choices: SYNC / KEY.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

FX

Set the focused part sends. Shared return processors are configured on later FX pages.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap FX > tap FX again until FX appears.

KNOB 1 - DST
Modulation destination for the selected matrix slot; engine targets follow the current engine.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 2 - CHO
Chorus amount/send; adds detuned movement and width.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

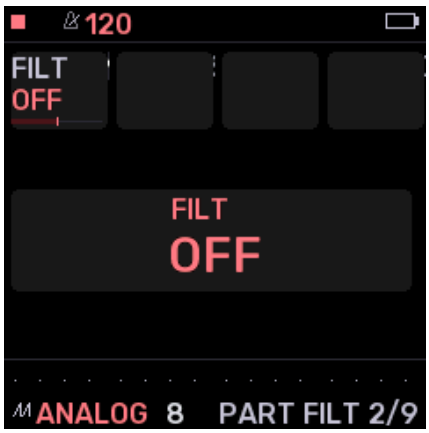
KNOB 3 - DLY
Delay send from this part.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 4 - REV
Reverb send from this part.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

PART FILTER

Apply a part-level DJ-style filter independently of the engine filter.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap FX > tap FX again until PART FILTER appears.

KNOB 1 - FILT
Part or master DJ-style filter: negative is low-pass, zero OFF, positive high-pass.
Range: LP100 % to HP100 %. Generic default: OFF. Stored steps: -64 to 63.

KNOB 2 - Unused
No assigned parameter on this page.

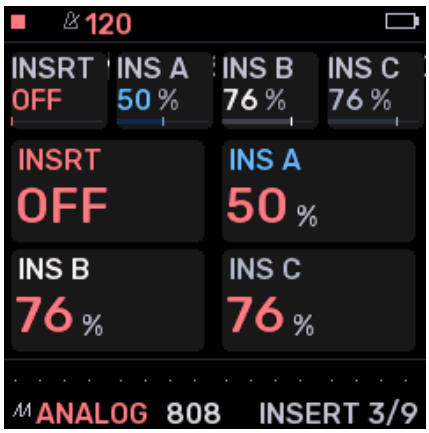
KNOB 3 - Unused
No assigned parameter on this page.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

INSERT / variant 1

Choose and shape the focused part insert processor.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap FX > tap FX again until INSERT appears.

Set INSERT TYPE to OFF.

KNOB 1 - INSRT
Select the per-part insert processor; its three following controls change with the type.
Range: OFF to JUN12. Generic default: OFF. Stored steps: 0 to 11.

Choices: OFF / SOFT / HARD / FOLD / FUZZ / CRUSH / PHASR / FLANG / CHOR / JUNO1 / JUNO2 / JUN12.

KNOB 2 - INS A
Inactive generic insert parameter when the insert is OFF. Select a processor to reveal its meaningful controls.
Range: 0 % to 100 %. Generic default: 50 %. Stored steps: 0 to 127.

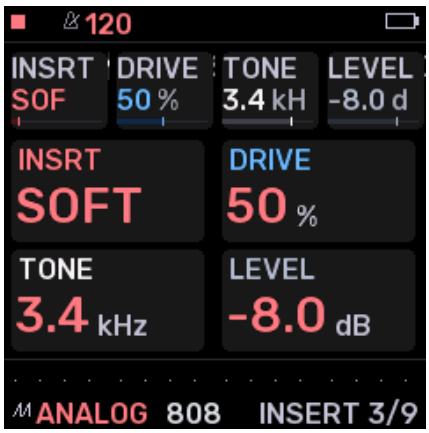
KNOB 3 - INS B
Inactive generic insert parameter when the insert is OFF. Select a processor to reveal its meaningful controls.
Range: 0 % to 100 %. Generic default: 76 %. Stored steps: 0 to 127.

KNOB 4 - INS C
Inactive generic insert parameter when the insert is OFF. Select a processor to reveal its meaningful controls.
Range: 0 % to 100 %. Generic default: 76 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

INSERT / variant 2

Choose and shape the focused part insert processor.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap FX > tap FX again until INSERT appears.

Set INSERT TYPE to SOFT.

KNOB 1 - INSRT

Select the per-part insert processor; its three following controls change with the type.

Range: OFF to JUN12. Generic default: OFF. Stored steps: 0 to 11.

Choices: OFF / SOFT / HARD / FOLD / FUZZ / CRUSH / PHASR / FLANG / CHOR / JUNO1 / JUNO2 / JUN12.

KNOB 2 - DRIVE

Drive into a saturation/filter stage. More drive can add harmonics and change level.

Range: 0 % to 100 %. Generic default: 50 %. Stored steps: 0 to 127.

KNOB 3 - TONE

Tone or spectral balance of this sound/processor.

Range: 30 Hz to 16 kHz. Generic default: 3.4 kHz. Stored steps: 0 to 127.

KNOB 4 - LEVEL

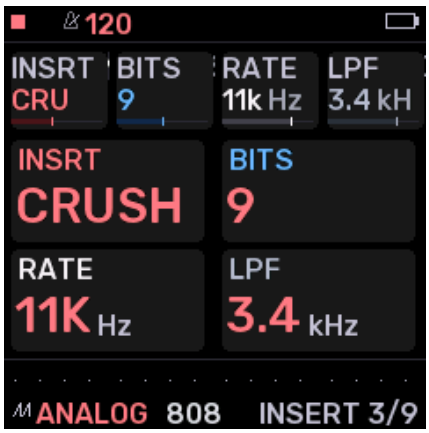
Level of the selected part/operator/processor; use the page context to identify which one. A zero part level silences that part.

Range: OFF to 7.5 dB. Generic default: -8.0 dB. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

INSERT / variant 6

Choose and shape the focused part insert processor.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap FX > tap FX again until INSERT appears.

Set INSERT TYPE to CRUSH.

KNOB 1 - INSRT
Select the per-part insert processor; its three following controls change with the type.
Range: OFF to JUN12. Generic default: OFF. Stored steps: 0 to 11.

Choices: OFF / SOFT / HARD / FOLD / FUZZ / CRUSH / PHASR / FLANG / CHOR / JUNO1 / JUNO2 / JUN12.

KNOB 2 - BITS
Bit depth of the crusher. Lower settings add quantization distortion.
Range: 1 to 16. Generic default: 9. Stored steps: 0 to 127.

KNOB 3 - RATE
Speed of the selected oscillator/modulator/effect. The displayed unit identifies Hz or a tempo division.
Range: 689 Hz to 44k Hz. Generic default: 11k Hz. Stored steps: 0 to 127.

KNOB 4 - LPF
Low-pass/filter cutoff: lower values remove more upper harmonics.
Range: 30 Hz to 16 kHz. Generic default: 3.4 kHz. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

INSERT / variant 7

Choose and shape the focused part insert processor.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap FX > tap FX again until INSERT appears.

Set INSERT TYPE to PHASR.

KNOB 1 - INSRT
Select the per-part insert processor; its three following controls change with the type.
Range: OFF to JUN12. Generic default: OFF. Stored steps: 0 to 11.

Choices: OFF / SOFT / HARD / FOLD / FUZZ / CRUSH / PHASR / FLANG / CHOR / JUNO1 / JUNO2 / JUN12.

KNOB 2 - RATE
Speed of the selected oscillator/modulator/effect. The displayed unit identifies Hz or a tempo division.
Range: .05 Hz to 40.0 Hz. Generic default: 1.45 Hz. Stored steps: 0 to 127.

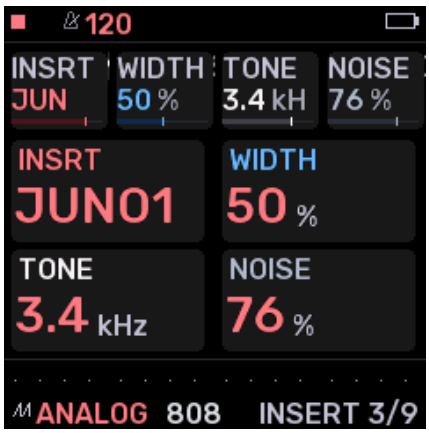
KNOB 3 - DEPTH
Modulation/chorus depth.
Range: 0 % to 100 %. Generic default: 76 %. Stored steps: 0 to 127.

KNOB 4 - FDBK
Feedback/recirculation. High values prolong the effect and can greatly increase level.
Range: 0 % to 100 %. Generic default: 76 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

INSERT / variant 10

Choose and shape the focused part insert processor.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap FX > tap FX again until INSERT appears.

Set INSERT TYPE to JUNO1.

KNOB 1 - INSRT
Select the per-part insert processor; its three following controls change with the type.
Range: OFF to JUN12. Generic default: OFF. Stored steps: 0 to 11.

Choices: OFF / SOFT / HARD / FOLD / FUZZ / CRUSH / PHASR / FLANG / CHOR / JUNO1 / JUNO2 / JUN12.

KNOB 2 - WIDTH
Stereo spread or oscillator/processor width. Check the result in mono.
Range: 0 % to 100 %. Generic default: 100 %. Stored steps: 0 to 127.

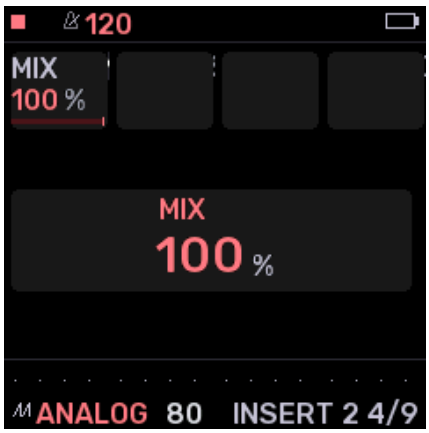
KNOB 3 - TONE
Tone or spectral balance of this sound/processor.
Range: 30 Hz to 16 kHz. Generic default: 4.2 kHz. Stored steps: 0 to 127.

KNOB 4 - NOISE
Noise oscillator/texture level.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

INSERT 2

Blend the insert with the dry part.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap FX > tap FX again until INSERT 2 appears.

KNOB 1 - MIX
Dry/wet balance or shared effect output level, as identified by the page.
Range: 0 % to 100 %. Generic default: 100 %. Stored steps: 0 to 127.

KNOB 2 - Unused
No assigned parameter on this page.

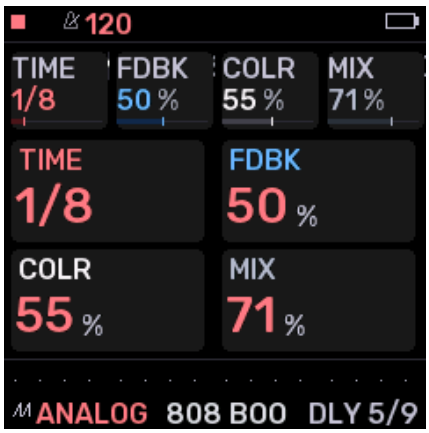
KNOB 3 - Unused
No assigned parameter on this page.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

DLY

Set the shared delay return.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap FX > tap FX again until DLY appears.

KNOB 1 - TIME
Delay time. Listen for pitch changes when changing tape-style delay time.
Range: 1/4 to 1/16D. Generic default: 1/8. Stored steps: 0 to 7.

Choices: 1/4 / 1/8 / 1/16 / 1/32 / 8T / 16T / 1/8D / 1/16D.

KNOB 2 - FDBK
Feedback/recirculation. High values prolong the effect and can greatly increase level.
Range: 0 % to 100 %. Generic default: 50 %. Stored steps: 0 to 120.

KNOB 3 - COLR
Tone or spectral balance of this sound/processor.
Range: 0 % to 100 %. Generic default: 55 %. Stored steps: 0 to 127.

KNOB 4 - MIX
Dry/wet balance or shared effect output level, as identified by the page.
Range: 0 % to 100 %. Generic default: 71 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

DLY 2

Select shared delay topology.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap FX > tap FX again until DLY 2 appears.

KNOB 1 - MODE
Choose the operating mode for this page: LFO restart, arp direction, delay style, controller protocol or operator frequency mode.
Range: MONO to TAPE. Generic default: MONO. Stored steps: 0 to 2.

Choices: MONO / PING / TAPE.

KNOB 2 - Unused
No assigned parameter on this page.

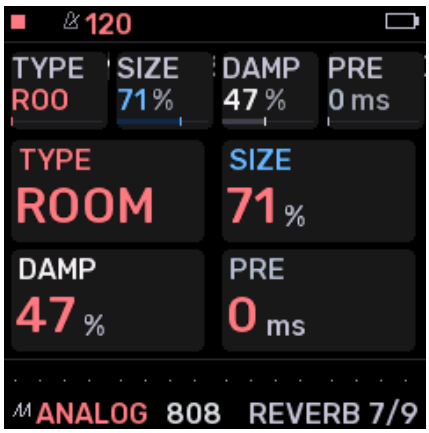
KNOB 3 - Unused
No assigned parameter on this page.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

REVERB

Set the shared reverb return.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap FX > tap FX again until REVERB appears.

KNOB 1 - TYPE
Select an effect or MIDI-message type.
Range: ROOM to SHIMR. Generic default: ROOM. Stored steps: 0 to 3.

Choices: ROOM / PLATE / HALL / SHIMR.

KNOB 2 - SIZE
Reverb space size/decay character.
Range: 0 % to 100 %. Generic default: 71 %. Stored steps: 0 to 127.

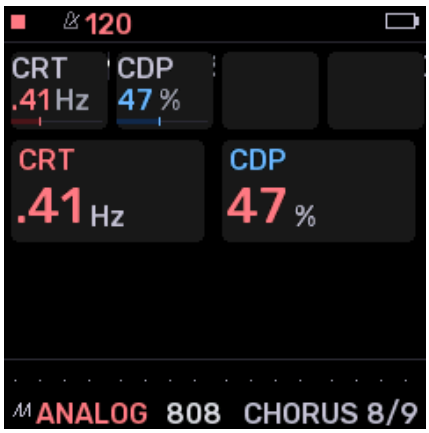
KNOB 3 - DAMP
High-frequency damping; darker reverberation or faster high-frequency decay as raised.
Range: 0 % to 100 %. Generic default: 47 %. Stored steps: 0 to 127.

KNOB 4 - PRE
Reverb pre-delay before the reverberant tail.
Range: 0 ms to 100 ms. Generic default: 0 ms. Stored steps: 0 to 100.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

CHORUS

Set shared chorus movement.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap FX > tap FX again until CHORUS appears.

KNOB 1 - CRT
Rate of the shared chorus modulation.
Range: .05 Hz to 40.0 Hz. Generic default: .41 Hz. Stored steps: 0 to 127.

KNOB 2 - CDP
Modulation/chorus depth.
Range: 0 % to 100 %. Generic default: 47 %. Stored steps: 0 to 127.

KNOB 3 - Unused
No assigned parameter on this page.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

EQ+COMP

Shape and compress the master bus.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap FX > tap FX again until EQ+COMP appears.

KNOB 1 - LOW
Master EQ gain for the indicated frequency band. Zero dB is neutral.
Range: -12 dB to +12 dB. Generic default: 0 dB. Stored steps: 88 to 112.

KNOB 2 - MID
Master EQ gain for the indicated frequency band. Zero dB is neutral.
Range: -10 dB to +10 dB. Generic default: 0 dB. Stored steps: 0 to 20.

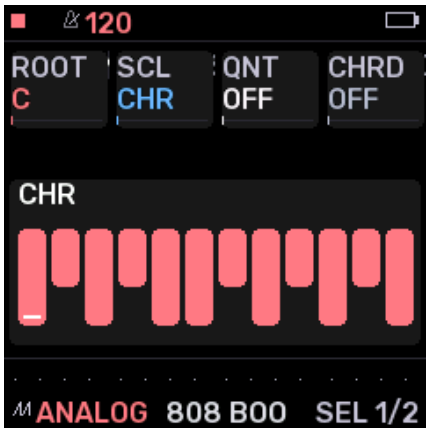
KNOB 3 - HIGH
Master EQ gain for the indicated frequency band. Zero dB is neutral.
Range: -12 dB to +12 dB. Generic default: 0 dB. Stored steps: 4 to 28.

KNOB 4 - COMP
Master bus compression amount; compare at similar listening volume.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

SEL

Choose note scale, root, quantization and chord behavior. The SEL tap opens a special scale browser over this page.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap SEL > tap SEL again until SEL appears.

KNOB 1 - ROOT

Root note of the selected scale. This transposes the scale relationship; it does not transpose every played note.
Range: C to B. Generic default: C. Stored steps: 0 to 11.

KNOB 2 - SCL

Keyboard scale used by note quantization and scale-aware chords.
Range: CHR to DIMWH. Generic default: CHR. Stored steps: 0 to 15.

Choices: CHR / MAJ / MIN / DOR / MIX / PEN / MPEN / HARM / PHRY / LYD / LOC / MEL / BLUES / WHOLE / DIMHW / DIMWH.

KNOB 3 - QNT

OFF leaves note selection chromatic; SNAP moves notes to the chosen scale; WHITE maps white-key playing into that scale. ALL modes apply the setting across all four parts and MIDI input.
Range: OFF to ALL. Generic default: OFF. Stored steps: 0 to 3.

Choices: OFF / SNAP / WHITE / ALL.

KNOB 4 - CHORD

Chord generator voicing. OFF plays individual notes. Available voicings are filtered for compatibility with the active microtonal scale.
Range: OFF to NEUTR. Generic default: OFF. Stored steps: 0 to 17.

Choices: OFF / TRIAD / 7TH / 9TH / SUS4 / POWER / MAJ / MIN / DOM7 / MAJ7 / MIN7 / SUS2 / DIM / AUG / AUTO / AUTO2 / OPEN 5 / NEUTRAL.

In the SEL-tap scale browser, SELECT chooses the scale and PRESET its root. The normal page controls shown above remain the underlying scale settings; a second SEL tap reaches SEL 2.

SEL 2

Set transposition, strum and voice leading.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap SEL > tap SEL again until SEL 2 appears.

KNOB 1 - TRN
Transpose this part by semitones; octave shifts are multiples of 12.
Range: -24 st to +24 st. Generic default: 0 st. Stored steps: -24 to 24.

KNOB 2 - STRUM
Spread chord notes in time. The upper values select tempo-synchronized direction/rate combinations.
Range: -60 to A32. Generic default: 0. Stored steps: -60 to 69.

KNOB 3 - VOIC
Chord voicing layout: CLOSE, SMOOTH, OPEN, first/second inversion or an octave-up layout.
Range: CLOSE to +OCT. Generic default: CLOSE. Stored steps: 0 to 5.

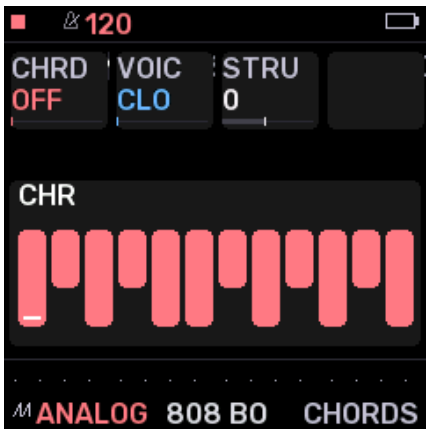
Choices: CLOSE / SMOOTH / OPEN / INV1 / INV2 / +OCT.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

CHORDS

Generate scale-aware chord voicings; opened by holding ENV for three seconds.



Native renderer example; current values may differ.

Open: HOME > select the part > hold ENV for 3 seconds > release.

KNOB 1 - CHORD

Chord generator voicing. OFF plays individual notes. Available voicings are filtered for compatibility with the active microtonal scale.

Range: OFF to NEUTR. Generic default: OFF. Stored steps: 0 to 17.

Choices: OFF / TRIAD / 7TH / 9TH / SUS4 / POWER / MAJ / MIN / DOM7 / MAJ7 / MIN7 / SUS2 / DIM / AUG / AUTO / AUTO2 / OPEN 5 / NEUTRAL.

KNOB 2 - VOIC

Chord voicing layout: CLOSE, SMOOTH, OPEN, first/second inversion or an octave-up layout.

Range: CLOSE to +OCT. Generic default: CLOSE. Stored steps: 0 to 5.

Choices: CLOSE / SMOOTH / OPEN / INV1 / INV2 / +OCT.

KNOB 3 - STRUM

Spread chord notes in time. The upper values select tempo-synchronized direction/rate combinations.

Range: -60 to A32. Generic default: 0. Stored steps: -60 to 69.

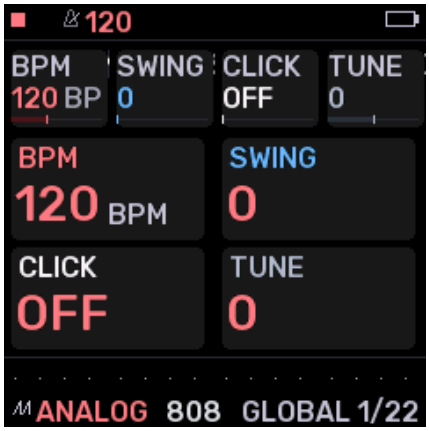
KNOB 4 - Unused

No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

GLOBAL

Set internal tempo, swing, metronome click and global pitch offset in cents.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap GLO > tap GLO again until GLOBAL appears.

KNOB 1 - BPM
Internal tempo in beats per minute.
Range: 40 BPM to 240 BPM. Generic default: 90 BPM. Stored steps: 40 to 240.

KNOB 2 - SWING
Off-beat timing offset. Zero is straight; higher settings delay alternating subdivisions.
Range: 0 to 100. Generic default: 0. Stored steps: 0 to 100.

KNOB 3 - CLICK
Metronome: OFF, REC only, or ON during playback.
Range: OFF to ON. Generic default: OFF. Stored steps: 0 to 2.

Choices: OFF / REC / ON.

KNOB 4 - TUNE
Global fine-tuning offset in cents (-50 to +50). This moves all notes in addition to the A4 reference and scale tuning.
Range: -50 to 50. Generic default: 0. Stored steps: -50 to 50.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

MASTER

Apply shared lo-fi texture and master DJ filter.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap GLO > tap GLO again until MASTER appears.

KNOB 1 - DUST
Master noise/lo-fi texture amount.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 2 - FILT
Part or master DJ-style filter: negative is low-pass, zero OFF, positive high-pass.
Range: LP100 % to HP100 %. Generic default: OFF. Stored steps: -64 to 63.

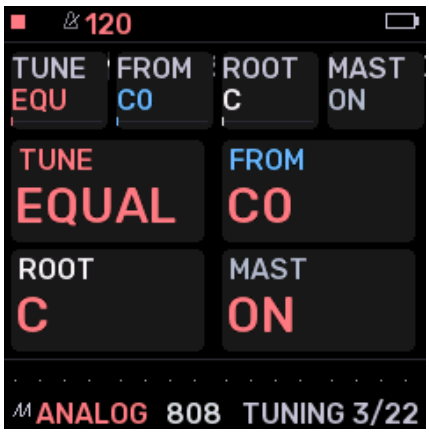
KNOB 3 - Unused
No assigned parameter on this page.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

TUNING

Choose the master scale, octave boundary, root and per-part tuning bypass.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap GLO > tap GLO again until TUNING appears.

KNOB 1 - TUNE

Select the master tuning system, including maqam and custom/Scala options.
Range: EQUAL to CUSTO. Generic default: EQUAL. Stored steps: 0 to 61.

Choices: EQUAL / JUST / PYTHA / MEANT / WERCK / KIRNB / RAST / BAYATI / SLEND / PELOG / 19EDO / 24EDO / 31EDO / USER / AJAM / AJAM USHAYRAN / ATHAR KURD / AWJ IRAQ.

Choices (continued): BASTANIKAR / BAYATI SHURI / DALANSHIN / FARAHFAZA / HIJAZ / HIJAZKAR / HIJAZKAR KURD / HUSAYNI / HUZAM / IRAQ / JIHARKAH / KIRDAN / KURD / LAMI / MAHUR / MUHAYYAR / MUSTAAR / NAHAWAND.

Choices (continued): NAHAWAND MURASSAA / NAIRUZ / NAWA ATHAR / NIKRIZ / RAHAT AL-ARWAH / SABA / SABA ZAMZAM / SAZKAR / SHADD ARABAN / SHAHNAZ / SHAWQ AFZA / SIKAH / SIKAH BALADI APPROX / SUZDALARA / SUZIDIL / SUZNAK / USHAQ MASRI / YAKAH.

Choices (continued): ZANJARAN / RAST DESCENDING / BAYATI UPPER RAST / HIJAZ UPPER RAST / CUSTOM 1 / CUSTOM 2 / CUSTOM 3 / CUSTOM 4.

KNOB 2 - FROM

On TUNING, select the lowest affected octave C0-C8. Notes below it remain equal temperament. On FMX ROUTE, choose the source operator.
Range: C0 to C8. Generic default: C0. Stored steps: 0 to 8.

Choices: C0 / C1 / C2 / C3 / C4 / C5 / C6 / C7 / C8.

KNOB 3 - ROOT

Root note of the selected scale. This transposes the scale relationship; it does not transpose every played note.
Range: C to B. Generic default: C. Stored steps: 0 to 11.

Choices: C / C# / D / D# / E / F / F# / G / G# / A / A# / B.

TUNING - continued

KNOB 4 - MASTER

Apply master tuning to the selected part; ON by default. OFF bypasses the shared maqam/microtuning for that part.

Range: OFF to ON. Generic default: ON. Stored steps: 0 to 1.

Choices: OFF / ON.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

TUNING 2

Set the A4 frequency reference.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap GLO > tap GLO again until TUNING 2 appears.

KNOB 1 - A4
Reference frequency of A4, in Hz. This is separate from the tuning-system octave boundary.
Range: 400 Hz to 480 Hz. Generic default: 440 Hz. Stored steps: 400 to 480.

KNOB 2 - Unused
No assigned parameter on this page.

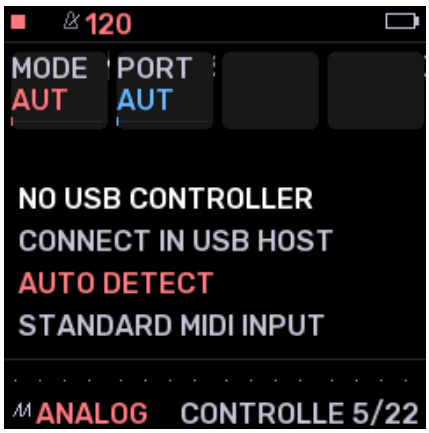
KNOB 3 - Unused
No assigned parameter on this page.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

CONTROLLER

Select automatic/MCU/HUI/controller handling and the MIDI cable/port.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap GLO > tap GLO again until CONTROLLER appears.

KNOB 1 - MODE
Choose the operating mode for this page: LFO restart, arp direction, delay style, controller protocol or operator frequency mode.
Range: AUTO to HUI. Generic default: AUTO. Stored steps: 0 to 3.

Choices: AUTO / MIDI / MCU / HUI.

KNOB 2 - PORT
MIDI transport/cable selection; AUTO uses automatic controller cable selection where offered.
Range: AUTO to 16. Generic default: AUTO. Stored steps: 0 to 16.

Choices: AUTO / 1 / 2 / 3 / 4 / 5 / 6 / 7 / 8 / 9 / 10 / 11 / 12 / 13 / 14 / 15 / 16.

KNOB 3 - Unused
No assigned parameter on this page.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

SYSTEM

Configure MIDI output, sync, routing and information display.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap GLO > tap GLO again until SYSTEM appears.

KNOB 1 - MIDI
Enable/route sequencer MIDI output according to the listed modes.
Range: KEYS to SEQ. Generic default: KEYS. Stored steps: 0 to 1.

Choices: KEYS / SEQ.

KNOB 2 - SYNC
Clock/synchronization source. External sync requires a running clock on the selected input.
Range: INT to TRS. Generic default: INT. Stored steps: 0 to 2.

Choices: INT / USB / TRS.

KNOB 3 - IN
MIDI input routing/filter setting.
Range: NOTES to CLOCK. Generic default: NOTES. Stored steps: 0 to 1.

Choices: NOTES / CLOCK.

KNOB 4 - CPU
CPU/engine implementation mode indicated by this model.
Range: 0 to 0. Generic default: 0. Stored steps: 0 to 0.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

MIDI

Configure channels, pitch bend and onboard local playing.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap GLO > tap GLO again until MIDI appears.

KNOB 1 - IN CH

Accepted MIDI input channel; OMNI accepts all channels, subject to the controller protocol/MPE settings.
Range: OMNI to 16. Generic default: OMNI. Stored steps: 0 to 16.

Choices: OMNI / 1 / 2 / 3 / 4 / 5 / 6 / 7 / 8 / 9 / 10 / 11 / 12 / 13 / 14 / 15 / 16.

KNOB 2 - OUT

On MIDI: output channel. On FMX ROUTE: selected operator output level.
Range: 1 to 16. Generic default: 1. Stored steps: 1 to 16.

KNOB 3 - BEND

Pitch-bend range in semitones; AUTO follows supported configuration messages.
Range: AUTO to 24. Generic default: AUTO. Stored steps: 0 to 24.

Choices: AUTO / 1 / 2 / 3 / 4 / 5 / 6 / 7 / 8 / 9 / 10 / 11 / 12 / 13 / 14 / 15 / 16 / 17.

Choices (continued): 18 / 19 / 20 / 21 / 22 / 23 / 24.

KNOB 4 - LOCAL

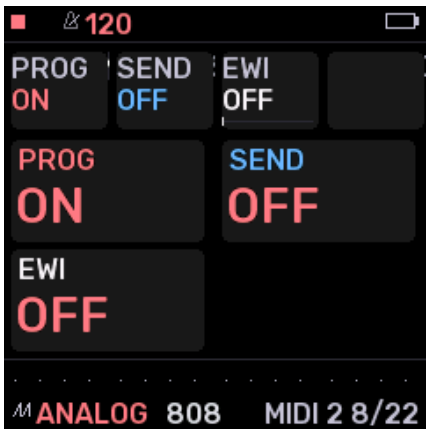
Whether the onboard keyboard drives the internal synth. MIDI input remains a separate path.
Range: OFF to ON. Generic default: ON. Stored steps: 0 to 1.

Choices: OFF / ON.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

MIDI 2

Configure program changes, CC output and breath control.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap GLO > tap GLO again until MIDI 2 appears.

KNOB 1 - PROG
Enable incoming program-change handling.
Range: OFF to ON. Generic default: ON. Stored steps: 0 to 1.

Choices: OFF / ON.

KNOB 2 - SEND
Enable supported parameter CC output.
Range: OFF to CC. Generic default: OFF. Stored steps: 0 to 1.

Choices: OFF / CC.

KNOB 3 - EWI
Breath-controller handling: OFF, volume, or volume plus pressure behavior.
Range: OFF to V+AT. Generic default: OFF. Stored steps: 0 to 2.

Choices: OFF / VOL / V+AT.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

PATCH MIDI

Enable incoming patch switching on its supported input.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap GLO > tap GLO again until PATCH MIDI appears.

KNOB 1 - PROG
Enable incoming program-change handling.
Range: OFF to ON. Generic default: ON. Stored steps: 0 to 1.

Choices: OFF / ON.

KNOB 2 - PORT
MIDI transport/cable selection; AUTO uses automatic controller cable selection where offered.
Range: USB to USB. Generic default: USB. Stored steps: 0 to 0.

Choices: USB.

KNOB 3 - Unused
No assigned parameter on this page.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

KEY MIDI

Choose keyboard input source and channel.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap GLO > tap GLO again until KEY MIDI appears.

KNOB 1 - KEY IN
Choose USB, TRS or both for keyboard MIDI input.
Range: USB to BOTH. Generic default: BOTH. Stored steps: 0 to 2.

Choices: USB / TRS / BOTH.

KNOB 2 - IN CH
Accepted MIDI input channel; OMNI accepts all channels, subject to the controller protocol/MPE settings.
Range: OMNI to 16. Generic default: OMNI. Stored steps: 0 to 16.

Choices: OMNI / 1 / 2 / 3 / 4 / 5 / 6 / 7 / 8 / 9 / 10 / 11 / 12 / 13 / 14 / 15 / 16.

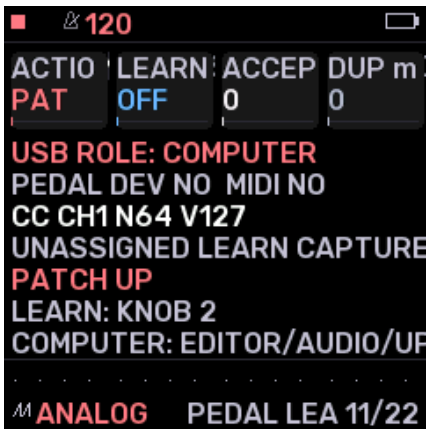
KNOB 3 - Unused
No assigned parameter on this page.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

PEDAL LEARN

Capture and confirm a pedal message for the selected relative action.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap GLO > tap GLO again until PEDAL LEARN appears.

KNOB 1 - ACTION

Select the pedal action to learn, edit or test.
Range: PATCH to SETLI. Generic default: PATCH. Stored steps: 0 to 9.

Choices: PATCH UP / PATCH DOWN / BANK UP / BANK DOWN / PROGRAM NEXT / PROGRAM PREV / SCENE NEXT / SCENE PREV / SETLIST NEXT / SETLIST PREV.

KNOB 2 - LEARN

Arm MIDI learn. Press the pedal once, inspect the captured message, then accept it.
Range: OFF to ON. Generic default: OFF. Stored steps: 0 to 1.

KNOB 3 - ACCEPT

Commit the captured MIDI message to the selected pedal action.
Range: 0 to 1. Generic default: 0. Stored steps: 0 to 1.

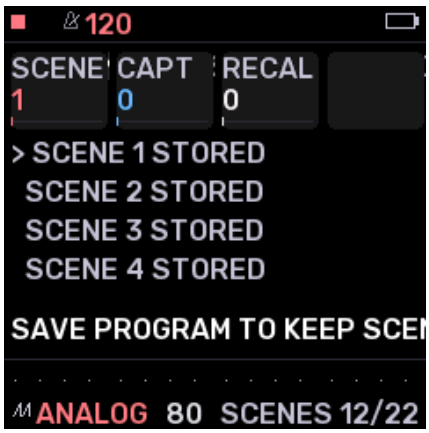
KNOB 4 - DUP ms

Duplicate-message suppression window in milliseconds. Zero disables suppression; use the smallest useful window so intentional taps remain distinct.
Range: 0 to 500. Generic default: 0. Stored steps: 0 to 500.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

SCENES

Capture and recall small changes inside the current program.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap GLO > tap GLO again until SCENES appears.

KNOB 1 - SCENE
Choose one of four compact scenes within the working program.
Range: 1 to 4. Generic default: 1. Stored steps: 1 to 4.

KNOB 2 - CAPTURE
Store the current supported mix, mutes, sends, macros and tuning settings into the selected scene.
Range: 0 to 1. Generic default: 0. Stored steps: 0 to 1.

KNOB 3 - RECALL
Apply the selected captured scene without loading another program.
Range: 0 to 1. Generic default: 0. Stored steps: 0 to 1.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

SETLIST

Build an ordered list of saved program references.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap GLO > tap GLO again until SETLIST appears.

KNOB 1 - ENTRY
Choose the position to edit in the ordered setlist.
Range: 1 to 32. Generic default: 1. Stored steps: 1 to 32.

KNOB 2 - PROGRAM
Assign a saved program number to this setlist position.
Range: 1 to 8. Generic default: 1. Stored steps: 1 to 8.

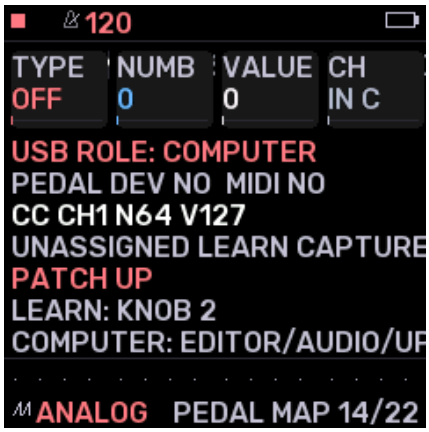
KNOB 3 - LENGTH
Length of the current sequence or setlist. Shorter lengths wrap sooner; a setlist length of zero disables the list.
Range: 0 to 32. Generic default: 0. Stored steps: 0 to 32.

KNOB 4 - LOAD
Load the selected saved program/setlist entry. Storage actions require their on-screen confirmation.
Range: 0 to 1. Generic default: 0. Stored steps: 0 to 1.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

PEDAL MAP

Edit a relative action assignment manually; use the action selector/status display to identify the assignment.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap GLO > tap GLO again until PEDAL MAP appears.

KNOB 1 - TYPE
Select an effect or MIDI-message type.
Range: OFF to PC. Generic default: OFF. Stored steps: 0 to 2.

Choices: OFF / CC / PC.

KNOB 2 - NUMBER
MIDI CC number or program-change number for this assignment.
Range: 0 to 127. Generic default: 0. Stored steps: 0 to 127.

KNOB 3 - VALUE
Set the selected step value, or the exact MIDI CC value required by a pedal assignment.
Range: 0 to 127. Generic default: 127. Stored steps: 0 to 127.

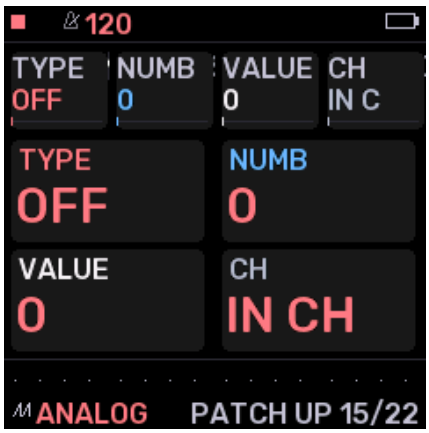
KNOB 4 - CH
Assignment channel. IN CH follows the configured input channel; numbered choices match that channel specifically.
Range: IN CH to 16. Generic default: IN CH. Stored steps: 0 to 16.

Choices: IN CH / 1 / 2 / 3 / 4 / 5 / 6 / 7 / 8 / 9 / 10 / 11 / 12 / 13 / 14 / 15 / 16.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

PATCH UP

Manually set the Patch Up MIDI assignment.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap GLO > tap GLO again until PATCH UP appears.

KNOB 1 - TYPE
Select an effect or MIDI-message type.
Range: OFF to PC. Generic default: OFF. Stored steps: 0 to 2.

Choices: OFF / CC / PC.

KNOB 2 - NUMBER
MIDI CC number or program-change number for this assignment.
Range: 0 to 127. Generic default: 0. Stored steps: 0 to 127.

KNOB 3 - VALUE
Set the selected step value, or the exact MIDI CC value required by a pedal assignment.
Range: 0 to 127. Generic default: 127. Stored steps: 0 to 127.

KNOB 4 - CH
Assignment channel. IN CH follows the configured input channel; numbered choices match that channel specifically.
Range: IN CH to 16. Generic default: IN CH. Stored steps: 0 to 16.

Choices: IN CH / 1 / 2 / 3 / 4 / 5 / 6 / 7 / 8 / 9 / 10 / 11 / 12 / 13 / 14 / 15 / 16.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

PATCH DOWN

Manually set the Patch Down MIDI assignment.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap GLO > tap GLO again until PATCH DOWN appears.

KNOB 1 - TYPE
Select an effect or MIDI-message type.
Range: OFF to PC. Generic default: OFF. Stored steps: 0 to 2.

Choices: OFF / CC / PC.

KNOB 2 - NUMBER
MIDI CC number or program-change number for this assignment.
Range: 0 to 127. Generic default: 0. Stored steps: 0 to 127.

KNOB 3 - VALUE
Set the selected step value, or the exact MIDI CC value required by a pedal assignment.
Range: 0 to 127. Generic default: 127. Stored steps: 0 to 127.

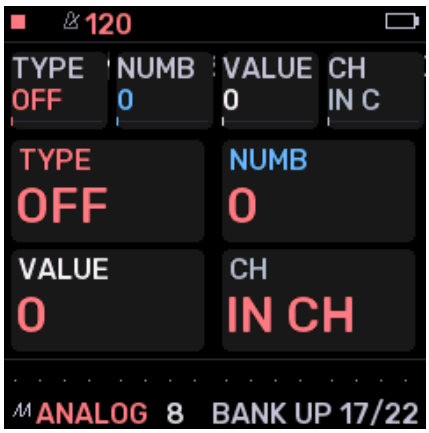
KNOB 4 - CH
Assignment channel. IN CH follows the configured input channel; numbered choices match that channel specifically.
Range: IN CH to 16. Generic default: IN CH. Stored steps: 0 to 16.

Choices: IN CH / 1 / 2 / 3 / 4 / 5 / 6 / 7 / 8 / 9 / 10 / 11 / 12 / 13 / 14 / 15 / 16.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

BANK UP

Manually set the Bank Up MIDI assignment.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap GLO > tap GLO again until BANK UP appears.

KNOB 1 - TYPE
Select an effect or MIDI-message type.
Range: OFF to PC. Generic default: OFF. Stored steps: 0 to 2.

Choices: OFF / CC / PC.

KNOB 2 - NUMBER
MIDI CC number or program-change number for this assignment.
Range: 0 to 127. Generic default: 0. Stored steps: 0 to 127.

KNOB 3 - VALUE
Set the selected step value, or the exact MIDI CC value required by a pedal assignment.
Range: 0 to 127. Generic default: 127. Stored steps: 0 to 127.

KNOB 4 - CH
Assignment channel. IN CH follows the configured input channel; numbered choices match that channel specifically.
Range: IN CH to 16. Generic default: IN CH. Stored steps: 0 to 16.

Choices: IN CH / 1 / 2 / 3 / 4 / 5 / 6 / 7 / 8 / 9 / 10 / 11 / 12 / 13 / 14 / 15 / 16.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

BANK DOWN

Manually set the Bank Down MIDI assignment.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap GLO > tap GLO again until BANK DOWN appears.

KNOB 1 - TYPE
Select an effect or MIDI-message type.
Range: OFF to PC. Generic default: OFF. Stored steps: 0 to 2.

Choices: OFF / CC / PC.

KNOB 2 - NUMBER
MIDI CC number or program-change number for this assignment.
Range: 0 to 127. Generic default: 0. Stored steps: 0 to 127.

KNOB 3 - VALUE
Set the selected step value, or the exact MIDI CC value required by a pedal assignment.
Range: 0 to 127. Generic default: 127. Stored steps: 0 to 127.

KNOB 4 - CH
Assignment channel. IN CH follows the configured input channel; numbered choices match that channel specifically.
Range: IN CH to 16. Generic default: IN CH. Stored steps: 0 to 16.

Choices: IN CH / 1 / 2 / 3 / 4 / 5 / 6 / 7 / 8 / 9 / 10 / 11 / 12 / 13 / 14 / 15 / 16.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

USB ROLE

Choose computer-device or pedal/controller-host operation.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap GLO > tap GLO again until USB ROLE appears.

KNOB 1 - USB ROLE
COMPUTER enables device/editor operation; PEDAL enables USB host operation. Stop notes/playback before changing roles and use powered host hardware.
Range: COMPU to PEDAL. Generic default: COMPU. Stored steps: 0 to 1.

Choices: COMPUTER / PEDAL.

KNOB 2 - Unused
No assigned parameter on this page.

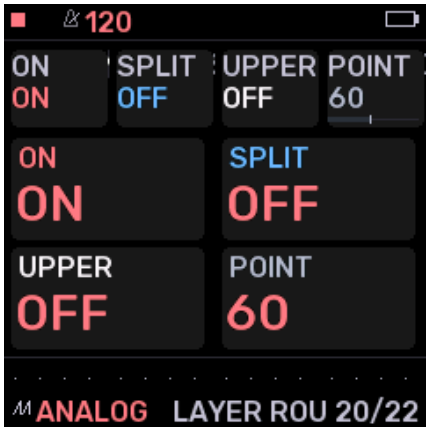
KNOB 3 - Unused
No assigned parameter on this page.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

LAYER ROUTE

Enable the part and choose its keyboard range.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap GLO > tap GLO again until LAYER ROUTE appears.

KNOB 1 - ON
Enable this part in layered playback.
Range: OFF to ON. Generic default: ON. Stored steps: 0 to 1.

Choices: OFF / ON.

KNOB 2 - SPLIT
Enable keyboard-range filtering for this part.
Range: OFF to ON. Generic default: OFF. Stored steps: 0 to 1.

Choices: OFF / ON.

KNOB 3 - UPPER
OFF uses the lower side, including the boundary. ON uses notes above the boundary.
Range: OFF to ON. Generic default: OFF. Stored steps: 0 to 1.

Choices: OFF / ON.

KNOB 4 - POINT
Split boundary as a MIDI note number. The held EDIT mixer presents it as a note/octave name.
Range: 0 to 127. Generic default: 60. Stored steps: 0 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

LAYER MIX

Set the part volume, pan, mute and solo.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap GLO > tap GLO again until LAYER MIX appears.

KNOB 1 - LEVEL

Level of the selected part/operator/processor; use the page context to identify which one. A zero part level silences that part.

Range: OFF to 7.5 dB. Generic default: -4.0 dB. Stored steps: 0 to 127.

KNOB 2 - PAN

Position of the selected part in the stereo field; center is zero.

Range: -100 % to +98 %. Generic default: 0 %. Stored steps: -64 to 63.

KNOB 3 - MUTE

Silence the selected part without deleting its sound.

Range: OFF to ON. Generic default: OFF. Stored steps: 0 to 1.

Choices: OFF / ON.

KNOB 4 - SOLO

Hear the soloed part(s), subject to layer routing/enabled state.

Range: OFF to ON. Generic default: OFF. Stored steps: 0 to 1.

Choices: OFF / ON.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

LAYER PITCH

Set part transpose and enable/disable four-part layering.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap GLO > tap GLO again until LAYER PITCH appears.

KNOB 1 - TRANS
Transpose this part by semitones; octave shifts are multiples of 12.
Range: -24 st to +24 st. Generic default: 0 st. Stored steps: -24 to 24.

KNOB 2 - LAYER
Toggle four-part performance. OFF plays the focused part; ON uses enabled parts and their routing.
Range: OFF to ON. Generic default: OFF. Stored steps: 0 to 1.

Choices: OFF / ON.

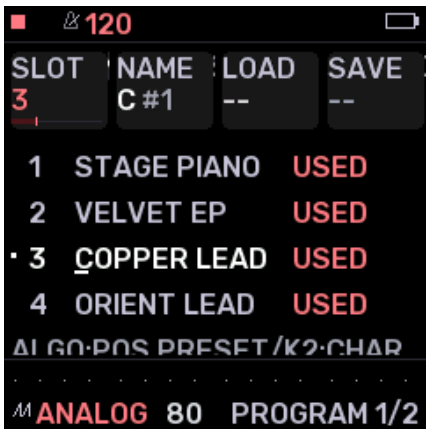
KNOB 3 - Unused
No assigned parameter on this page.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

PROGRAM

Name, load and save one of eight complete four-part programs.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap SAVE > tap SAVE again until PROGRAM appears.

KNOB 1 - SLOT
Select the item being edited; on PROGRAM this is the saved program slot, on MATRIX it is one of the 12 modulation connections.
Range: 1 to 8. Generic default: 1. Stored steps: 1 to 8.

KNOB 2 - NAME
Edit the selected saved-program name. ALGORITHM chooses a character position; PRESET/KNOB 2 changes its character.
Range: -- to --. Generic default: --. Stored steps: 0 to 0.

Choices: --.

KNOB 3 - LOAD
Load the selected saved program/setlist entry. Storage actions require their on-screen confirmation.
Range: -- to GO. Generic default: --. Stored steps: 0 to 1.

Choices: -- / GO.

KNOB 4 - SAVE
Write the working four-part program to the selected saved slot after confirmation.
Range: -- to GO. Generic default: --. Stored steps: 0 to 1.

Choices: -- / GO.

SELECT / KNOB 1: slot 1-8. ALGORITHM: character position 1-12. PRESET / KNOB 2: character. KNOB 3: LOAD, twice to confirm. KNOB 4: SAVE, twice to confirm. Browsing a slot never loads or overwrites it.

TOOLS

Clear or initialize working data after confirmation.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap SAVE > tap SAVE again until TOOLS appears.

KNOB 1 - CLRSQ
Clear sequence data using the requested confirmation; this is destructive to the working pattern.
Range: -- to GO. Generic default: --. Stored steps: 0 to 1.

Choices: -- / GO.

KNOB 2 - INIT
Initialize the selected working sound. Save a copy before using it.
Range: -- to GO. Generic default: --. Stored steps: 0 to 1.

Choices: -- / GO.

KNOB 3 - Unused
No assigned parameter on this page.

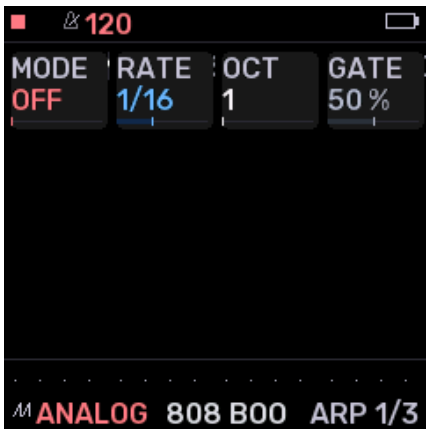
KNOB 4 - NEW
Initialize the working project/program as requested by the confirmation. Back up wanted edits first.
Range: -- to GO. Generic default: --. Stored steps: 0 to 1.

Choices: -- / GO.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

ARP

Generate ordered note patterns from held keys.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap ARP > tap ARP again until ARP appears.

KNOB 1 - MODE
Choose the operating mode for this page: LFO restart, arp direction, delay style, controller protocol or operator frequency mode.
Range: OFF to TURNG. Generic default: OFF. Stored steps: 0 to 15.

Choices: OFF / UP / DN / UPDN / RND / ORD / REPEAT / DNUP / UP+8 / CONV / DIVG / PINKY / THUMB / WALK / CHORD / TURNG.

KNOB 2 - RATE
Speed of the selected oscillator/modulator/effect. The displayed unit identifies Hz or a tempo division.
Range: 1/4 to 16T. Generic default: 1/16. Stored steps: 0 to 5.

Choices: 1/4 / 1/8 / 1/16 / 1/32 / 8T / 16T.

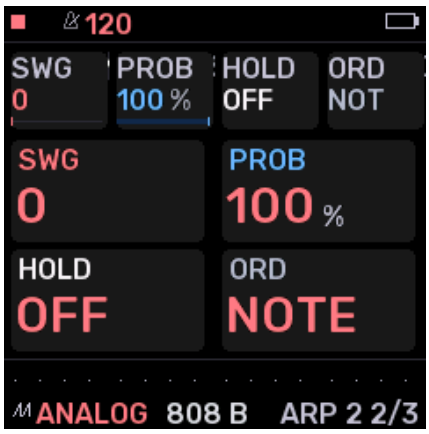
KNOB 3 - OCT
Arpeggiator octave range or part octave shift, depending on the page.
Range: 1 to 4. Generic default: 1. Stored steps: 1 to 4.

KNOB 4 - GATE
Duration of each generated arp/sequence note as a fraction of its step.
Range: 1 % to 100 %. Generic default: 50 %. Stored steps: 1 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

ARP 2

Set variation, hold and note ordering.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap ARP > tap ARP again until ARP 2 appears.

KNOB 1 - SWG
Off-beat timing offset. Zero is straight; higher settings delay alternating subdivisions.
Range: 0 to 100. Generic default: 0. Stored steps: 0 to 100.

KNOB 2 - PROB
Probability of playing a generated note. Lower settings deliberately omit some notes.
Range: 0 % to 100 %. Generic default: 100 %. Stored steps: 0 to 127.

KNOB 3 - HOLD
Keep the arpeggiated/held notes after key-up. Turn off to release the latched note set.
Range: OFF to ON. Generic default: OFF. Stored steps: 0 to 1.

Choices: OFF / ON.

KNOB 4 - ORD
Order in which held notes enter the arpeggiator.
Range: NOTE to PLAY. Generic default: NOTE. Stored steps: 0 to 1.

Choices: NOTE / PLAY.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

SPICE

Add controlled variation to generated notes. TURNG changes SPICE/DICE to MUTAT/SEED.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap ARP > tap ARP again until SPICE appears.

KNOB 1 - SPICE
Amount of controlled variation applied to generated notes.
Range: 0 % to 100 %. Generic default: 0 %. Stored steps: 0 to 127.

KNOB 2 - DICE
Variation seed/reroll control. TURNG mode changes its interpretation to a repeating random sequence seed.
Range: 0 to 127. Generic default: 0. Stored steps: 0 to 127.

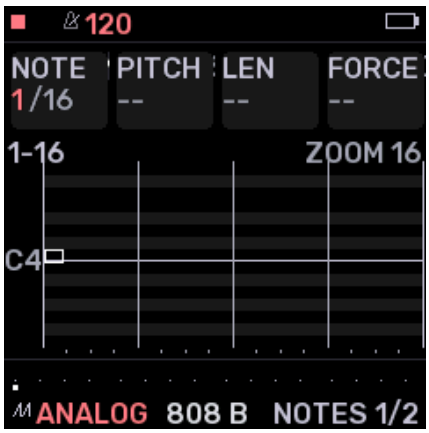
KNOB 3 - LOOP
Repeat/loop behavior. On ENV 2, ONE is a single contour and LOOP repeats it; on SPICE/TURNG it repeats generated variation.
Range: 1 to 16. Generic default: 16. Stored steps: 1 to 16.

KNOB 4 - Unused
No assigned parameter on this page.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

NOTES

Edit notes in the focused part pattern.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap SEQ > tap SEQ again until NOTES appears.

KNOB 1 - NOTE
Move between notes/chord members and empty steps.

KNOB 2 - PITCH
Edit the selected MIDI pitch (1-127). On an empty step, the first turn inserts the last-used note.

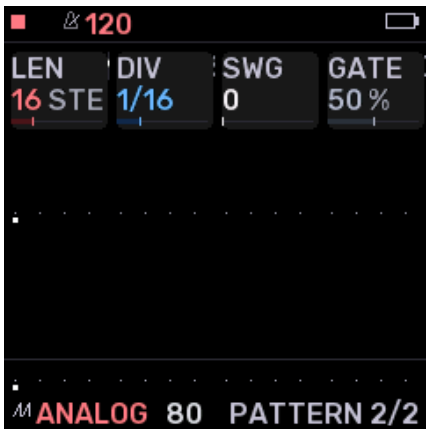
KNOB 3 - LEN
Extend/shorten the selected note in steps, stopping at an occupied note or pattern end.

KNOB 4 - FORCE
Choose GHOST, SOFT, NORM, HARD or ACC articulation.

PRESET changes the visible zoom (8-64 steps). Hold OCT+ for about 0.4 seconds to open the six-row note sheet. Step Advance determines whether releasing played keys advances the cursor.

PATTERN

Set pattern timing and length.



Native renderer example; current values may differ.

Open: HOME > SELECT the part > tap SEQ > tap SEQ again until PATTERN appears.

KNOB 1 - LEN
Length of the current sequence or setlist. Shorter lengths wrap sooner; a setlist length of zero disables the list.
Range: 1 STEP to 64 STEP. Generic default: 16 STEP. Stored steps: 1 to 64.

KNOB 2 - DIV
Pattern step division relative to tempo.
Range: 1/4 to 2BAR. Generic default: 1/16. Stored steps: 0 to 8.

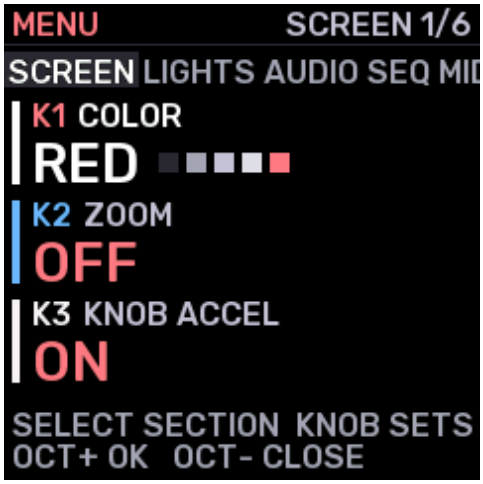
Choices: 1/4 / 1/8 / 1/16 / 1/32 / 8T / 16T / 1/2 / 1BAR / 2BAR.

KNOB 3 - SWG
Off-beat timing offset. Zero is straight; higher settings delay alternating subdivisions.
Range: 0 to 100. Generic default: 0. Stored steps: 0 to 100.

KNOB 4 - GATE
Duration of each generated arp/sequence note as a fraction of its step.
Range: 1 % to 100 %. Generic default: 50 %. Stored steps: 1 to 127.

Use the four knobs in displayed order. Tap the family button for its next page. Return with HOME; to change the focused part, use SELECT on HOME, then reopen this family. Save the program after wanted sound/performance edits.

System - SCREEN



Hold HOME for 3 seconds, then release.

01 Turn SELECT to choose the SCREEN section. PRESET cycles the row cursor.

02 Turn KNOBS 1-3 to change the corresponding rows. OCT+ steps/opens the highlighted setting; OCT- closes the menu.

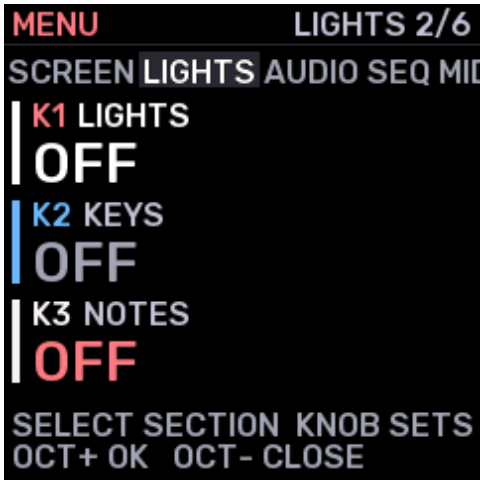
03 Settings are saved when the menu closes if stopped; a save may wait until playback stops. Restart when the screen requests it.

COLOR - RED / WHITE / BLUE / COOL / GREY device palette; use a high-contrast option for stage visibility.

ZOOM - OFF/ON: enlarge the active parameter display while editing.

KNOB ACCEL - OFF gives fine detents; ON makes fast turns change values faster.

System - LIGHTS



Hold HOME for 3 seconds, then release.

01 Turn SELECT to choose the LIGHTS section. PRESET cycles the row cursor.

02 Turn KNOBS 1-3 to change the corresponding rows. OCT+ steps/opens the highlighted setting; OCT- closes the menu.

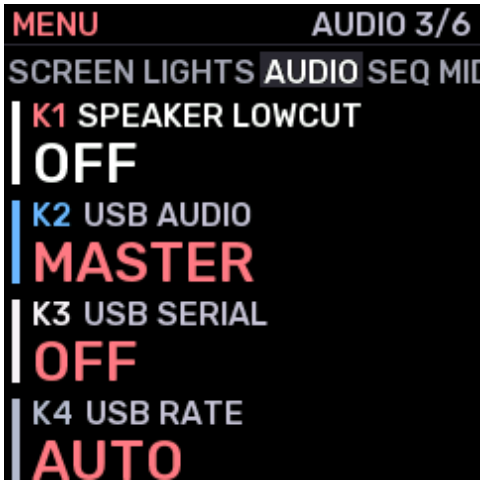
03 Settings are saved when the menu closes if stopped; a save may wait until playback stops. Restart when the screen requests it.

LIGHTS - OFF/LOW/MID/HIGH label backlight brightness.

KEYS - OFF/C KEYS/WHITE KEYS/ALL KEYS landmarks. Requires LIGHTS above OFF.

NOTES - OFF/ON: illuminate sounding notes on the keyboard.

System - AUDIO



Hold HOME for 3 seconds, then release.

01 Turn SELECT to choose the AUDIO section. PRESET cycles the row cursor.

02 Turn KNOBS 1-4 to change the corresponding rows. OCT+ steps/opens the highlighted setting; OCT- closes the menu.

03 Settings are saved when the menu closes if stopped; a save may wait until playback stops. Restart when the screen requests it.

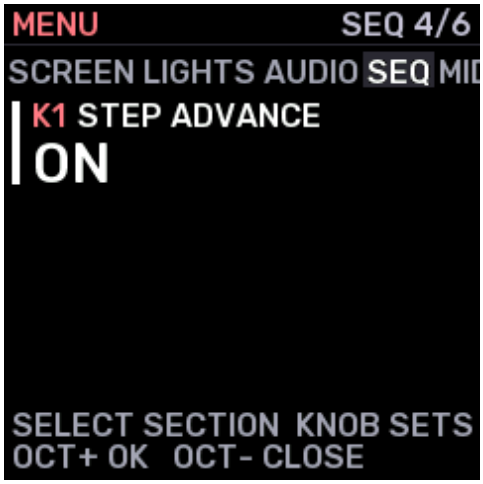
SPEAKER LOWCUT - OFF/ON high-pass protection/tonal adjustment for the internal speaker path.

USB AUDIO - MASTER follows the master output; FULL supplies the full-level USB recording path.

USB SERIAL - OFF/ON USB serial console; a changed setting takes effect after restart.

USB RATE - AUTO / 44.1 KHZ / 48 KHZ USB audio rate; restart after changing.

System - SEQ



Hold HOME for 3 seconds, then release.

01 Turn SELECT to choose the SEQ section. PRESET cycles the row cursor.

02 Turn KNOBS 1-1 to change the corresponding rows. OCT+ steps/opens the highlighted setting; OCT- closes the menu.

03 Settings are saved when the menu closes if stopped; a save may wait until playback stops. Restart when the screen requests it.

STEP ADVANCE - ON moves the NOTES cursor after key release; OFF keeps it in place for manual editing.

System - MIDI



Hold HOME for 3 seconds, then release.

01 Turn SELECT to choose the MIDI section. PRESET cycles the row cursor.

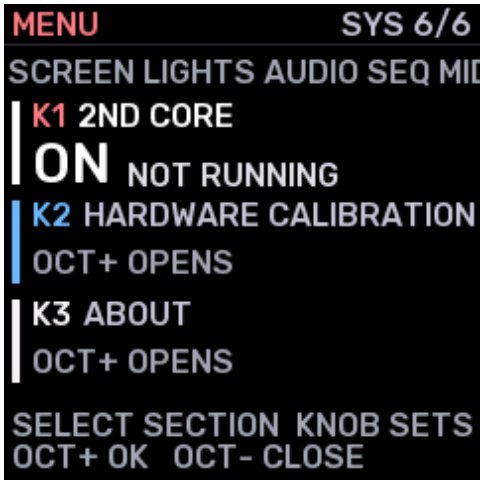
02 Turn KNOBS 1-2 to change the corresponding rows. OCT+ steps/opens the highlighted setting; OCT- closes the menu.

03 Settings are saved when the menu closes if stopped; a save may wait until playback stops. Restart when the screen requests it.

CLOCK OUT (USB) - OFF/ON MIDI clock transmission over USB while internal synchronization is selected.

MPE (CH 2-16) - OFF / 48 ST / 2 ST / 12 ST / 24 ST / 96 ST. Match the member-channel bend range to the controller; CH 2-16 provide independent expression.

System - SYS



Hold HOME for 3 seconds, then release.

01 Turn SELECT to choose the SYS section. PRESET cycles the row cursor.

02 Turn KNOBS 1-3 to change the corresponding rows. OCT+ steps/opens the highlighted setting; OCT- closes the menu.

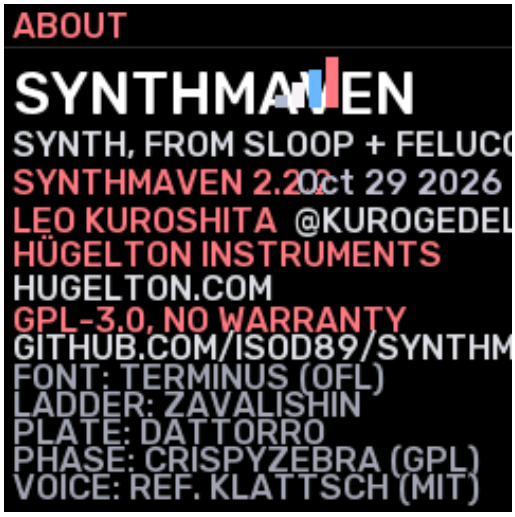
03 Settings are saved when the menu closes if stopped; a save may wait until playback stops. Restart when the screen requests it.

2ND CORE - OFF/ON the second DSP core. NOT RUNNING means the current execution environment has no active second core.

HARDWARE CALIBRATION - OCT+ starts teaching the physical buttons and encoder directions. Only needed for an incorrect panel map.

ABOUT - OCT+ opens version, build/core and attribution information; OCT- returns.

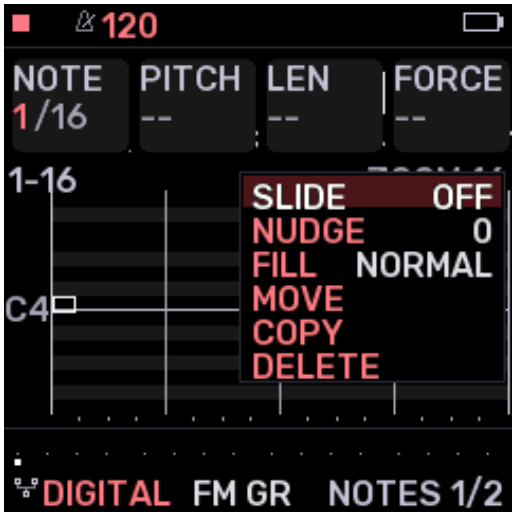
ABOUT



SYS > ABOUT > OCT+.

Use this screen when reporting a problem. Record the firmware version and build/core information. OCT- returns to SYS; OCT- again closes settings. ABOUT is read-only.

Note sheet - SLIDE



SEQ > NOTES > hold OCT+ about 0.4 seconds.

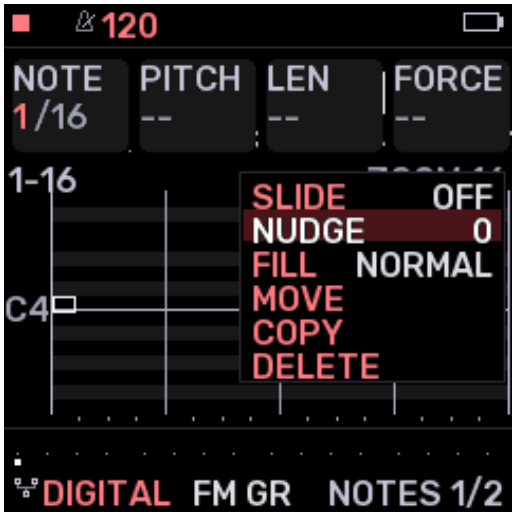
01 Use KNOB 1 on NOTES to choose the note to edit, then hold OCT+ to open its sheet.

02 Use KNOB 1 to highlight SLIDE. KNOB 2 sets OFF/ON, or tap OCT+ to toggle. A slide changes the transition into the note.

03 Use OCT- to close the sheet. Return to HOME when finished, and save the program to retain the edited pattern.

These OCT+ and REC actions have their own timing; they are not the two-second preview / three-second latch used by the main function buttons.

Note sheet - NUDGE



SEQ > NOTES > hold OCT+ about 0.4 seconds.

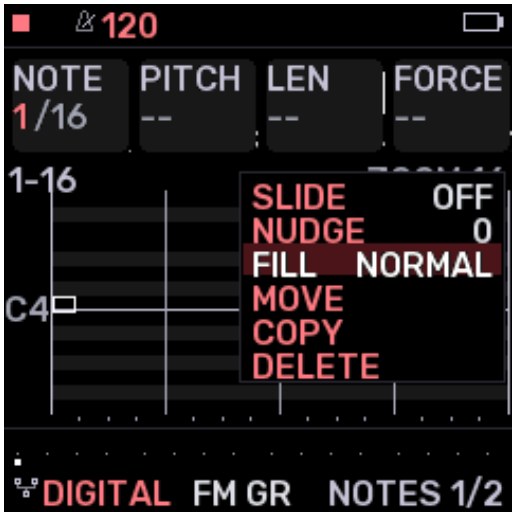
01 Use KNOB 1 on NOTES to choose the note to edit, then hold OCT+ to open its sheet.

02 Use KNOB 1 to highlight NUDGE. KNOB 2 shifts the selected step earlier/later within the allowed microtiming range; zero is on the grid.

03 Use OCT- to close the sheet. Return to HOME when finished, and save the program to retain the edited pattern.

These OCT+ and REC actions have their own timing; they are not the two-second preview / three-second latch used by the main function buttons.

Note sheet - FILL



SEQ > NOTES > hold OCT+ about 0.4 seconds.

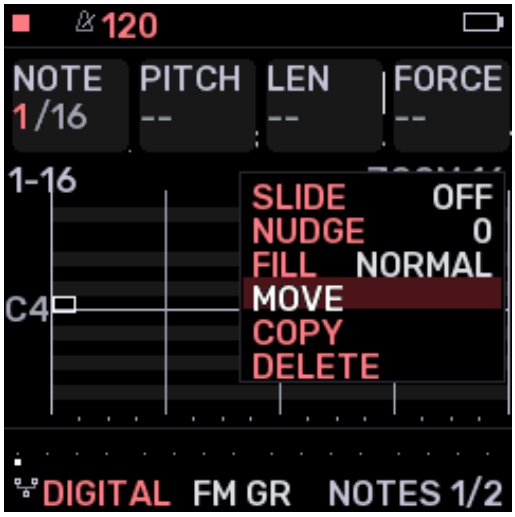
01 Use KNOB 1 on NOTES to choose the note to edit, then hold OCT+ to open its sheet.

02 Use KNOB 1 to highlight FILL. KNOB 2 chooses normal, fill only or no fill. This determines when the note plays.

03 Use OCT- to close the sheet. Return to HOME when finished, and save the program to retain the edited pattern.

These OCT+ and REC actions have their own timing; they are not the two-second preview / three-second latch used by the main function buttons.

Note sheet - MOVE



SEQ > NOTES > hold OCT+ about 0.4 seconds.

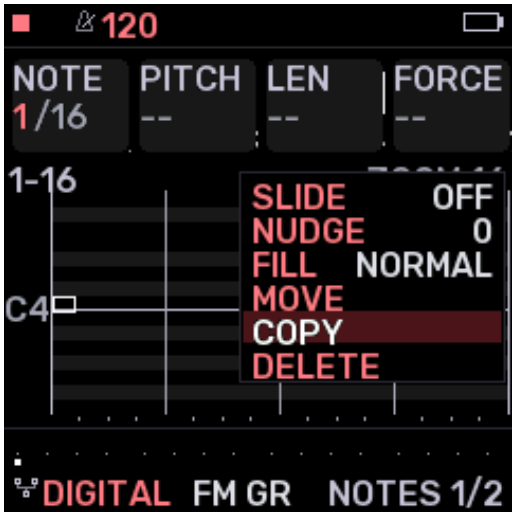
01 Use KNOB 1 on NOTES to choose the note to edit, then hold OCT+ to open its sheet.

02 Use KNOB 1 to highlight MOVE. KNOB 2 moves the note by steps, stopping when there is no room.

03 Use OCT- to close the sheet. Return to HOME when finished, and save the program to retain the edited pattern.

These OCT+ and REC actions have their own timing; they are not the two-second preview / three-second latch used by the main function buttons.

Note sheet - COPY



SEQ > NOTES > hold OCT+ about 0.4 seconds.

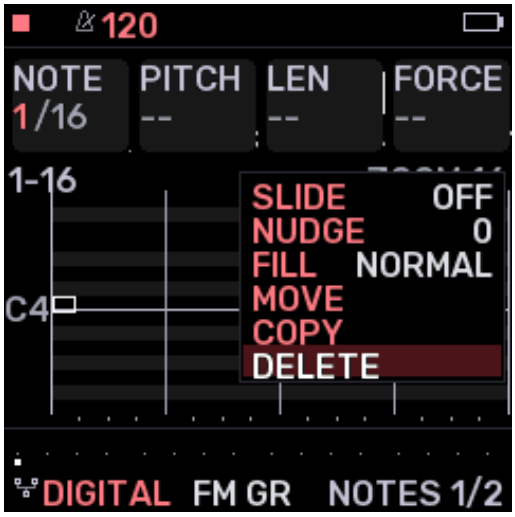
01 Use KNOB 1 on NOTES to choose the note to edit, then hold OCT+ to open its sheet.

02 Use KNOB 1 to highlight COPY. Tap OCT+ to copy immediately after the note. The sheet then selects MOVE so KNOB 2 can reposition the copy.

03 Use OCT- to close the sheet. Return to HOME when finished, and save the program to retain the edited pattern.

These OCT+ and REC actions have their own timing; they are not the two-second preview / three-second latch used by the main function buttons.

Note sheet - DELETE



SEQ > NOTES > hold OCT+ about 0.4 seconds.

01 Use KNOB 1 on NOTES to choose the note to edit, then hold OCT+ to open its sheet.

02 Use KNOB 1 to highlight DELETE. Tap OCT+ to delete the selected note. The sheet closes after deletion.

03 Use OCT- to close the sheet. Return to HOME when finished, and save the program to retain the edited pattern.

These OCT+ and REC actions have their own timing; they are not the two-second preview / three-second latch used by the main function buttons.

EDIT mixer

Short taps and timed holds have different outcomes.



layer mixer

01 Hold EDIT for 2 seconds for the mixer preview; keep holding to 3 seconds, then release to latch it.

02 KNOBS 1-4 change the four part levels. SELECT chooses the focused part. PRESET sets its pan. ALGORITHM sets the split boundary from A1 to A6.

03 Use GLO > LAYER ROUTE to choose lower/upper/full range. Tap EDIT or HOME to leave. Save the program after changes.

GLO layer toggle

Short taps and timed holds have different outcomes.



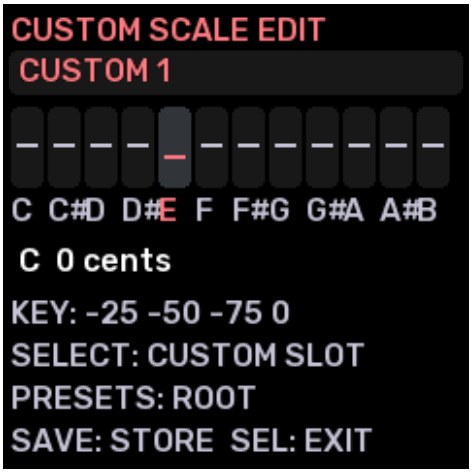
layered toggle

01 Hold GLO for 2 seconds. The screen explicitly shows Layered ON or Layered OFF.

02 The toggle fires once per hold. Release to return; GLO never latches. A short tap opens GLO pages.

SEL custom tuning

Short taps and timed holds have different outcomes.



custom scale

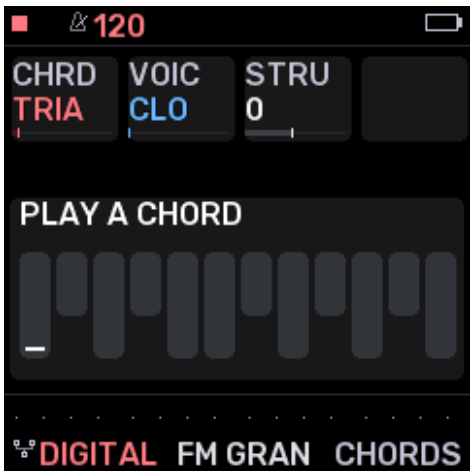
01 Hold SEL for 3 seconds, then release to latch the custom-scale editor.

02 Use SELECT to choose CUSTOM 1-4 and PRESET to set its root. Press a note class repeatedly: -25, -50, -75, then 0 cents. It affects that class wherever master tuning applies.

03 Press SAVE to store the scale (the screen may report SCALE SAVE PENDING until playback stops), then select its root and octave boundary under TUNING. Tap SEL or HOME to leave.

ENV chords

Short taps and timed holds have different outcomes.



chords

01 Hold ENV for 3 seconds, then release to open CHORDS. A tap opens amplitude-envelope pages.

02 KNOB 1 chooses a compatible voicing, KNOB 2 chord voicing, KNOB 3 strum. Play notes across the selected scale.

03 For maqam/microtonal tuning, incompatible choices are hidden. Turn chords OFF for ordinary single-note playing.

SEQ step layer

Short taps and timed holds have different outcomes.



hold seq

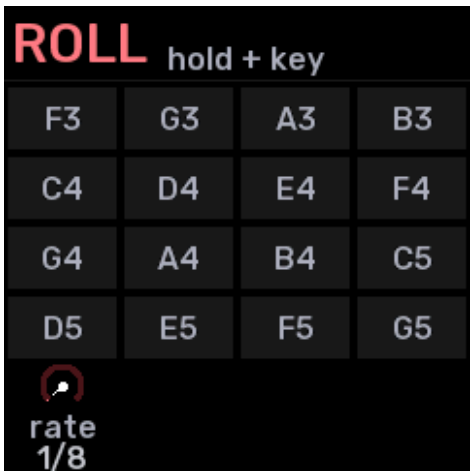
01 Hold SEQ for 2 seconds to preview the step keyboard, or 3 seconds to latch.

02 The sixteen white keys address steps in the current page; the black page keys select the step group. Hold a step and use KNOBS 1-4 for note, level, ratchet and nudge.

03 With a step held, ALGORITHM chooses a lock parameter and PRESET changes its locked value. OCT- clears its nudge/locks/fill condition; OCT+ cycles its fill condition. Tap SEQ or HOME to exit.

ARP repeat layer

Short taps and timed holds have different outcomes.



hold arp

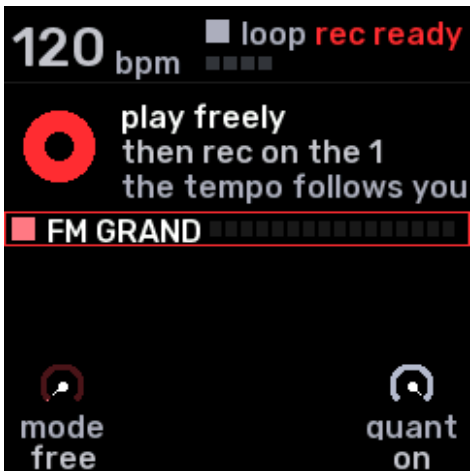
01 Hold ARP for 2 seconds for the repeat layer, or 3 seconds to latch.

02 Hold a playing key to repeat it; KNOB 1 selects the repeat rate. Release the key to stop its repeat.

03 Tap ARP or HOME to leave the layer. The normal ARP pages configure patterned note generation separately.

REC setup

Short taps and timed holds have different outcomes.



record armed

01 Tap REC to arm recording or begin recording in the current transport context.

02 KNOB 1 selects FREE/TEMPO when the project is empty. KNOB 2 chooses 1, 2 or 4 bars for tempo recording. KNOB 3 chooses note-start/count-in. KNOB 4 switches record timing quantization.

03 Play to start when note-start is selected, or follow the count-in. Tap REC to finish recording. Use PLAY for transport and NOTES to edit the result.

REC clear confirmation

Short taps and timed holds have different outcomes.



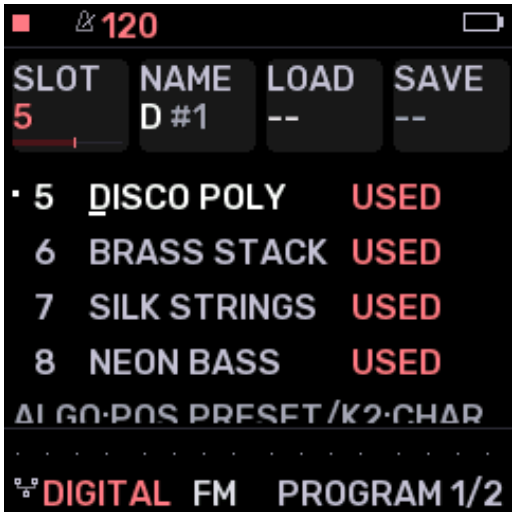
hold rec clear

01 Hold REC. After about 0.7 seconds the initial recording press is undone and the clear ring begins.

02 Release before the ring completes to cancel. Continue for another 1.3 seconds to clear the selected track/pattern.

03 This is a destructive clear, not a three-second menu latch. Save a copy or export a backup before clearing wanted material.

Saved slots 5-8

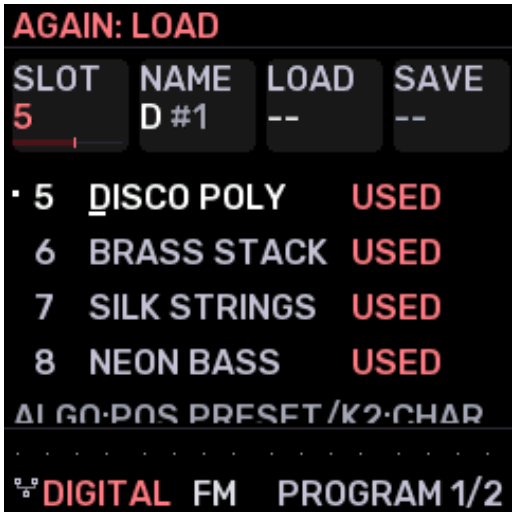


Production-rendered example state.

On SAVE > PROGRAM, turn SELECT past slot 4. The list follows the selected group of four. Slot selection only browses; LOAD or SAVE is a separate confirmed action.

Read the on-screen state before acting. A status/confirmation screen is contextual; it does not create an extra permanent menu page.

Load confirmation

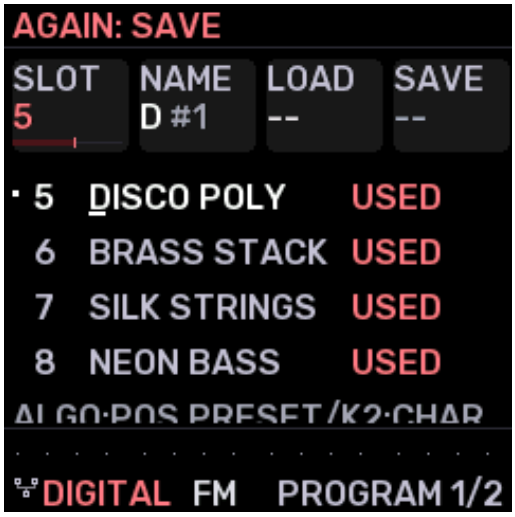


Production-rendered example state.

Turn KNOB 3 once. AGAIN: LOAD asks for a second turn within about 1.5 seconds. If you wait or change the slot, the pending action cancels.

Read the on-screen state before acting. A status/confirmation screen is contextual; it does not create an extra permanent menu page.

Save confirmation



Production-rendered example state.

Turn KNOB 4 once. Verify the slot and name, then turn it again within about 1.5 seconds to overwrite that slot. Wait without confirming to cancel.

Read the on-screen state before acting. A status/confirmation screen is contextual; it does not create an extra permanent menu page.

Clear pattern confirmation



Production-rendered example state.

SAVE > TOOLS > CLRSQ: a first turn arms; a second confirms. This clears the working sequence, so save or back up first.

> Read the on-screen state before acting. A status/confirmation screen is contextual; it does not create an extra permanent menu page.

Initialize sound confirmation



Production-rendered example state.

SAVE > TOOLS > INIT: confirm with a second turn only if you want to reset the focused working sound.

Read the on-screen state before acting. A status/confirmation screen is contextual; it does not create an extra permanent menu page.

New working program confirmation

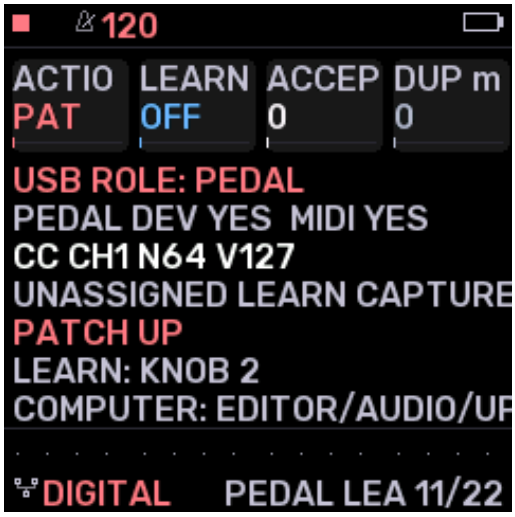


Production-rendered example state.

SAVE > TOOLS > NEW: confirm with a second turn to initialize the working project/program. Saved versions and an external backup are the recovery path.

Read the on-screen state before acting. A status/confirmation screen is contextual; it does not create an extra permanent menu page.

Pedal detected

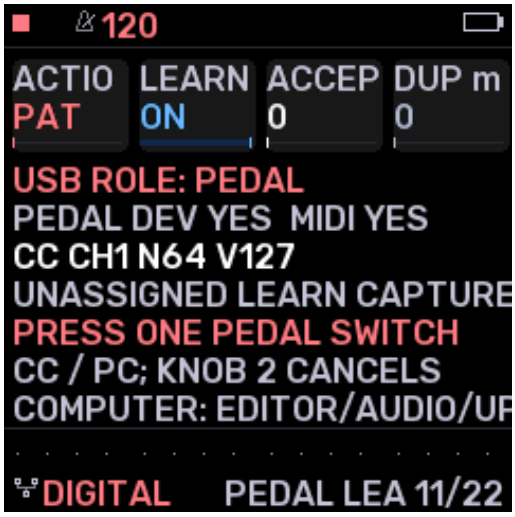


Production-rendered example state.

GLO > PEDAL LEARN separates USB device presence from MIDI-interface presence. PEDAL DEV YES / MIDI NO means enumeration alone has not supplied a usable MIDI interface.

Read the on-screen state before acting. A status/confirmation screen is contextual; it does not create an extra permanent menu page.

Learn armed

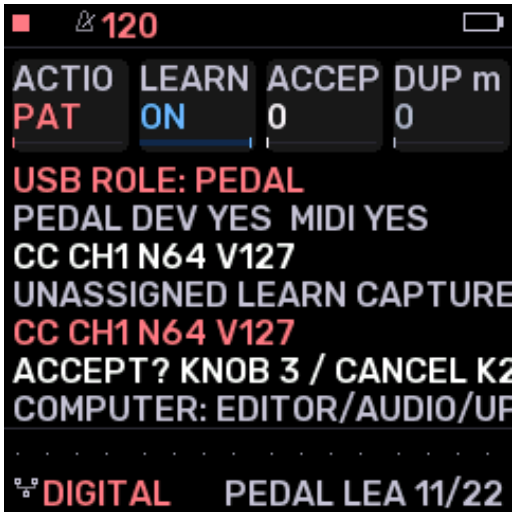


Production-rendered example state.

Choose ACTION with KNOB 1, then arm with KNOB 2. Press one pedal switch. KNOB 2 cancels without committing an assignment.

> Read the on-screen state before acting. A status/confirmation screen is contextual; it does not create an extra permanent menu page.

Accept captured MIDI



Production-rendered example state.

Check the captured CC/PC type, channel, number and value. KNOB 3 accepts; KNOB 2 cancels. An already-assigned conflicting message is rejected.

Read the on-screen state before acting. A status/confirmation screen is contextual; it does not create an extra permanent menu page.

Test pedal action

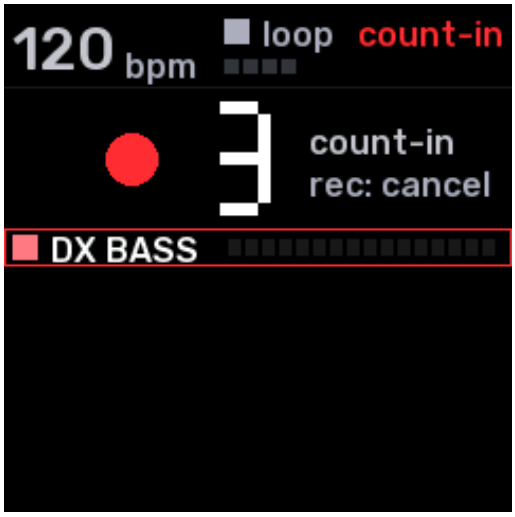


Production-rendered example state.

Press the assigned switch again. The display reports the last message, matched action and trigger result. DUP SUPPRESSED means repeat suppression discarded it; NO MATCH means there was no matching assignment.

> Read the on-screen state before acting. A status/confirmation screen is contextual; it does not create an extra permanent menu page.

Recording count-in

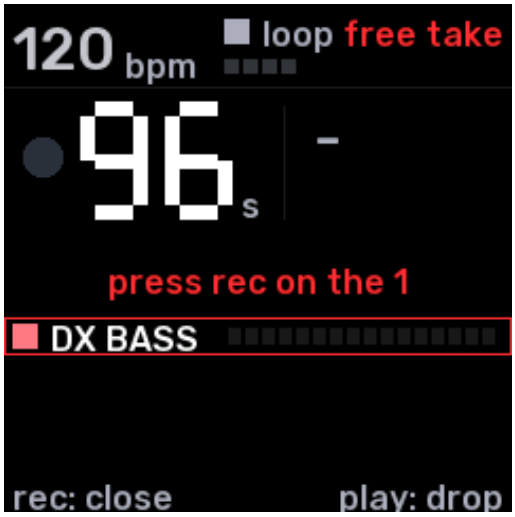


Production-rendered example state.

Choose TEMPO and COUNT on the REC setup. Start the take and wait for the count-in. The selected bar length and clock source determine the recorded loop.

Read the on-screen state before acting. A status/confirmation screen is contextual; it does not create an extra permanent menu page.

Free take recording



Production-rendered example state.

In an empty project, choose FREE and play without a fixed tempo. REC on the next downbeat closes the take and fits a loop; PLAY discards the take. Review the result before saving.

Read the on-screen state before acting. A status/confirmation screen is contextual; it does not create an extra permanent menu page.

Hardware calibration

01 Open HOME hold > SYS > HARDWARE CALIBRATION > OCT+, or hold both octave buttons during power-up. Release the buttons when asked.

02 Press the exact printed button requested at each prompt. All fourteen buttons are learned once.

03 Turn each requested encoder clockwise. The firmware learns both its physical position and its direction.

04 Finish all prompts. If you leave the setup idle, it cancels and restores the old map. Do not use calibration to repair unrelated USB or audio problems.

Calibration prompts

Press only the physical control named on the screen.



Press FX



Press SEL



Press ENV



Press LFO

Each prompt advances after the requested input. Incorrect mappings change which physical control performs an action; repeat calibration if the panel behaves inconsistently.

Calibration prompts

Press only the physical control named on the screen.



Press EDIT



Press GLO



Press HOME



Press SAVE

Each prompt advances after the requested input. Incorrect mappings change which physical control performs an action; repeat calibration if the panel behaves inconsistently.

Calibration prompts

Press only the physical control named on the screen.



Press ARP



Press SEQ



Press PLAY



Press REC

Each prompt advances after the requested input. Incorrect mappings change which physical control performs an action; repeat calibration if the panel behaves inconsistently.

Calibration prompts

Press only the physical control named on the screen.



Press OCT-



Press OCT+

Each prompt advances after the requested input. Incorrect mappings change which physical control performs an action; repeat calibration if the panel behaves inconsistently.

Calibration prompts

Turn clockwise only the physical control named on the screen.



Turn clockwise SELECT



Turn clockwise ALGORITHM



Turn clockwise PRESET



Turn clockwise KNOB 1

Each prompt advances after the requested input. Incorrect mappings change which physical control performs an action; repeat calibration if the panel behaves inconsistently.

Calibration prompts

Turn clockwise only the physical control named on the screen.



Turn clockwise KNOB 2



Turn clockwise KNOB 3



Turn clockwise KNOB 4

Each prompt advances after the requested input. Incorrect mappings change which physical control performs an action; repeat calibration if the panel behaves inconsistently.

Reference coverage and page finder

This edition documents 65 menu page definitions and 151 captured layouts. Additional screens cover six system sections, ABOUT, all six note-sheet rows, secondary functions, recording and every hardware-calibration prompt.

ENV

ENV - 37

ENV DEST - 38

ENV 2 - 39

ENV 2 LOOP - 40

OP ENV - 41

EDIT

FILTER - 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57

FILTER GAIN - 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73

EDIT 1 - 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85, 86, 87, 88, 89, 90, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101

EDIT 2 - 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113, 114, 115, 116, 117, 118, 119, 120, 121, 122, 123, 124, 125, 126, 127, 128

OP - 129

OP 2 - 130

ROUTE - 131

FMX - 132

VOICE - 133

CHARACTER - 134

LFO

LFO - 135

LFO DEST - 136

MOD PRESET - 137

MOD - 138

LFO 2 - 139

LFO 3 - 140

MATRIX - 141

MOD SEQ - 142

Reference coverage and page finder - continued

MOD SEQ 2 - 143

FX

FX - 144

PART FILTER - 145

INSERT - 146, 147, 148, 149, 150

INSERT 2 - 151

DLY - 152

DLY 2 - 153

REVERB - 154

CHORUS - 155

EQ+COMP - 156

SEL

SEL - 157

SEL 2 - 158

CHORDS

CHORDS - 159

GLO

GLOBAL - 160

MASTER - 161

TUNING - 162

TUNING 2 - 164

CONTROLLER - 165

SYSTEM - 166

MIDI - 167

MIDI 2 - 168

PATCH MIDI - 169

KEY MIDI - 170

Reference coverage and page finder – continued

PEDAL LEARN - 171

SCENES - 172

SETLIST - 173

PEDAL MAP - 174

PATCH UP - 175

PATCH DOWN - 176

BANK UP - 177

BANK DOWN - 178

USB ROLE - 179

LAYER ROUTE - 180

LAYER MIX - 181

LAYER PITCH - 182

SAVE

PROGRAM - 183

TOOLS - 184

ARP

ARP - 185

ARP 2 - 186

SPICE - 187

SEQ

NOTES - 188

PATTERN - 189

Editor session and sound tools

These controls live in the web editor. The preceding device-screen reference shows the corresponding on-device controls where available. Editor-only panels are described here; native LCD captures are not presented as browser screenshots.

Connect / Read

Use COMPUTER USB role, choose the FM-1 ports and allow SysEx. Read the current state before editing an existing sound. Mock mode previews the interface and does not contact hardware.

Part / engine / preset

Choose the intended part before editing. The four-patch browser browses sounds; a complete factory program is recalled separately from Programs. Engine-specific cards appear only for the matching engine/protocol.

Undo / Redo

Undo returns through sound-edit history; Redo reapplies an undone edit. Ctrl+Z and Ctrl+Shift+Z are the editor shortcuts. Save wanted versions before large edits/imports.

Compare

Temporarily compare with the loaded/original sound. Return to the edited state before continuing work so you know which state is being changed.

Set A / Set B / A-B / Morph

Capture two working sound states, switch between them, or move Morph from 0 to 127. Continuous values interpolate; discrete choices switch. Across different engines, the engine changes rather than both running as extra layers.

Scope / Amount / Random

Choose which section to randomize and how far (0-127; initial 40). EDIT, +ENV, ALL, ENV, MOD, FX and OPS restrict the operation. Keep a saved starting version and use Undo if the result is unsuitable.

What changed

Expand the differences view to compare edited parameter values with the reference. This is an edit aid, not a device save or backup.

HYBRID: oscillator settings

These controls live in the web editor. The preceding device-screen reference shows the corresponding on-device controls where available. Editor-only panels are described here; native LCD captures are not presented as browser screenshots.

Read / Send / Live

Read the selected part patch. With Live ON, supported changes send immediately; with Live OFF they remain staged until Send. Init and factory Load replace the staged patch; Export/Import use patch JSON. Name holds up to ten characters.

OSC 1-3 Type

OFF, SAW, PULSE, TRI, SINE, MORPH, SUPER, FM, SYNC, FOLD or NOISE. OFF removes that source; the three oscillator levels form the source mix.

Coarse / Fine / Level

Coarse -48 to +48 semitones; Fine -64 to +63 cents; Level 0-127. Coarse/fine change source pitch relative to the key, not the selected master scale.

Shape

0-127, dependent on type: PULSE width; SINE feedback; MORPH sine/triangle/saw/square position; SUPER detune; FM index; SYNC pitch over oscillator 1; FOLD amount; NOISE darkness. Some types have no shape behavior.

Reset / Fixed

Reset restarts phase for repeatable attacks. Fixed makes oscillator frequency independent of ordinary note tracking; use it deliberately for fixed resonances/percussion.

Ring 1 x 2 / Drift / Drive

Each 0-127: ring modulation between sources 1 and 2, analog-style variation, and input saturation. Start with modest levels and compare at matched output volume.

HYBRID: filters and routing

These controls live in the web editor. The preceding device-screen reference shows the corresponding on-device controls where available. Editor-only panels are described here; native LCD captures are not presented as browser screenshots.

Route

SER sends through filters in series; PAR processes parallel paths; SPLIT uses the engine split routing. These are internal synthesis routes, distinct from the four-part keyboard split.

Balance F1-F2

0-127, shown for PAR. Adjusts the relative parallel filter contribution.

Filter 1 / Filter 2 Type

OFF, LP12, LP24, BP, HP, NOTCH, PEAK or MOOG. The compact firmware FILTER page addresses filter 1; the editor exposes both and their route.

Cutoff / Resonance

Both 0-127 internally, cutoff displayed in Hz. Cutoff sets the spectral boundary; resonance emphasizes it. High resonance/drive can change level sharply.

ENV2

-64 to +63: signed depth from HYBRID envelope 2 to cutoff. Negative values reverse the contour.

Key / Vel

Key -64 to +63: cutoff tracking across the keyboard. Vel 0-127: velocity influence on the filter. Test low/high notes and soft/hard playing.

Output compensation

The separate FILTER GAIN compensation switch estimates attenuation for resonance/drive changes. It does not measure integrated LUFS or replace a deliberate mixer balance.

HYBRID: modulation and global settings

These controls live in the web editor. The preceding device-screen reference shows the corresponding on-device controls where available. Editor-only panels are described here; native LCD captures are not presented as browser screenshots.

ENV2 / ENV3

Each has A, D, S, R stored 0-127. A/D/R display time; S displays percent. ENV2 is available directly to filter modulation; both can be matrix sources.

LFO2 / LFO3 Wave

SINE, TRI, SAW, SQR, S&H or SMTH. KEY restarts with a note; SYNC changes rate to a tempo division; FADE delays the full modulation depth.

LFO Rate / Fade

0-127 each. Synced rate choices cover 1/32, triplets, 1/16 through whole notes, dotted/long divisions and up to eight bars. LFO1 remains on the shared LFO page.

16-slot matrix Source

ENV2, ENV3, AENV, LFO1-3, VEL, KEY, MW, AT, BEND, EXPR, RAND, M1-4 or ONE. An empty source/destination leaves a slot inactive.

Via / Destination / Amount

Via scales a source by another controller; blank leaves it unscaled. Destinations include global/per-oscillator pitch, shape and levels, ring, cutoff(s), resonance(s), drive, amp, LFO2/3 and filter balance. Amount -64 to +63 sets signed depth.

Bend / Level

Bend 0-24 semitones; patch Level 0-127. These belong to the custom HYBRID patch and are separate from the program mixer fader.

Save a custom patch

After sending and checking the sound, save the containing program so its patch bytes travel with it. Also export the patch JSON and editor library entry for reuse.

FM6: patch and voice controls

These controls live in the web editor. The preceding device-screen reference shows the corresponding on-device controls where available. Editor-only panels are described here; native LCD captures are not presented as browser screenshots.

Read / Send / Live

Read and edit the selected FM6 patch. Live controls immediate sending. Bank slot Load/Store/Erase operates FM6 patch storage; it is different from saving a four-part program or a playable user sound.

Name / ALG / FB / OKS / TRNSP

Name up to ten characters. ALG chooses 1-32 algorithms; the display identifies carriers/feedback operator. FB 0-7 sets feedback. OKS 0/1 sets oscillator key synchronization. TRNSP 0-48 is the patch transpose encoding with center at 24.

LFO Wave / Speed / Delay

Wave TRI, SAW-, SAW+, SQR, SIN or S&H. Speed and Delay 0-99 control rate and delayed onset.

PMD / AMD / PMS / Sync

Pitch-mod depth and amplitude-mod depth 0-99; pitch-mod sensitivity 0-7; key sync 0/1. Operator AMS also determines amplitude-mod response.

Pitch envelope

Four rates PR1-PR4 and four levels PL1-PL4, each 0-99. These are DX7-style rate/level stages, not ordinary ADSR time values.

Import / Export / Init

Import supported DX7 single/bank SysEx or binary data, inspect a voice, then send/store it deliberately. Export patch or bank keeps a reusable copy. Init replaces the current patch with a starting voice.

FM6: every operator field

These controls live in the web editor. The preceding device-screen reference shows the corresponding on-device controls where available. Editor-only panels are described here; native LCD captures are not presented as browser screenshots.

R1-R4 / L1-L4

Each 0-99: four envelope rates and four target levels. Rates control speed; levels control stage amplitude. A carrier envelope shapes loudness, a modulator envelope shapes brightness.

BP / LD / RD

BP 0-99 is keyboard-scaling breakpoint; LD/RD 0-99 set scaling depth below/above it.

LC / RC

Left/right scaling curves: -LIN, -EXP, +EXP, +LIN. These determine how operator level changes away from the breakpoint.

RS / AMS / KVS

Rate scaling 0-7 changes envelope speed with key pitch. Amplitude-mod sensitivity 0-3 controls LFO amplitude response. Key-velocity sensitivity 0-7 controls response to playing force.

OL

Operator output level 0-99. A carrier contributes to output; a modulator changes the spectrum of the operators it feeds.

MODE / FC / FF / DET

MODE is RATIO or FIXED. FC 0-31 sets coarse frequency; FF 0-99 fine frequency; DET 0-14 is centered at 7. The frequency column displays the resulting ratio/frequency.

Imported bank selection

Click a voice to inspect/edit it. Tick voices to assign bank order B1 onward; Select first / None changes the selection. Choose bank-only or playable user-preset destinations before Store cartridge. Keep the source SysEx file.

FMX: routing and operator settings

These controls live in the web editor. The preceding device-screen reference shows the corresponding on-device controls where available. Editor-only panels are described here; native LCD captures are not presented as browser screenshots.

Read / Send / Live / Init

Same staged-versus-live patch workflow as HYBRID. Load a factory patch or Init, then edit. Import/Export use FMX JSON. Name is up to ten characters; Store To writes the selected FMX bank slot.

Algorithm template / routing grid

A template supplies starting routes. Rows are TO operators; columns are FROM operators. Each matrix amount 0-127 sends the column operator into the row operator. Diagonal cells are feedback. Manual edits clear the template selection.

OUT

Each operator output amount 0-127. At least one audible operator must reach output; routing modulation alone does not guarantee an audible sound.

Wave / Mode

Wave SINE, TRI, SAW, SQR, HALF, ABS, QUART, PULSE, SIN8 or NOISE. Mode PM, FM or RING changes the modulation method.

Fixed / Ratio / Fine

Fixed OFF uses ratios 0.5, 1-31. Fixed ON uses bases 1, 10, 100 or 1000 Hz. Fine 0-99 adds a ratio offset or a fixed-frequency multiplier.

Level / ADSR / Vel / KSC / DTN

Level, ADSR stages, velocity and key scaling 0-127; detune stored 0-14, displayed -7 to +7. Each of six operators has these controls.

PEG / PTIM / Level

Patch pitch-envelope amount, pitch-envelope time and output level (0-127). PEG is centered at 64 and displayed in semitones. Save the program after sending the patch.

Layers, scales and modulation sequencer in the editor

These controls live in the web editor. The preceding device-screen reference shows the corresponding on-device controls where available. Editor-only panels are described here; native LCD captures are not presented as browser screenshots.

Read mixer / part strips

Read the four-part mixer, then set each part level, pan, mute/solo, enabled state, keyboard split side/boundary and transpose. Layer program sounds ON combines enabled routed parts; OFF uses the focused part.

Tuning system / Root / Apply from octave

Choose master temperament/maqam/custom tuning, its root, and C0-C8 lower boundary. Notes below the boundary retain equal-tempered intervals. This does not limit ordinary note quantization or chord generation.

Apply master tuning / A4

Per-part checkbox ON by default; OFF bypasses master tuning for that sound. A4 400-480 Hz is the shared reference. QNT ALL applies the focused ROOT/SCL across all four parts and MIDI independently.

Load .scl / Send

Choose a Scala tuning file, review the imported scale information and send it. Keep the original file and save the program/backup containing your setup. A Scala file provides tuning, not a new sound engine.

Mod-sequencer Read / Send / Live

Read four 16-step lanes; edit their step values -8 to +7. Live sends edits, otherwise Send commits the staged lane data.

Shape / Apply

Apply RAMP+, RAMP-, TRI, SINE, SQR, RND, FLAT, INVRT or SHIFT to the chosen modulation lane. Rate, length, smoothing and sync/key restart are set on MOD SEQ 2; route SEQ1-4 through the matrix.

Sequencer editor: notes and performance lanes

These controls live in the web editor. The preceding device-screen reference shows the corresponding on-device controls where available. Editor-only panels are described here; native LCD captures are not presented as browser screenshots.

Reload / Clear / Length

Reload reads the selected part pattern; Clear removes its notes. Length is 1-64 steps. Keep a saved version before destructive transformations.

Note grid / piano roll

Select a note or step, adjust pitch and duration, and use the displayed selection/roll controls to choose the editing range. Read the current step detail before replacing a chord member.

Copy/paste / shift / transpose / reverse / double

These transform the selected pattern/range. Double repeats/extends material within the 64-step limit. Check ties and pattern length after transformations.

Chord / accent / slide tools

Choose a chord transform for selected notes, or switch accent/slide flags on/off for the selected range. These edit pattern data, separate from the live CHORDS generator.

Velocity / nudge / ratchet / hit level lanes

Velocity sets note force; nudge offsets timing; ratchet repeats hits; hit level shapes individual step strength. Draw the lane or enter a range value and Apply.

Step list

Expand the list for a numerical view of notes/step attributes. It is another view of the same working pattern, not a second saved sequence.

Sequencer generators and pattern library

These controls live in the web editor. The preceding device-screen reference shows the corresponding on-device controls where available. Editor-only panels are described here; native LCD captures are not presented as browser screenshots.

Key / scale / base

Generator root, scale and base register shape generated note choices. These are generation settings, not an automatic replacement for master maqam tuning.

Euclidean

Hits 0-64, steps 1-64, rotation 0-63, note text (initial C3), accent and merge options. Make distributes hits; Merge adds to existing notes instead of replacing the selected material.

Random melody

Density, motion, ties, accent, slide and chords each 0-100; octaves 1-4; seed 0-999999. Make uses those settings; Dice changes the random seed/result.

Text/pattern presets

Choose a preset or enter supported pattern text, then Make. Check the preview/result before saving the program.

Humanize

Velocity variation up to +/-64 and nudge up to +/-32 in the generator controls. Apply introduces bounded variation; it does not make all notes equally loud.

Pattern library

Name up to 40 characters, Save/Load/Delete, export one pattern or the bank, and import pattern JSON. This browser library is separate from named device programs and should be exported for backup.

Library, user bank, imports and backups

These controls live in the web editor. The preceding device-screen reference shows the corresponding on-device controls where available. Editor-only panels are described here; native LCD captures are not presented as browser screenshots.

Search / engine / tags / sort

Filter and order browser library entries. Rename changes its display name; Tags classify it; Duplicate makes another library record; Delete removes that record.

Audition / To Slot / Save to Library

Audition loads a working sound for listening. To Slot writes the chosen device user-sound destination. Save to Library captures a browser entry; it is not a four-part program save.

Import / Export patch / Export library

Patch JSON and full library JSON support reuse. Importing files into the browser does not automatically populate every device storage area.

Device bank Load / Store / Get / Erase

Load recalls a user sound; Store writes the working sound to that slot; Get copies it into the editor library; Erase clears the selected device user sound. Refresh rereads the bank. Export Bank keeps an external copy.

Vital / minilogue xd / DX7

Vital and minilogue xd imports translate supported synthesis settings into HYBRID approximations. Review unsupported-feature reports before sending. Korg custom-unit binaries cannot execute on FM-1. DX7 import targets FM6.

Wavetables

Upload supported user wavetable data, then select the corresponding user table in WAVE. USER WT intentionally stays silent until data has been uploaded; it has no automatic audible factory fallback.

Programs / full backup

Recall a read-only factory program or load/save a named user program. Full backup exports/restores device storage; do not confuse this with Export Library. Stop playback/release notes and keep the original backup file.

Controller and pedal setup: complete workflow

These controls live in the web editor. The preceding device-screen reference shows the corresponding on-device controls where available. Editor-only panels are described here; native LCD captures are not presented as browser screenshots.

Automatic controller identity

Host mode reads vendor/product/interface identity. Alesis V/VI/V Mini, Novation Launchkey/FLkey/Impulse/modern Launchpad, Arturia MiniLab/KeyLab/KeyStep/BeatStep/MicroLab, Akai MPK/MPD/LPD8/LPK25/MIDImix/APC and Mackie families have documented handling.

Standard / MCU / HUI override

Automatic selects the profile; Standard MIDI disables surface decoding; MCU or HUI forces the supported control-surface interpretation. DAW port AUTO or 1-16 selects its USB MIDI cable. Musical and DAW traffic remain separate.

MCU/HUI supported controls

First four channel strips map faders to part volume, encoders to pan, and switches to mute/solo/select/record arm. Play/Stop/Record controls transport. Feedback updates supported controls; touched motor faders are not overwritten.

Pad mappings / limits

Supported modern Launchpads use a chromatic 8x8 C2-D#7 layout and feedback. APC Mini uses chromatic pads and the first four volume faders. Unknown devices keep class-compliant standard MIDI. Extra strips, scribble strips and arbitrary DAW scripts are not implemented.

Pedal Learn / Accept / test

Enable patch switching, choose the action, arm Learn, press the pedal, inspect CC/PC/channel/number/value, Accept, then press again to test. Cancel discards the candidate. Duplicate/conflicting assignments are refused.

Repeat suppression / USB role

Choose 0-500 ms suppression only for unwanted duplicates; zero is OFF. CC release re-arms a tap; identical PC taps inside the window can be suppressed. Return to COMPUTER for editing/updates; host mode needs powered hardware and physical compatibility validation.